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Preface

This textbook was written to fill a gap that became apparent in teaching ECON 204 at the University of Northern British Columbia: most available economics textbooks treat contemporary issues through an American or generic international lens, leaving Canadian students without a primary reference that speaks directly to their economy, their institutions, and their policy debates.

The Canadian economy has its own character. It is deeply integrated with the United States yet markedly different in its institutional architecture—universal healthcare, a federal carbon price, a publicly owned central bank with an explicit inflation target, and a tradition of provincially diverse labour markets. Housing affordability in Vancouver and Toronto is not the same problem as in Houston. Canada’s experience with post-COVID inflation, while similar in some dimensions to the American experience, has its own drivers, timeline, and policy response. Understanding these differences requires a textbook that takes Canada seriously as its own economic case study.

Who This Book Is For

This textbook is written for second-year undergraduate students in economics or related social science disciplines who have completed introductory microeconomics and macroeconomics. It assumes you know what supply and demand curves are, understand the basics of GDP, and have been introduced to the concepts of inflation and unemployment. It does not assume you have studied advanced mathematics or econometrics.

The goal is to help you move from understanding basic economic concepts to using them as analytical tools for reasoning about real, complex, messy issues—the kind you read about in the news every day.

What Makes This Textbook Different

Three features distinguish this textbook:

First, it is **Canadian**. More than 80% of the examples, case studies, and data refer to Canada. When international comparisons are useful, they are made explicitly and for a reason.

Second, it is **contemporary**. Each chapter addresses an issue that is actively shaping Canada’s economic future: the inflation surge of 2022–2023, the housing crisis, artificial

intelligence and the future of work, and climate policy. Economic theory is the tool; contemporary Canada is the subject.

Third, it is **honest about uncertainty**. Economics does not always deliver clean answers. Every chapter includes a “Policy Debate” section that presents multiple perspectives, weighs the evidence, and acknowledges what we do not yet know. This is closer to how economics is actually practised than many textbooks suggest.

Acknowledgements

I am grateful to students in ECON 204 at UNBC whose questions and discussions have shaped the structure and tone of every chapter. Northern British Columbia, with its resource economy, Indigenous communities, and distinctive labour markets, provides a uniquely rich context for studying contemporary economic issues, and I have tried to honour that richness throughout.

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Prince George, British Columbia
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How to Use This Textbook

Each chapter in this textbook follows a consistent structure designed to help you build understanding progressively. Here is a guide to the features you will encounter.

Chapter Features

Learning Objectives

Every chapter opens with a list of what you should be able to do after studying the chapter. Return to this list when reviewing for exams.

Definition Boxes

Key terms are defined in gray boxes the first time they appear. These terms form the foundation of economic reasoning and will appear in exam questions.

Canadian Example Boxes

Teal boxes highlight real Canadian applications of the theory being discussed. These help connect abstract concepts to recognizable events and institutions.

Think Like an Economist

Purple boxes invite you to apply economic reasoning to a question or scenario. These are excellent preparation for analytical exam questions.

Policy Debate Boxes

Blue boxes present competing perspectives on a policy question. They show that economics involves genuine disagreement and trade-offs, not just formulas.

Key Takeaway

Green summary boxes at strategic points in the chapter capture the most important insight from the preceding section.

Data Spotlight

Orange boxes present important Canadian economic data, explain how to read it, and discuss what it tells us.

End-of-Chapter Material

Each chapter ends with:

- **Chapter Summary** — a concise review of the main ideas.
- **Review Questions** — concept and comprehension questions aligned with the learning objectives.
- **Discussion Questions** — open-ended questions for seminar discussion or short written responses.
- **Exercises** — quantitative, graphical, and applied problems similar to those on midterm and final exams.

Figures

Every figure includes a descriptive title, labeled axes, and an explanatory caption. Before looking at the caption, try to interpret the figure on your own. Reading graphs is a skill that develops with practice and is tested in exams.

A Note on Data

Economic data change over time. The figures and statistics in this textbook reflect data available at the time of writing (late 2026). For the most current data, consult Statistics Canada (www.statcan.gc.ca) and the Bank of Canada (www.bankofcanada.ca).

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CHAPTER 1

The Canadian Economy: Structure, Institutions, and Analytical Tools

In the spring of 2024, four people in Prince George, British Columbia faced economic decisions shaped by forces none of them fully understood.

Diane, 52, received her mortgage renewal notice in March. Her monthly payment was rising from \$1,840 to \$2,710 — an increase of \$870, or more than \$10,000 per year. Nothing had changed in her household except one number: the Bank of Canada's overnight interest rate, which had climbed from 0.25% in early 2022 to 5.00% by mid-2023. She had not overspent. She had not lost her job. But a decision made in an Ottawa boardroom had fundamentally altered her monthly budget.

Her son Marcus, 24, had just graduated from UNBC with a degree in business administration. He was earning \$22 per hour at a local firm — a wage that would have seemed generous five years ago, but which, after two years of inflation averaging over 5%, felt significantly less secure. He was trying to save for a down payment on an apartment, but rental costs in Prince George had risen nearly 30% since 2020. His parents' house had doubled in value, but that wealth would not be his for decades.

At the same sawmill where Marcus's uncle worked, the plant manager was navigating a \$3 million investment decision: automate the log-grading line with computer vision technology, or keep 18 workers. The system would pay for itself in four years. The workers earned above the minimum wage, had families in the community, and spent their incomes at local businesses. The economics pointed one direction; the community consequences pointed another.

And at the Northern Health Authority, administrator Priya was reviewing a budget that had grown 14% in two years — driven by an aging patient population, nurse recruitment costs, and pharmaceutical price increases — while the provincial government's transfer payment had grown by only 6%. The gap had to come from somewhere.

These four people live in the same economy. They are affected by the same interest rates, the same inflation, the same labour market pressures, and the same fiscal constraints. None of their situations can be understood without understanding how the Canadian economy works, how its institutions make decisions, and how those decisions

ripple outward into ordinary lives.

That is what this textbook is about.

Learning Objectives

- Describe the scale and structure of the Canadian economy using key aggregate statistics, including GDP, GDP per capita, and sectoral composition.
- Apply the expenditure approach to GDP and calculate nominal GDP growth, real GDP growth, and GDP per capita from raw data.
- Identify Canada's major economic institutions and explain the mandate, tools, and independence of each.
- Distinguish between monetary policy and fiscal policy, explain the transmission channels of each, and identify which institution is responsible for each in Canada.
- Apply the core tools of economic reasoning — models, opportunity cost, marginal analysis, and the positive–normative distinction — to real-world Canadian policy questions.
- Situate Canada within an international context by comparing key macroeconomic indicators with OECD peers.
- Preview the ten major contemporary economic issues addressed in this textbook and explain how they are interconnected.

1.1 Why Study Contemporary Economic Issues?

Economics is often introduced as the study of scarcity: because resources are limited and human wants are not, societies must make choices about how to allocate what they have. That definition is correct, but it understates the subject's reach. A more precise framing for this course is that economics is the study of *incentives*, *trade-offs*, and *unintended consequences* — the three forces that shape virtually every economic outcome a society produces.

Contemporary economic issues are economic problems that are actively unfolding, contested among reasonable people, and consequential for the well-being of Canadians. They differ from purely theoretical questions because they involve real institutions making real decisions under genuine uncertainty. They differ from purely political questions because the empirical evidence matters — how markets actually behave, how people actually respond to incentives, and what policies have actually worked in

comparable contexts.

Consider the four people from our opening vignette. Diane's mortgage crisis cannot be understood without knowing why the Bank of Canada raised interest rates, what caused inflation to spike in 2022, and how monetary policy transmits through the financial system to household balance sheets. Marcus's wage frustration cannot be understood without knowing how labour markets determine compensation, why real wages have lagged productivity growth since the 1980s, and how automation changes the demand for different kinds of work. The sawmill manager's dilemma cannot be understood without the economics of technological adoption, capital investment, and labour market adjustment. And Priya's budget cannot be balanced without understanding fiscal federalism, demographic aging, and the political economy of healthcare spending.

Economics provides the analytical tools to work through each of these problems systematically. It does not always deliver clean, comfortable answers. But it imposes a discipline on reasoning that is absent from purely ideological or intuitive approaches: it forces you to trace consequences, identify trade-offs, acknowledge trade-offs on the other side, and ask whether the evidence supports your conclusions.

Think Like an Economist

When you encounter a policy proposal in this course — a carbon tax, a rent control ordinance, an immigration target, a minimum wage increase — train yourself to ask three questions before forming an opinion:

- (1) **What incentives does this policy create?** Who responds, how, and in what direction?
- (2) **What are the trade-offs?** What must be given up to achieve this goal?
- (3) **What might happen that the policymaker did not intend?** Economics is full of well-intentioned policies that produced surprising consequences.

These questions do not determine whether a policy is good or bad. They are the beginning of a serious analysis, not its conclusion.

1.2 The Canadian Economy at a Glance

Before analysing specific contemporary issues, we need a clear statistical portrait of the economy we are studying. This section establishes the foundational facts of Canadian economic structure that every subsequent chapter will draw upon.

1.2.1 Scale and Aggregate Output

Canada is the world's ninth-largest economy measured by nominal gross domestic product. In 2024, Canada's GDP reached approximately \$3.0 trillion in Canadian dollars — a figure that places Canada ahead of Australia, South Korea, and Spain, but significantly behind the United States (\$29 trillion USD) and far smaller than China and the European Union collectively. With a population of approximately 41.5 million people, Canada's GDP per capita — our first approximation of average living standards — was roughly \$72,000 CAD, or approximately \$53,000 USD at prevailing exchange rates.

Definition: Gross Domestic Product (GDP)

GDP is the total market value of all final goods and services produced within a country's borders during a given period, typically one calendar year or one fiscal quarter.

Economic intuition: GDP measures the size of the economy's productive output. It captures what the economy *produces* — not what it owns, not what it earns abroad, and not what flows from one person to another without new production (such as transfer payments or the sale of existing assets).

What GDP excludes: Household production (cooking, childcare at home), the underground economy, environmental degradation, the distribution of income, and non-material well-being.

Canadian example: In 2020, Canada's real GDP fell 5.2% — the steepest single-year contraction since the Great Depression — as COVID-19 forced businesses to close and household spending collapsed. By 2021, GDP rebounded by 5.0% as restrictions lifted. The swing of more than 10 percentage points in two years illustrates both the fragility and the resilience of a modern market economy.

1.2.2 The Expenditure Approach to GDP

There are three equivalent ways to measure GDP: the expenditure approach, the income approach, and the production approach. In practice, national statistical agencies use all three and reconcile any discrepancies. For macroeconomic analysis, the *expenditure approach* is the most widely used because it decomposes GDP by who is spending.

The fundamental expenditure identity is:

$$Y = C + I + G + (X - M) \quad (1.1)$$

where:

- Y is nominal GDP (total output, measured in current dollars)
- C is *household consumption expenditure* — spending by households on goods (food, clothing, vehicles) and services (healthcare, education, financial services). This is typically the largest component, accounting for roughly 55–60% of GDP in Canada.
- I is *gross investment* (or gross capital formation) — business spending on plant, equipment, and inventory, plus residential construction. This is the most volatile component of GDP.
- G is *government expenditure on goods and services* — note this includes only direct government purchases, *not* transfer payments such as Employment Insurance or the Canada Child Benefit, which are redistribution rather than new production.
- X is exports — goods and services produced in Canada and sold abroad.
- M is imports — goods and services produced abroad and purchased in Canada.
- $(X - M) = NX$ is *net exports* (the trade balance). A positive NX means Canada sells more abroad than it buys — a trade surplus. A negative NX means a trade deficit.

Data Spotlight: Canada's GDP by Expenditure Component, 2024 (Approximate)

Component	Value (\$B CAD)	Share of GDP (%)
Household Consumption (C)	1,652	55.1
Gross Investment (I)	652	21.7
Government Expenditure (G)	718	23.9
Net Exports ($NX = X - M$)	-21	-0.7
GDP ($Y = C + I + G + NX$)	3,001	100.0

Note: Values are approximate and illustrative, consistent with Statistics Canada national accounts data. Figures may not sum to exactly 3,001 due to rounding.

Source: Statistics Canada, Table 36-10-0104-01 (National Accounts, expenditure approach), 2024 estimates.

Worked Example 1.1 — Calculating GDP from Components

Suppose you are given the following data for a simplified economy:

Item	Value (\$B)
Household spending on food, housing, and services	1,200
Business investment in machinery and factories	400
Residential housing construction	150
Federal, provincial, and municipal government purchases	500
Exports of goods and services	620
Imports of goods and services	670

Step 1: Identify the components of Equation (1.1):

$$C = 1,200$$

$$I = 400 + 150 = 550 \quad (\text{business investment} + \text{residential construction})$$

$$G = 500$$

$$NX = 620 - 670 = -50 \quad (\text{a trade deficit of \$50 billion})$$

Step 2: Apply the identity:

$$Y = 1,200 + 550 + 500 + (-50) = \mathbf{2,200 \text{ billion}}$$

Interpretation: This economy produces \$2.2 trillion of output. The trade deficit (\$50B) means the economy is spending more than it produces — the excess is financed by borrowing from abroad or by foreigners purchasing domestic assets.

Key Takeaway

GDP measures production, not welfare. A society can have high GDP and high inequality, high GDP and poor health outcomes, or high GDP and severe environmental degradation. GDP is an indispensable starting point for macroeconomic analysis, but it is not the endpoint for evaluating economic well-being.

1.2.3 Real GDP, Nominal GDP, and the GDP Deflator

When we compare GDP across years, we face a measurement challenge: prices change over time. If Canadian GDP rose from \$2.8 trillion in 2022 to \$3.0 trillion in 2024, some of

that increase reflects more goods and services being produced, but some reflects higher prices for the same goods and services. Only the first part represents genuine economic growth.

Nominal GDP measures output at the *current year's prices*. **Real GDP** measures output at the prices of a fixed *base year*, removing the effect of inflation.

The conversion between them uses the **GDP deflator**:

$$\text{GDP Deflator}_t = \frac{\text{Nominal GDP}_t}{\text{Real GDP}_t} \times 100 \quad (1.2)$$

Rearranging, real GDP can be obtained from nominal GDP and the deflator:

$$\text{Real GDP}_t = \frac{\text{Nominal GDP}_t}{\text{GDP Deflator}_t/100} \quad (1.3)$$

The **real GDP growth rate** — the standard measure of economic expansion or contraction — is:

$$g_t = \frac{\text{Real GDP}_t - \text{Real GDP}_{t-1}}{\text{Real GDP}_{t-1}} \times 100 \quad (1.4)$$

Worked Example 1.2 — Real vs. Nominal GDP Growth, Canada 2022–2023

Year	Nominal GDP (\$B)	GDP Deflator	Real GDP (\$B, 2017\$)	Real Growth (%)
2022	2,721	110.3	2,467	3.4
2023	2,879	115.2	2,499	1.3

Calculation for 2023 real GDP:

$$\text{Real GDP}_{2023} = \frac{2,879}{115.2/100} = \frac{2,879}{1.152} \approx 2,499 \text{ billion (2017 dollars)}$$

Calculation for real growth rate:

$$g_{2023} = \frac{2,499 - 2,467}{2,467} \times 100 = \frac{32}{2,467} \times 100 \approx 1.3\%$$

Interpretation: Nominal GDP rose 5.8% between 2022 and 2023, but more than four percentage points of that increase reflected higher prices rather than more output. Real growth was only 1.3% — an economy moving at a moderate pace, consistent with Canada's estimated potential growth rate of approximately 1.5–2.0% per year in the mid-2020s.

1.2.4 GDP Per Capita: A Measure of Living Standards

Total GDP tells us the size of the economy; it tells us very little about the typical person’s standard of living. An economy of 1.4 billion people will have a larger GDP than one of 40 million people, even if each of its citizens is far poorer. For cross-country and cross-time comparisons of living standards, economists use *GDP per capita* — total output divided by population:

$$\text{GDP per capita}_t = \frac{Y_t}{N_t} \tag{1.5}$$

where N_t is the total population.

Worked Example 1.3 — Canada vs. United States GDP per Capita, 2024

Country	Nominal GDP (USD B)	Population (M)	GDP per Capita (USD)
United States	28,781	337.0	85,406
Canada	2,221	41.5	53,518
Canada–US Gap			–\$31,888 (–37.3%)

Note: Canada’s nominal GDP is converted to USD at approximate 2024 exchange rate of 0.74 USD/CAD. GDP in USD thousands: $\$2,221\text{B} \times 0.74 = \$1,643\text{B USD}$; $\$1,643\text{B} / 41.5\text{M} = \$39,590 \text{ USD}$. For PPP-adjusted comparison (which removes exchange rate distortions), the gap narrows but Canada still trails the US by approximately 20–25%.

This income gap with the United States is one of the most important and persistent facts about the Canadian economy. It is not primarily explained by natural resources, education, or geography. A major contributor is a productivity gap — Canadians produce less output per hour worked than Americans — and that productivity puzzle is the subject of Chapter 9.

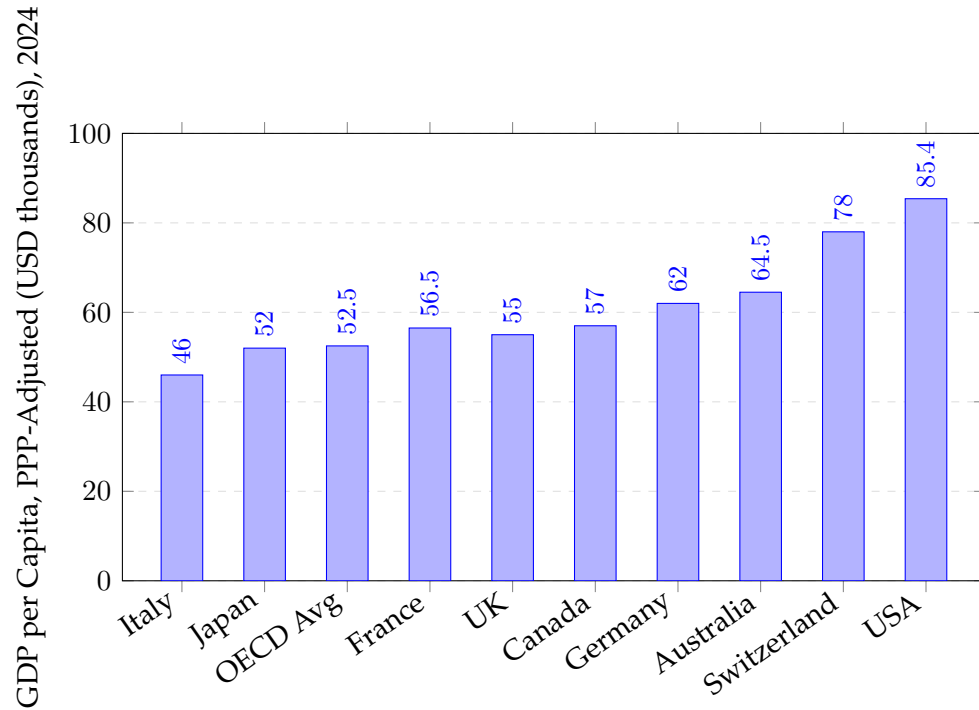


Figure 1.1. GDP per Capita (PPP-Adjusted, USD thousands), Selected OECD Countries, 2024. Canada’s living standard exceeds the OECD average and surpasses France, the UK, Japan, and Italy, but trails Germany, Australia, Switzerland, and the United States. The large Canada–US gap of approximately 37%, despite similar geography, institutions, and education levels, reflects a structural productivity difference explored in Chapter 9.

Source: OECD National Accounts at a Glance, 2024; World Bank World Development Indicators.

1.2.5 Sectoral Structure: The Service Economy

Canada’s economy is predominantly a **service economy**. Services — finance, real estate, retail, healthcare, professional services, education, and public administration — account for approximately 77% of GDP. This pattern is common among high-income countries: as economies develop, labour and capital shift from agriculture to manufacturing to services.

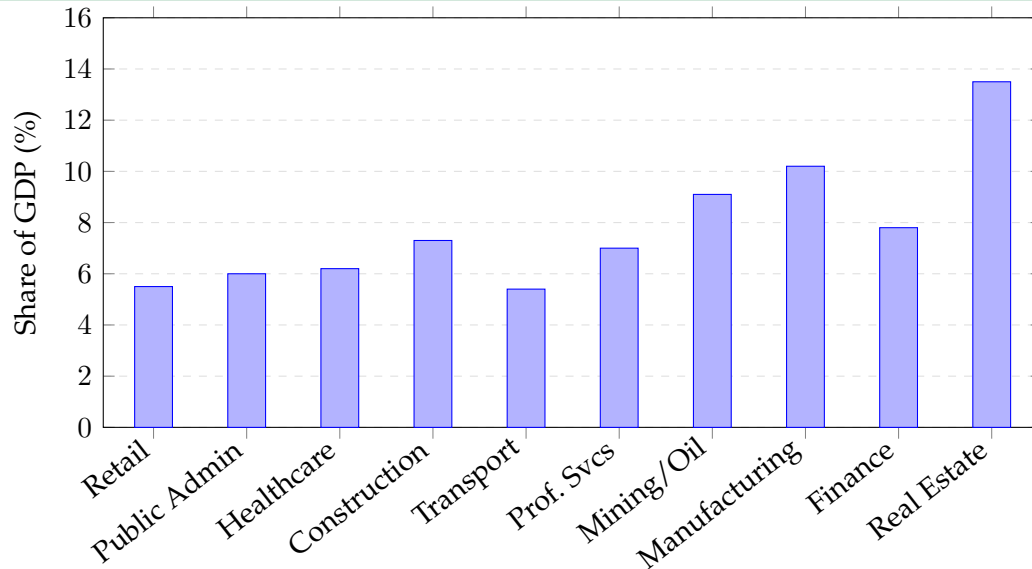


Figure 1.2. Canada’s Major Industries as a Share of GDP, 2024. Real estate dominates — reflecting high property values and the construction and rental sectors — while manufacturing and mining/oil and gas follow. Mining and oil and gas (9.1%) punches above its weight politically relative to its actual GDP share, a tension explored in Chapter 10. The service-heavy structure means Canada’s business cycle is largely driven by consumer spending, housing, and financial conditions — all channels through which monetary policy operates powerfully.

Source: Statistics Canada, Table 36-10-0401-01, GDP at basic prices by industry, 2024.

Critically, within the goods-producing sector, **natural resources** (oil and gas, mining, forestry, and agriculture) play a disproportionately large role relative to most comparable advanced economies. Canada’s oil sands in Alberta produce over 3 million barrels of crude oil per day. British Columbia’s forest products sector exports billions of dollars of softwood lumber annually. Saskatchewan supplies approximately 40% of the world’s potash. These resources generate export revenues, government royalties, and employment — but they also make Canada’s economy sensitive to global commodity price cycles and they are central to the climate transition challenge explored in Chapter 10.

1.2.6 Regional Economic Geography

Canada is a vast country, and its economy is not evenly distributed. The regional concentration of economic activity creates sharp differences in employment conditions, wages, housing markets, and fiscal capacity across provinces — differences that make national averages sometimes misleading for understanding local economic conditions.

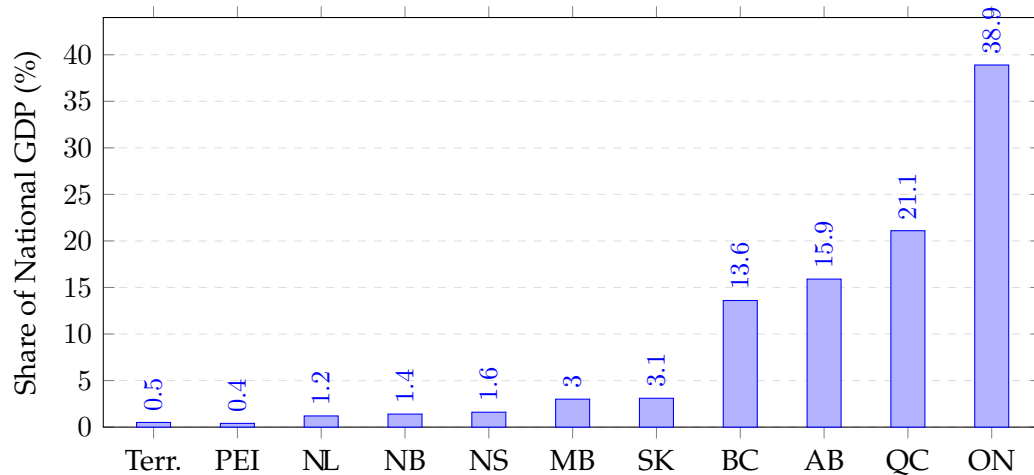


Figure 1.3. Share of National GDP by Province and Territory, 2024. Ontario and Quebec together account for 60% of Canadian economic output. Alberta, despite having one-quarter of Ontario's population, generates 15.9% of GDP — reflecting the outsized economic weight of the oil sands. The three territories combined produce 0.5% of GDP despite covering 40% of Canada's land mass. This extreme geographic concentration of economic activity motivates the equalization payment system examined in Chapter 7.

Source: Statistics Canada, Provincial and Territorial Economic Accounts, 2024.

The regional diversity of Canada's economy creates significant policy challenges. When oil prices fall, Alberta's provincial revenue collapses — but Ontario's economy is largely unaffected. When the Bank of Canada raises interest rates to combat national inflation, the impact on heavily indebted households in Toronto and Vancouver is far more severe than on renters in Moncton or resource workers in Fort McMurray. National policy instruments applied to a regionally diverse economy inevitably produce unequal regional consequences.

Key Takeaway

Canada's economy is large in aggregate but regionally unequal: Ontario and Quebec dominate output; Alberta dominates resources; and the Atlantic provinces and territories depend heavily on federal transfers. National economic statistics always need to be disaggregated by region for a complete picture.

1.3 Canada's Trade Dependence

Canada is one of the most trade-dependent large economies in the world. *Trade openness* — defined as total trade (exports plus imports) as a share of GDP — measures how

integrated an economy is with the global trading system:

$$\text{Trade Openness} = \frac{X + M}{Y} \times 100 \quad (1.6)$$

For Canada in 2024:

$$\text{Trade Openness} = \frac{897 + 918}{3,001} \times 100 = \frac{1,815}{3,001} \times 100 \approx 60.5\%$$

This means that the sum of Canada's exports and imports is equivalent to more than 60% of total GDP — an extraordinary degree of integration for an economy of Canada's size. By comparison, the United States, with a much larger domestic market, has a trade openness ratio of approximately 27%.

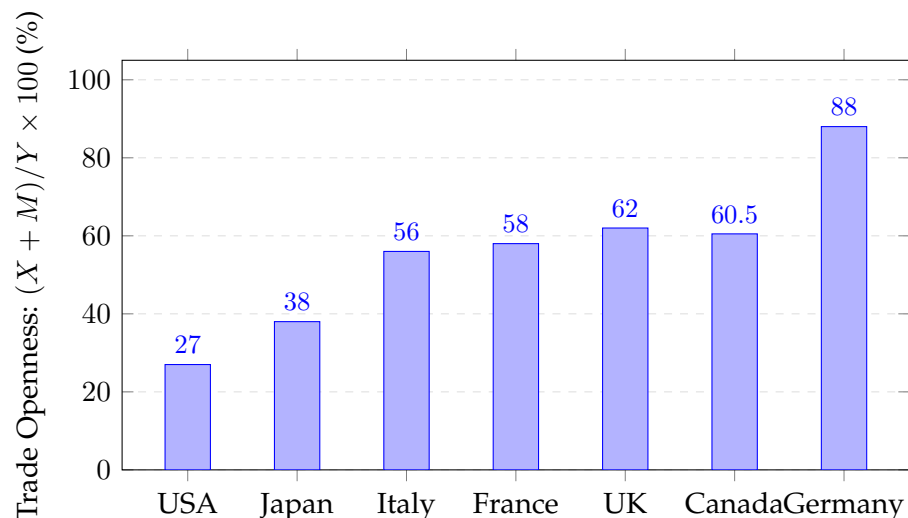


Figure 1.4. Trade Openness — $(X + M)/Y \times 100$ — G7 Countries, 2024. Canada's trade openness (60.5%) is among the highest in the G7, surpassed only by Germany. The US is far less trade-dependent because its enormous domestic market satisfies most of its own demand. Canada's high trade dependence means it is especially vulnerable to US tariff policy, exchange rate movements, and global commodity price shocks — themes developed in Chapter 8.

Source: World Bank World Development Indicators; Statistics Canada, Table 12-10-0121-01.

The most critical structural fact about Canada's trade is its **extreme concentration** in a single market. The United States absorbs approximately 75% of Canada's merchandise exports, making Canada more dependent on one trading partner than almost any other advanced economy in the world. This concentration is a source of enormous prosperity — geographic proximity, a shared language and legal tradition, and preferential market access under CUSMA lower transaction costs dramatically — but it is also a source of

strategic vulnerability.

Table 1.1. Canada's Top 10 Trading Partners, 2024

Country	Export Share (%)	Import Share (%)	Trade Balance (\$B CAD)
United States	75.4	51.2	+130.2
China	4.2	12.8	−37.9
United Kingdom	2.8	2.1	+3.1
Japan	2.1	2.8	−4.0
Germany	0.8	2.8	−7.1
Mexico	0.8	4.4	−13.8
South Korea	0.8	1.5	−2.3
Netherlands	1.2	0.6	+2.2
India	0.7	1.0	−1.1
France	0.6	1.2	−2.4
All others	10.6	19.6	−30.5

Source: Statistics Canada, Table 12-10-0121-01; Global Affairs Canada, Trade Data Online, 2024.

Table 1.1 reveals two important asymmetries. First, Canada runs a substantial trade surplus with the United States (\$130B), driven almost entirely by energy exports and auto-sector trade. Second, Canada runs trade deficits with most other major partners, including China and Germany. The concentration of the surplus in one country and one set of sectors creates the vulnerability analysed in Chapter 8: a shift in US trade policy — such as the imposition of tariffs on Canadian steel, lumber, or automobiles — can rapidly transform Canada's trade balance from surplus to deficit.

1.4 Canada's Major Economic Institutions

Economic outcomes do not emerge spontaneously from markets alone. They are shaped by institutions — the formal rules, organizations, and governance frameworks within which economic decisions are made. Understanding Canada's key economic institutions is a prerequisite for understanding how economic policy is made and why its effects are what they are.

1.4.1 The Bank of Canada

The *Bank of Canada* is Canada's central bank. Established under the *Bank of Canada Act* in 1934, it is structured as a Crown corporation that operates at formal arm's length from the elected federal government. In practice, this independence is one of the most consequential institutional features of the Canadian economy.

The Bank of Canada's primary mandate is to **promote the economic and financial welfare of Canada**. Since 1991, this mandate has been operationalized through a formal *inflation-targeting framework*, renewed most recently in 2021: the Bank commits to keeping CPI inflation within a 1–3% **control range**, with a 2% **midpoint target**.

The Bank conducts monetary policy primarily through its *policy interest rate* — the target for the *overnight rate*, the interest rate at which major financial institutions lend to each other on an overnight basis. When the Bank raises the overnight rate, it becomes more expensive to borrow throughout the financial system. Mortgage rates rise. Business loans cost more. The exchange rate tends to appreciate. All of these effects reduce spending and investment, which eventually reduces inflation. When the Bank lowers the overnight rate, the opposite occurs.

Canadian Example: The 2022–2024 Rate-Hike Cycle: The Fastest in Modern History

Between March 2022 and July 2023, the Bank of Canada raised its policy interest rate from 0.25% to 5.00% — an increase of 475 basis points (4.75 percentage points) in just 16 months. This was the most aggressive monetary tightening cycle in the Bank of Canada's modern history, undertaken to combat inflation that had reached 8.1% in June 2022, the highest reading since September 1983.

The rate hike cycle illustrates the transmission mechanism of monetary policy in practice. By late 2023, inflation had fallen to approximately 3%. By mid-2024, it had approached the 2% target. But the rate hikes also raised mortgage payments for millions of Canadian households — an estimated \$700 on average for variable-rate mortgage holders — and contributed to a sharp slowdown in housing market activity. The Bank began cutting rates in June 2024.

This episode is the central case study of Chapter 2. It also directly shapes the housing analysis in Chapter 3 and the distributional analysis in Chapter 5.

1.4.2 Department of Finance Canada

The *Department of Finance Canada* is the federal government ministry responsible for *fiscal policy* — the use of government spending and taxation to influence the economy. The Minister of Finance presents an annual federal budget, typically in March, which outlines revenue plans, spending priorities, and projections for the federal deficit or surplus.

Unlike the Bank of Canada, the Department of Finance operates under direct democratic accountability: its decisions are made by elected politicians and must pass through Parliament. This creates a fundamentally different set of incentives than those governing the Bank of Canada, and it is one reason why monetary and fiscal policy are deliberately separated in Canada's institutional design.

Key agencies operating alongside Finance Canada include:

- **Parliamentary Budget Officer (PBO):** An independent officer of Parliament established in 2006, mandated to provide non-partisan economic and fiscal analysis. The PBO has become an important check on government budget projections, regularly producing independent assessments that sometimes differ substantially from government forecasts.
- **Canada Revenue Agency (CRA):** Administers tax collection and enforces tax law. Revenue collection is the CRA's core function; it does not make tax policy, which remains with Finance Canada.
- **Employment and Social Development Canada (ESDC):** Administers transfer programs including Employment Insurance (EI), the Canada Pension Plan (CPP), the Canada Child Benefit (CCB), and Old Age Security (OAS/GIS). These programs collectively transfer hundreds of billions of dollars per year from taxpayers to households.

1.4.3 Statistics Canada

Statistics Canada (*Statistique Canada*) is the national statistical agency, established under the *Statistics Act*. It is the primary data source for the analysis conducted throughout this textbook. Its publications include:

- **Consumer Price Index (CPI):** Published monthly; the primary measure of inflation (Table 18-10-0004-01). Analysed in depth in Chapter 2.
- **Labour Force Survey (LFS):** Published monthly; measures the employment rate, unemployment rate, and labour force participation rate (Table 14-10-0287-01). Central

to Chapter 4.

- **National Income and Expenditure Accounts:** Quarterly GDP by expenditure, income, and industry (Table 36-10-0104-01). The foundation for Chapters 1 through 7.
- **Survey of Consumer Finances / Canadian Income Survey:** Annual data on household income distribution, poverty, and inequality (Table 11-10-0190-01). Central to Chapter 5.
- **Census of Population:** Every five years; provides comprehensive demographic, labour market, and income data. The 2021 Census is heavily used in Chapters 5 and 6.

Statistics Canada operates with formal independence from political direction in its data collection and methodological decisions. The integrity of its data is a fundamental public good: without reliable statistics, evidence-based policymaking is impossible.

1.4.4 Canada Mortgage and Housing Corporation (CMHC)

The *Canada Mortgage and Housing Corporation* (CMHC) is a federal Crown corporation with two distinct functions. First, it operates a *mortgage insurance* program: Canadians who purchase a home with a down payment of less than 20% must purchase CMHC mortgage insurance, which protects the lender against default. This insurance program allows lenders to extend mortgages to buyers with limited savings, making homeownership accessible to a broader population. Second, CMHC is the primary source of Canadian housing market data — including housing starts, rental vacancy rates, affordability indices, and housing market outlooks — and it is the implementing agency for the federal government’s housing programs. CMHC is central to the analysis in Chapter 3.

1.4.5 Provincial Governments

In Canada’s federal system, provincial governments bear primary responsibility for several of the largest areas of economic policy:

- **Healthcare:** Provinces operate and fund the publicly administered insurance plans mandated by the *Canada Health Act*. Healthcare is typically the single largest provincial spending item — consuming 35–45% of provincial budgets.
- **Education:** Provinces regulate and largely fund education from kindergarten through post-secondary.

- **Labour market regulation:** Minimum wages, occupational safety, labour relations, and most employment standards are set provincially. This is why minimum wages vary from \$13.85 in some Atlantic provinces to \$17.40 in British Columbia.
- **Natural resources:** Under the Constitution Act, provinces own the natural resources within their borders. Provincial royalty regimes for oil, gas, and minerals are major determinants of resource sector economics.
- **Housing regulation:** Zoning, land use planning, and building codes are primarily municipal functions delegated by provinces. This makes provinces central players in the housing affordability crisis analysed in Chapter 3.

Table 1.2. Canada's Major Economic Institutions: Mandate, Tools, and Independence

Institution	Primary Mandate	Main Tools	Independence Level
Bank of Canada	Low and stable inflation (2% target); financial stability	Overnight rate; quantitative easing; financial regulation	High — Crown corporation, arm's length from government
Department of Finance Canada	Fiscal policy; budget; taxation	Government spending; tax rates; transfer programs	Low — elected minister, accountable to Parliament
Statistics Canada	Collect, produce, and disseminate national statistics	Surveys; censuses; administrative data	High — statistical independence protected by law
CMHC	Housing finance; mortgage insurance; housing policy	Mortgage insurance; housing programs; market data	Moderate — Crown corporation with federal mandate
Parliamentary Budget Officer	Independent fiscal and economic analysis	Economic modelling; budget assessments; reports	High — independent officer of Parliament
Provincial Governments	Healthcare; education; labour; natural resources	Spending; regulation; taxation; royalties	Full — constitutionally sovereign within their jurisdiction

Source: *Bank of Canada Act; Statistics Act; Canada Mortgage and Housing Corporation Act; Constitution Act, 1867.*

1.5 The Policy Toolkit: Monetary and Fiscal Policy

Two broad categories of macroeconomic policy tools are the primary instruments through which governments and central banks attempt to influence the performance of the economy. Understanding the distinction between them — and the interaction between them — is foundational for every topic in this course.

1.5.1 Monetary Policy

Definition: Monetary Policy

Monetary policy refers to the actions taken by a central bank to influence the money supply, interest rates, and credit conditions, with the ultimate goals of maintaining price stability and supporting sustainable economic growth.

Economic intuition: When the central bank raises its policy interest rate, borrowing becomes more expensive throughout the economy. Households take out smaller mortgages. Firms postpone capital investment. The exchange rate tends to appreciate, making imports cheaper and exports less competitive. These effects reduce aggregate demand and, over time, inflation. The reverse occurs when the Bank cuts rates.

Key constraint — the zero lower bound: The policy rate cannot be reduced significantly below zero (depositors would withdraw cash and hold it physically rather than accept negative returns on deposits). This creates an asymmetry: the Bank can always raise rates to fight inflation, but its ability to cut rates to fight recession is constrained at zero. After 2008 and again in 2020, the Bank cut its rate to 0.25% — its effective lower bound — and was forced to rely on other tools (forward guidance and, to a limited extent, quantitative easing) to provide additional stimulus.

Canadian example: The Bank cut the overnight rate from 1.75% to 0.25% in March 2020 in response to COVID-19, then raised it to 5.00% between March 2022 and July 2023 to fight inflation — the full range of monetary policy in action within a four-year period.

The primary transmission channels through which a Bank of Canada interest rate change affects the broader economy are:

1. **Credit cost channel:** Higher overnight rates are passed through to mortgage rates, car loan rates, and business borrowing costs, reducing household and business spending.

2. **Asset price channel:** Higher interest rates reduce the present value of future asset cash flows, lowering stock prices and real estate values, which reduces household wealth and consumption (the “wealth effect”).

3. **Exchange rate channel:** Higher Canadian interest rates attract foreign capital seeking better returns, increasing demand for Canadian dollars and appreciating the exchange rate. A stronger Canadian dollar makes imports cheaper (reducing inflation) and exports more expensive (reducing demand for Canadian goods abroad).

4. **Expectations channel:** Perhaps most importantly, when the Bank credibly commits to fighting inflation, households and firms adjust their price-setting and wage demands downward, reducing inflationary momentum even before interest rates fully transmit through the system.

1.5.2 Fiscal Policy

Definition: Fiscal Policy

Fiscal policy refers to the use of government spending and taxation to influence macroeconomic conditions.

Economic intuition: When aggregate demand is insufficient to support full employment, the government can increase its own spending (shifting the aggregate demand curve rightward) or cut taxes (increasing household disposable income and thus consumption). Conversely, when the economy is overheating, fiscal restraint — cutting spending or raising taxes — can reduce inflationary pressure.

Key constraint — the deficit and debt: When the government spends more than it collects in taxes, it runs a **budget deficit**, which must be financed by borrowing. The accumulation of past deficits is the **national debt**. Sustained high deficits raise the stock of debt relative to GDP, increasing interest payments and potentially crowding out private investment. The fiscal arithmetic of debt sustainability is explored fully in Chapter 7.

Canadian example: In 2020–21, the federal government ran a deficit of \$327.7 billion — 14.9% of GDP — as it deployed the Canada Emergency Response Benefit (CERB), the Canada Emergency Wage Subsidy (CEWS), and other programs to protect household incomes during the pandemic shutdown. This was the largest single-year deficit in Canadian history. It was deliberate and effective in preventing a financial collapse, but it also raised Canada’s federal debt-to-GDP ratio from approximately 31% to 49% within two years.

1.5.3 The Interaction Between Monetary and Fiscal Policy

Monetary and fiscal policy are operated by different institutions with different mandates, but they interact in important ways. When fiscal policy is highly expansionary — as it was in 2020–21 — it can increase aggregate demand and put upward pressure on inflation, requiring a tighter monetary policy response. Conversely, when monetary policy is very tight — as it was in 2022–23 — higher interest rates increase the government’s debt servicing costs, creating fiscal pressure.

The period 2020–2023 illustrates this interaction vividly. Emergency fiscal spending in 2020–21 was appropriate and necessary. But the combination of massive fiscal stimulus and near-zero interest rates through 2021 contributed to the demand-side component of the inflation surge that the Bank of Canada then had to combat aggressively in 2022–

23. Understanding this interaction is essential for evaluating both the monetary policy response (Chapter 2) and the fiscal policy choices (Chapter 7).

CANADA'S MACROECONOMIC POLICY ARCHITECTURE

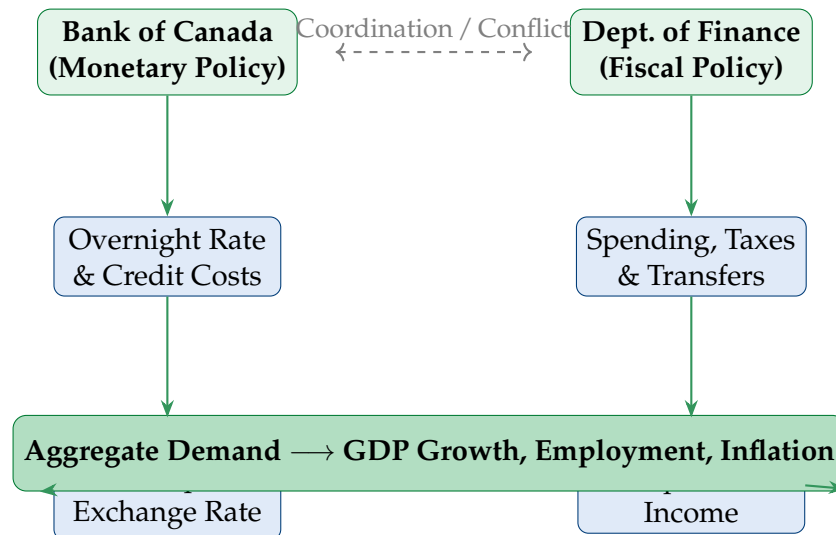


Figure 1.5. Canada's Macroeconomic Policy Architecture. Monetary policy (left branch) and fiscal policy (right branch) each affect aggregate demand through distinct channels, ultimately influencing GDP growth, employment, and inflation. The dashed bidirectional arrow indicates that the two policy levers can work in the same direction (as during the COVID stimulus) or in opposing directions (as in 2022–23, when fiscal policy remained expansionary even as the Bank of Canada was tightening). Understanding this interaction is one of the central analytical tasks of contemporary macroeconomics.

Policy Debate: Should the Bank of Canada Be Independent from the Federal Government?

The Issue: Central bank independence — the insulation of monetary policy decisions from direct political control — is a cornerstone of modern macroeconomic governance. But independence is not constitutionally guaranteed in Canada. The *Bank of Canada Act* allows the federal government, in exceptional circumstances, to direct the Bank. Should this arm's-length relationship be strengthened, weakened, or left as is?

The Case for Independence: Economic theory and empirical evidence both support central bank independence. The fundamental problem is *time inconsistency*: elected governments have incentives to expand the economy before elections, even at the cost of higher future inflation. If markets expect governments to exploit mone-

tary policy for electoral gain, they will build that expectation into wages and prices, requiring permanently higher unemployment to achieve any given inflation rate. An independent central bank can credibly commit to price stability in a way that elected governments cannot — and that credibility is itself enormously valuable. Cross-country evidence consistently shows that countries with more independent central banks have lower average inflation without worse unemployment or growth outcomes (see ??).

The Case Against Full Independence: Critics argue that central bank independence raises a *democratic accountability* problem. When the Bank of Canada raises interest rates by 475 basis points in 16 months — dramatically increasing mortgage costs for millions of Canadians — this is a consequential distributional decision that arguably deserves more democratic oversight than it currently receives. Furthermore, the separation of monetary and fiscal policy can make macroeconomic management incoherent: if the elected government wishes to support employment and the central bank is tightening to control inflation, the two policies work against each other, potentially prolonging both the high interest rates and the high unemployment.

The Canadian Position: Canada’s current arrangement is a hybrid. The Bank of Canada is independent in its day-to-day operations, but its inflation target is set jointly with the Government of Canada in a periodic agreement (most recently renewed in 2021). This “joint mandate” approach attempts to combine the credibility benefits of independence with some degree of democratic input into the overarching framework.

Your Assessment: After completing Chapter 2, you will be equipped to evaluate the Bank’s rate-hike cycle on its merits. Did the Bank move too aggressively? Too slowly? At the right pace? These are exactly the kinds of questions where the distinction between positive analysis (what happened and what were the effects?) and normative judgement (was this the right policy?) matters most.

1.6 How Economists Think

Developing the analytical habits of economic reasoning takes time and deliberate practice. This section introduces the core tools that you will use throughout the course.

1.6.1 Economic Models and Assumptions

Economists use *models* — carefully constructed simplifications of reality — to generate predictions and organize thinking about complex phenomena. A model deliberately omits details in order to focus on the mechanisms that most matter for a particular question.

The *supply and demand model*, for example, abstracts away from the unique characteristics of millions of individual buyers and sellers and focuses instead on the aggregate relationship between price and quantity. This abstraction is a feature, not a limitation: it allows economists to make clear, testable predictions about how markets respond to changes in supply, demand, and policy.

Every model rests on assumptions. Evaluating a model requires understanding which assumptions are load-bearing (if they are wrong, the model's predictions fail) and which are harmless simplifications. The assumption that economic agents “respond to incentives” is load-bearing for almost all of economics. The assumption of perfect information is often harmless as a first approximation but becomes crucial in analyses of insurance, financial markets, and healthcare.

Think Like an Economist

A map is a model. A road map of Canada omits trees, buildings, and terrain because those details are irrelevant to the task of navigating from Prince George to Edmonton. A topographic map of the same region omits roads but captures elevation, because that information matters for a hiker. Neither map is “more correct” than the other — they are optimized for different questions. Economic models work the same way. A model that is useful for understanding inflation may be useless for understanding income inequality. The skill is in choosing the right model for the question at hand.

1.6.2 Positive and Normative Economics

One of the most important distinctions in economic methodology is between *positive economics* and *normative economics*.

Definition: Positive and Normative Economics

Positive economics describes and explains economic phenomena as they are, without making value judgements. It answers “what is” or “what will happen if.” Positive claims are, in principle, testable against empirical evidence.

Normative economics makes value judgements about what *should* happen. It answers “what ought to be.” Normative claims cannot be resolved by evidence alone — they also depend on values and priorities.

Examples in the Canadian context:

- *Positive*: “Canada’s CPI inflation peaked at 8.1% in June 2022.” [Observable fact.]
- *Positive*: “A 25% US tariff on Canadian auto exports would reduce Canadian GDP by an estimated 0.3–0.7 percentage points.” [Testable empirical claim, though uncertain.]
- *Normative*: “The federal carbon tax should be increased to \$250 per tonne.” [Value judgement about the right balance between climate and economic costs.]
- *Normative*: “Canada should prioritize reducing inequality over maximizing GDP growth.” [Value judgement about priorities.]

Note: many policy debates conflate positive and normative claims. Part of the discipline of economics is keeping them separate.

Policy debates are often confusing precisely because positive and normative claims are mixed together. When a politician argues that the carbon tax “kills jobs” (positive claim — testable) and “should be abolished” (normative claim — based on values), they are making two distinct types of argument that require two distinct types of response. Throughout this textbook, we will be explicit about which claims are positive (and what the evidence says) and which are normative (and what the competing values are).

1.6.3 Opportunity Cost and Trade-offs

A fundamental principle of economic reasoning is that *all choices involve trade-offs*. Resources — time, money, productive capacity, policy attention — allocated to one purpose cannot simultaneously be used for another. The **opportunity cost** of any choice is the value of the next-best alternative foregone.

$$\text{Opportunity Cost of Choice A} = \text{Value of the best foregone alternative to A} \quad (1.7)$$

Opportunity cost is not the same as monetary cost. When a university student works part-time during the school year, the opportunity cost of those hours worked is not just the forgone leisure but the forgone study time and the potential impact on grades. When the federal government allocates \$15 billion to the National Housing Strategy, the opportunity cost is not just the \$15 billion in budget space but whatever the next-best use of that money would have been — whether tax cuts, healthcare investment, debt repayment, or climate adaptation spending.

Canadian Example: The Opportunity Cost of Canada's COVID Fiscal Response

In 2020–21, the federal government spent approximately \$350 billion in emergency COVID-19 support. This was financed by borrowing, raising the federal debt-to-GDP ratio by nearly 18 percentage points. The opportunity cost of this spending was not only the future interest payments (approximately \$43 billion per year by 2024–25), but also the fiscal space foregone for other priorities: a national pharmacare program, a substantial climate investment fund, or a faster post-pandemic deficit reduction path. The emergency spending was arguably necessary and worth its cost — but a complete analysis must acknowledge the trade-offs. Chapter 7 revisits this question with full fiscal data.

1.6.4 Marginal Analysis

Economists typically analyse decisions not as binary choices (do this or do not do this) but at the *margin*: how much of something should be done? **Marginal cost** (MC) is the additional cost of one more unit of an activity. **Marginal benefit** (MB) is the additional benefit. The economically optimal decision equates the two:

$$\text{Optimal decision: } MB = MC \quad (1.8)$$

The marginal analysis framework applies broadly. Should the Bank of Canada raise interest rates another 25 basis points? The marginal benefit is a further reduction in inflation; the marginal cost is a further increase in mortgage payments and a risk of over-tightening into recession. Should Canada raise the carbon tax from \$65 to \$80 per tonne? The marginal benefit is additional emissions reduction; the marginal cost

is additional burden on households and firms. In both cases, the right question is not “should we do more?” in the abstract, but “does the marginal benefit exceed the marginal cost at this particular point?”

1.6.5 Correlation and Causation

Perhaps the most important methodological lesson in empirical economics is the distinction between *correlation* and *causation*. Two variables are correlated if they tend to move together. But correlation does not establish causation: variable A may cause B, B may cause A, or a third variable C may cause both A and B.

Think Like an Economist

Ice cream sales and drowning deaths are positively correlated — both rise in summer. Does eating ice cream cause drowning? Of course not: the third variable, warm weather, drives both. Identifying this kind of spurious correlation is easy. In economics, the confounding variables are often far less obvious.

Consider: provinces with higher minimum wages tend to have higher unemployment rates. Does this prove that minimum wages cause unemployment? Not necessarily — provinces with historically weaker labour markets may have adopted higher minimum wages as a compensatory policy, or other factors (industry mix, demographics) may explain both. Establishing *causal* effects from observational data requires careful research design, and the results are often contested. We will encounter this challenge repeatedly throughout the course.

1.6.6 Efficiency and Equity

A final pair of concepts that runs through every chapter is the distinction between *economic efficiency* and *economic equity* (or fairness). An allocation of resources is *efficient* if it is impossible to make someone better off without making someone else worse off — a *Pareto optimal* state. Markets, under the right conditions, tend to produce efficient outcomes. But efficient outcomes need not be equitable: a distribution in which one person owns everything and everyone else owns nothing is technically Pareto optimal (you cannot help the poor without taking from the rich).

Many of the most important policy debates in this course — should Canada have a carbon tax? should it have rent control? should it have a wealth tax? — involve precisely this tension. A policy that increases efficiency (total GDP) may worsen equity

(distribution of that GDP), or vice versa. Evaluating these trade-offs requires clear thinking about both what the evidence says (positive) and what values should guide our priorities (normative).

1.7 Canada in International Context

Understanding Canada's contemporary economic situation requires situating it within the broader international environment. Canada's economic performance is shaped by global forces — commodity prices, US monetary policy, global trade flows, and climate commitments — that no Canadian institution can fully control.

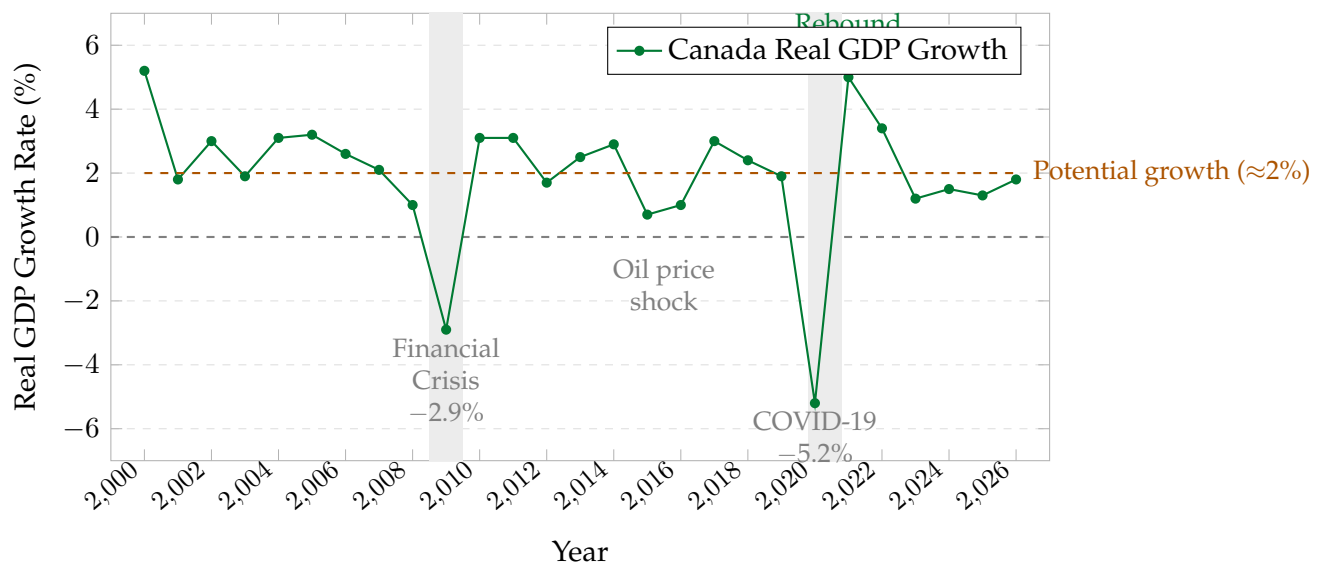


Figure 1.6. Canada's Real GDP Growth Rate, 2000–2026. The shaded bands mark recessionary years. Three episodes stand out: the 2009 financial crisis contraction (−2.9%), driven by the global banking collapse and the near-simultaneous US recession; the 2015–16 slow-down driven by the oil price collapse to below \$30/barrel (US); and the 2020 COVID-19 collapse (−5.2%), the steepest in peacetime Canadian history. The dashed orange line at 2% represents Canada's approximate *potential* growth rate — the rate consistent with stable unemployment and inflation. Years above this line indicate excess demand; years below it indicate slack.

Source: Statistics Canada, Table 36-10-0104-01, Chained (2017) dollars.

Table 1.3. Key Canadian Macroeconomic Indicators, 2015–2026

Year	Real GDP Growth (%)	CPI Infl. (%)	Unemp. Rate (%)	Bank Rate (%)	Fed. Balance (\$B)	Fed. Debt (% GDP)
2015	0.7	1.1	6.9	0.50	−2.9	31.0
2016	1.0	1.4	7.0	0.50	−17.8	31.5
2017	3.0	1.6	6.3	1.00	−19.0	30.8
2018	2.4	2.3	5.8	1.75	−14.0	30.1
2019	1.9	1.9	5.7	1.75	−39.4	30.8
2020	−5.2	0.7	9.5	0.25	−327.7	49.1
2021	5.0	3.4	7.5	0.25	−113.8	45.2
2022	3.4	6.8	5.3	4.25	−36.0	42.4
2023	1.2	3.9	5.4	5.00	−40.0	42.0
2024	1.5	2.6	6.3	4.25	−46.8	42.3
2025	1.3	2.1	6.7	3.00	−48.0	42.5
2026	1.8	2.0	6.5	2.75	−30.0	42.0

Source: Statistics Canada, Tables 36-10-0104-01, 18-10-0004-01, 14-10-0287-01; Bank of Canada Key Interest Rates; Department of Finance Canada, Fiscal Reference Tables, 2026. Values for 2025–2026 are preliminary or projected. Federal balance figures in nominal Canadian dollars; negative values indicate a deficit.

Table 1.3 is the single most important reference table in this textbook. It captures the macroeconomic story of Canada across 12 years in six dimensions. Study it carefully. Notice that 2020 is the anomaly year on every dimension simultaneously: growth collapses, inflation vanishes, unemployment spikes, the Bank rate hits its floor, and the budget deficit reaches a historic extreme. Notice that 2022 is the year of the great inflation — 6.8%, the highest since 1991. Notice that the Bank rate and CPI inflation move together through 2022–2024, reflecting the monetary policy response. And notice that the federal debt-to-GDP ratio rose by nearly 18 percentage points in a single year (2019 to 2020) and has not fully recovered — a fiscal legacy that Chapter 7 examines in depth.

1.8 Case Studies

Case Study: COVID-19's Economic Shock and Canada's Recovery

Background: On March 11, 2020, the World Health Organization declared COVID-19 a global pandemic. Within days, Canadian provinces ordered the closure of non-essential businesses, schools, and public spaces. Economic activity collapsed at an unprecedented speed.

The Economic Shock: In April 2020, Canada lost approximately 2 million jobs — more than were lost in the entire 2008–09 recession. GDP fell at an annualized rate exceeding 40% in the second quarter of 2020. The unemployment rate reached 13.7% in May 2020. These numbers, while startling, understate the disruption: millions of additional Canadians were employed but working zero hours (classified as employed but absent by the LFS methodology), and the labour force participation rate fell sharply as many stopped looking for work.

Policy Response: The federal government deployed the Canada Emergency Response Benefit (CERB), which paid \$2,000 per month to approximately 9 million workers who lost employment income — more than one in four Canadian workers. The Canada Emergency Wage Subsidy (CEWS) paid up to 75% of the wages of employees at firms whose revenues had fallen, at a total cost exceeding \$100 billion. The Bank of Canada cut its policy rate to 0.25% and committed to holding it there until the recovery was “well underway.” Combined, the fiscal and monetary response was the most aggressive in Canadian peacetime history.

Recovery: Canada's recovery was faster than most projections. By early 2022, employment had fully recovered to pre-pandemic levels. GDP recovered its lost ground by early 2022 as well. But the recovery was uneven: high-contact service sectors (restaurants, hotels, live entertainment) recovered more slowly; women, racialized Canadians, and low-wage workers experienced longer-lasting earnings impacts; and the combination of massive fiscal stimulus, low interest rates, and supply chain disruptions created the inflationary surge of 2022 that reshaped the economic environment for years afterward.

Evaluation: The COVID episode illustrates both the power and the limits of macroeconomic policy. The fiscal and monetary response prevented a financial collapse and protected millions of households. But it also contributed to inflationary dynamics that the Bank of Canada then had to combat at significant cost to mortgage-holding

households. Understanding this sequence of events — pandemic shock, policy response, inflationary consequence, monetary tightening, housing market correction — provides the narrative architecture for Chapters 2 through 3.

Data reference: Statistics Canada, Table 14-10-0287-01 (employment); Table 36-10-0104-01 (GDP); Parliamentary Budget Officer, COVID-19 Fiscal Analysis, 2021.

Case Study: Northern British Columbia as a Regional Economy

Background: The University of Northern British Columbia sits in Prince George — the largest city in northern BC, with a population of approximately 75,000. The regional economy of northern BC illustrates, in compressed form, many of the structural features and contemporary challenges of the Canadian economy as a whole.

Economic Structure: Northern BC's economy is built on natural resources: forestry, mining (gold, copper, coal), and increasingly, natural gas from the Montney formation and the Peace River region. The completion of the Trans Mountain Pipeline Expansion (2024) and the LNG Canada facility near Kitimat (under construction) represent billion-dollar investments that will shift the regional resource mix toward hydrocarbons. The forest sector, historically the dominant employer, has been reshaped by mill closures, automation, and the mountain pine beetle epidemic that devastated the lodgepole pine forests of the 1990s and 2000s.

Labour Market Features: Northern BC labour markets are characterized by: above-average wages in resource sectors (unionized sawmill and mine workers earn substantially more than provincial minimum wage); persistent labour shortages in healthcare and social services (geographic remoteness makes recruitment difficult); a significant Indigenous population (First Nations account for approximately 16% of the regional population) whose economic participation and outcomes reflect the historical legacy of dispossession analysed in Chapter 5; and a mismatch between the skills demanded by resource employers and those produced by the local education system.

Contemporary Challenges: The three most acute economic challenges in northern BC today mirror those of the national economy: a housing market that has become significantly less affordable as in-migration from Metro Vancouver and Alberta has driven up prices; a healthcare system under fiscal pressure (Northern Health Authority); and the question of how a resource-dependent regional economy navigates

the clean energy transition while maintaining employment and fiscal revenues. These regional challenges appear in the course not as peripheral “local colour” but as concrete examples through which national economic forces become tangible.

Case Study: Canada’s 2022 Inflation Surge: A Turning Point

Background: Canada had not experienced inflation above 5% since 1991. A generation of economists, policymakers, and households had come of age in a low-inflation environment. The 2021–2022 surge was therefore not merely an economic event but a psychological and institutional stress test.

What Happened: Consumer price inflation rose from 0.7% in 2020 (deflated by pandemic-related demand collapse) to 3.4% in 2021 and then to a peak of 8.1% in June 2022. By that point, food prices were rising at over 10% year-over-year. Shelter costs were rising at over 7%. Energy prices — driven by Russian sanctions following the Ukraine invasion — were rising at over 30% for gasoline and over 20% for natural gas.

Why It Happened: The inflation surge had multiple, simultaneous causes. On the demand side: the COVID fiscal stimulus had placed over \$350 billion into Canadian household bank accounts and business balance sheets, creating excess demand when supply chains could not expand fast enough to meet it. Low interest rates through 2021 added further fuel. On the supply side: global shipping costs rose tenfold; semiconductor shortages constrained automobile and electronics production; pandemic-related labour shortages reduced output in food processing, transportation, and construction. The Russian invasion of Ukraine (February 2022) added an energy and food commodity price shock on top of already-elevated inflation.

Policy Response: The Bank of Canada raised its policy rate 10 times between March 2022 and July 2023, from 0.25% to 5.00%. By end-2023, inflation had fallen to 3.4%; by end-2024, it was approaching 2%. But the rate hikes came at significant cost: mortgage payments for variable-rate holders increased by hundreds of dollars per month; housing affordability deteriorated further; and economic growth slowed sharply.

Lesson: The 2022 episode illustrates that macroeconomic stability cannot be taken for granted. Institutional frameworks (like the inflation-targeting regime) that functioned smoothly for three decades can be tested by the right combination of

shocks. It also illustrates the limits of monetary policy: the Bank of Canada can raise rates to reduce demand, but it cannot directly fix supply chain disruptions, energy prices determined in global markets, or geopolitical shocks. These limits are a recurring theme in Chapter 2.

1.9 Economics in Everyday Life

Economics in Everyday Life: From the Gas Station to the Central Bank: Tracing Economic Forces in Your Daily Life

The next time you fill up your car, consider the chain of economic forces determining the number on the pump.

The **global oil price** (set in markets in Houston and Amsterdam) determines the base cost of crude. Canada's **exchange rate** — the value of the Canadian dollar relative to the US dollar — translates that global price into Canadian dollars: a 10-cent drop in the loonie raises the price of a barrel of oil (priced in USD) by approximately 7–8% in Canadian terms. The **Bank of Canada's interest rate decisions** affect the exchange rate by influencing capital flows into and out of Canada. Federal and provincial **excise taxes** add approximately 40–50 cents per litre at the pump — of which about 17 cents is the federal carbon levy (at the 2024 rate of \$65 per tonne). **Refinery margins**, influenced by the number and capacity of Canadian refineries (a market structure question), add another layer of cost.

Every chapter in this textbook touches on one of these forces. Chapter 2 examines how energy prices drive CPI inflation. Chapter 8 examines the exchange rate. Chapter 7 examines excise taxes. Chapter 10 examines the carbon levy. By the end of this course, the number on that gas pump will be transparent to you in a way it is not for most Canadians — not because you will agree with every policy that produced it, but because you will understand the economic forces behind each component.

1.10 Preview of Course Themes

The remaining nine chapters of this textbook address the major contemporary economic issues in the order determined by the pedagogical logic explained in the preface. Each

chapter builds on the institutional and analytical foundations established here.

Chapter 2: Inflation, Monetary Policy, and the Cost of Living. The immediate crisis of 2022–2023 examined in full: CPI measurement and its biases, the aggregate demand–aggregate supply framework, the Phillips Curve, the Bank of Canada’s transmission mechanism, and the distributional consequences of both inflation and the interest rate response.

Chapter 3: Canada’s Housing Crisis. The economics of housing markets, the user cost model, the supply constraints that prevent new construction from responding to demand, the role of immigration and interest rates on the demand side, and the full range of federal and provincial policy responses. This chapter flows directly from Chapter 2: the interest rate hikes needed to control inflation were the same rate hikes that deepened the mortgage crisis.

Chapter 4: Labour Markets, Employment, and Demographic Change. Labour supply and demand, labour force accounting, the Beveridge Curve, the wage-productivity gap, minimum wage economics, non-standard employment, and the relationship between an aging workforce and labour market tightness.

Chapter 5: Income Inequality, Poverty, and Indigenous Economic Development. The Lorenz Curve and Gini coefficient, trends in Canadian income distribution since 1980, intergenerational mobility, the gender wage gap, poverty measurement, and the economic consequences of the historical dispossession of Indigenous peoples.

Chapter 6: Immigration, Population Growth, and the Canadian Economy. Canada’s immigration system, the demographic arithmetic of the dependency ratio, the labour market effects of immigration, the housing demand effects, immigrant economic integration, and the policy debate over Canada’s historically high immigration targets.

Chapter 7: Fiscal Policy, Government Debt, and Public Finance. The federal budget process, deficit and debt arithmetic, the fiscal multiplier, the COVID-era fiscal expansion, healthcare spending and its demographic drivers, fiscal federalism and equalization, and the question of long-run fiscal sustainability.

Chapter 8: International Trade, Supply Chains, and Canada-US Relations. Comparative advantage, the gains from trade, Canada’s merchandise trade patterns, CUSMA and its predecessors, the exchange rate and its effects, the costs of tariff protection, supply chain vulnerabilities, and the catastrophic potential of a US-Canada trade conflict.

Chapter 9: Technology, Automation, Artificial Intelligence, and Productivity. The Solow growth model, Canada’s productivity puzzle, the task model of automation, which Canadian jobs are at risk, the economics of AI adoption, and what productivity growth requires in terms of investment, competition policy, and innovation incentives.

Chapter 10: Energy, Climate Change, and Canada's Green Transition. The carbon externality as a market failure, the Pigouvian tax rationale for carbon pricing, Canada's carbon price schedule, the oil sands and their fiscal importance, the electricity grid transition, the distributional incidence of carbon pricing, and the economic challenge of achieving net-zero by 2050 while managing a responsible transition for fossil-fuel dependent workers and communities.

Chapter Summary

- Contemporary economic issues are actively unfolding, contested, and consequential problems that require both theoretical tools and empirical evidence to analyse rigorously.
- Canada is the world's ninth-largest economy with a GDP of approximately \$3.0 trillion (2024), a population of 41.5 million, and a GDP per capita of roughly \$72,000 CAD (\$53,000 USD PPP-adjusted) — placing it above the OECD average but significantly below the United States.
- GDP is measured using the expenditure approach: $Y = C + I + G + (X - M)$. Real GDP adjusts nominal GDP for changes in the price level, measured by the GDP deflator. Real GDP growth is the standard measure of economic expansion or contraction.
- Canada's economy is predominantly service-based (77% of GDP), but with a disproportionately important natural resource sector (oil and gas, mining, forestry). Economically, Canada is dominated by Ontario and Quebec (60% of national GDP), with Alberta as the third-largest provincial economy due to oil sands.
- Canada is highly trade-dependent (trade openness $\approx 60.5\%$ of GDP) and extremely concentrated in the US market (75% of merchandise exports go to the United States).
- Canada's major economic institutions include the Bank of Canada (monetary policy, inflation targeting), the Department of Finance (fiscal policy), Statistics Canada (data), CMHC (housing), and provincial governments (healthcare, education, labour, natural resources).
- Monetary policy works through the overnight rate and its transmission channels (credit cost, asset prices, exchange rate, expectations). Fiscal policy works through government spending and taxation. The two interact: loose fiscal policy may require tight monetary policy, and vice versa.

- Economic reasoning requires: models (deliberate simplifications), positive–normative distinctions (what is vs. what should be), opportunity cost thinking, marginal analysis, and careful attention to the difference between correlation and causation.
- Canada’s macroeconomic history from 2015 to 2026 — summarized in Table 1.3 — encompasses a commodity price shock (2015–16), a historic pandemic shock (2020), a rapid fiscal and monetary policy response, and an inflationary surge and tightening cycle (2022–2024) that will shape Canada’s economy for years.

Key Terms

Aggregate demand

Total spending on domestically produced goods and services in the economy ($C + I + G + NX$).

Bank of Canada

Canada’s central bank; sets the overnight rate to achieve the 2% inflation target.

Budget deficit The excess of government spending over government revenues in a given fiscal year.

Comparative advantage

The ability to produce a good at a lower opportunity cost than another producer.

Contemporary economic issues

Actively unfolding economic problems that are contested and consequential for social well-being.

Correlation A statistical relationship in which two variables tend to move together; does not imply causation.

Fiscal policy The use of government spending and taxation to influence macroeconomic outcomes.

GDP deflator The ratio of nominal GDP to real GDP multiplied by 100; a broad measure of economy-wide price level changes.

Gross Domestic Product (GDP)

The total market value of all final goods and services produced within a country's borders in a given period.

GDP per capita

GDP divided by population; a rough measure of average material living standards.

Inflation targeting

A monetary policy framework in which a central bank commits to keeping inflation within a specified range.

Marginal cost and benefit

The additional cost or benefit of one more unit of an activity.

Model

A deliberate simplification of reality used to focus analysis on the mechanisms that matter for a specific question.

Monetary policy

Actions by a central bank to influence interest rates and credit conditions, aiming at price stability and sustainable growth.

Net exports (NX)

Exports minus imports; the trade balance component of the GDP expenditure identity.

Nominal GDP GDP measured in current-year prices.

Opportunity cost

The value of the next-best alternative foregone when making a choice.

Overnight rate The interest rate targeted by the Bank of Canada; the rate at which major financial institutions lend to each other overnight.

Pareto efficiency

An allocation in which no one can be made better off without making someone else worse off.

Positive economics

Description and explanation of economic phenomena without value judgements; testable against evidence.

Normative economics

Value judgements about what economic outcomes *should* be.

Real GDP GDP measured in the prices of a fixed base year; removes the effect of inflation from output comparisons.

Trade openness

$(X + M)/Y \times 100$; the ratio of total trade to GDP.

Transfer payments

Government payments to individuals that do not correspond to the production of new goods or services (e.g. EI, CPP, OAS).

Review Questions

1. **(Concept)** Define GDP and explain what it measures and what it does not measure. Give two examples of economically important activities that GDP omits, and explain why each omission might matter for policy.
2. **(Concept)** Explain the difference between nominal GDP growth and real GDP growth. If Canada's nominal GDP rose from \$2.72 trillion in 2022 to \$2.88 trillion in 2023, but the GDP deflator rose from 110.3 to 115.2, what was Canada's real GDP growth rate in 2023? Show all steps.
3. **(Institutions)** Explain the difference between the Bank of Canada and the Department of Finance. Why does Canada have two separate institutions managing monetary and fiscal policy rather than one? What problem does this separation solve?
4. **(Concept)** Classify each of the following statements as positive or normative, and explain your classification:
 - (a) "A 25% US tariff on Canadian auto exports would cost approximately 80,000 Canadian jobs."
 - (b) "Canada should implement a national childcare program to increase female labour force participation."
 - (c) "Canada's GDP per capita is approximately 37% lower than that of the United States."
 - (d) "The Bank of Canada should have started raising interest rates in late 2021 rather than waiting until March 2022."

5. **(Application)** Use Table 1.3 to answer the following:

- (a) In which year was Canada's unemployment rate highest? What event caused this?
- (b) In which year was Canada's inflation rate highest? What was the Bank of Canada's policy rate in that year?
- (c) What happened to the federal debt-to-GDP ratio between 2019 and 2020? Express the change in percentage points and explain what drove it.

6. **(Critical Thinking)** Canada's trade openness ratio is approximately 60.5%, while the United States' is approximately 27%. Using the concept of opportunity cost, explain why a high trade openness ratio creates both benefits and risks. Is there a "right" level of trade openness for a country like Canada?

Quantitative Problems

1. **GDP Calculation.** You are given the following data for a hypothetical Canadian province:

Item	Value (\$B)
Household consumption expenditure	480
Business investment in fixed capital	120
Change in business inventories	8
Provincial government spending on goods/services	140
Exports to other provinces and countries	210
Imports from other provinces and countries	195

- (a) Calculate nominal GDP using the expenditure approach (Equation 1.1). Show all steps.
- (b) What is the trade balance for this province? What does its sign indicate?
- (c) If the population is 4.8 million, calculate GDP per capita.

2. **Real vs. Nominal GDP.** The following data describe a simplified economy producing only two goods: wheat and lumber.

Good	Year 1 (Base)		Year 2	
	Price	Quantity	Price	Quantity
Wheat	\$4.00/bushel	500	\$5.50/bushel	520
Lumber	\$8.00/board-ft	300	\$9.00/board-ft	330

- Calculate nominal GDP in Year 1 and Year 2.
- Calculate real GDP in Year 2 using Year 1 prices.
- Calculate the nominal GDP growth rate and the real GDP growth rate between Year 1 and Year 2.
- What is the GDP deflator in Year 2 (with Year 1 = 100)? What is the implied inflation rate between Year 1 and Year 2?
- If the nominal growth rate was 17.8% but the real growth rate was only 6.5%, what does the difference represent?

3. **Trade Openness.** Using Table 1.1 and the following data:

- Canada's total merchandise exports, 2024: \$897 billion CAD
- Canada's total merchandise imports, 2024: \$918 billion CAD
- Canada's nominal GDP, 2024: \$3,001 billion CAD

- Calculate Canada's trade openness ratio (Equation 1.6).
- Calculate Canada's merchandise trade balance.
- What share of Canada's total exports goes to the United States? (Use Table 1.1.)
- If the United States were to impose a 25% tariff on all Canadian goods, and this caused Canada's US-bound exports to fall by 30%, what would be the new trade balance? (Assume all other trade is unchanged.)

4. **GDP per Capita Growth.** The relationship between GDP per capita growth, GDP growth, and population growth is approximately:

$$g_{\text{per capita}} \approx g_{\text{GDP}} - g_{\text{population}}$$

- In 2023, Canada's real GDP grew by 1.2% and its population grew by approximately 3.2% (driven largely by immigration). What was the approximate growth

- rate of real GDP *per capita*? What does this imply about average living standards?
- (b) If Canada's potential GDP growth rate is 2.0% and population growth is 3.0%, what is the implied trend in GDP per capita? Is this sustainable?
 - (c) Suggest two economic policies that could raise Canada's GDP growth rate so that it exceeds its population growth rate. For each policy, identify a potential trade-off.
5. **Opportunity Cost.** The federal government has \$20 billion in additional fiscal capacity. Three competing proposals are on the table:
- Proposal A: Invest in new public housing construction.
 - Proposal B: Cut the corporate income tax rate by 2 percentage points.
 - Proposal C: Fund a national pharmacare program for uninsured drugs.
- (a) For each proposal, identify one major benefit and one major cost (the opportunity cost of choosing that proposal over the others).
 - (b) Which economic concept (efficiency, equity, or both) is most relevant for evaluating each proposal?
 - (c) Is the choice among these three proposals a positive economic question, a normative question, or both? Explain.
6. **Data Exercise.** Visit the Statistics Canada website (www.statcan.gc.ca) and locate Table 36-10-0104-01 (GDP by expenditure, seasonally adjusted, at annual rates).
- (a) Report the real GDP growth rate (annualized) for the most recent available quarter.
 - (b) Break down GDP by component (C, I, G, NX) for the most recent quarter. Which component is largest? Which is most volatile over the past four quarters?
 - (c) Calculate the percentage change in each component over the past four quarters.
 - (d) Based on the current composition of growth, assess whether Canada's economic expansion is likely to be sustainable. Which components would you expect to change if interest rates rise further?

Essay Questions

1. **(500–700 words)** Using data from Table 1.3, write an essay describing the macroeconomic history of Canada between 2019 and 2026. Your essay should identify the major shocks, explain the policy responses, assess their effectiveness, and discuss the economic legacy of this period for Canada going forward. Use at least three specific data points from the table to support your argument.
2. **(400–600 words)** Canada runs an enormous merchandise trade surplus with the United States but trade deficits with most other countries (see Table 1.1). Using the concepts of comparative advantage and opportunity cost, explain what this pattern tells us about Canada's productive specialization. Discuss one risk and one benefit of Canada's trade concentration in the US market.
3. **(400–500 words)** The opening vignette of this chapter describes four people — Diane, Marcus, the sawmill manager, and Priya — each facing an economic challenge shaped by macroeconomic forces. Choose two of these four people. For each, identify the specific macroeconomic force most relevant to their situation, explain the economic mechanism at work, and identify one policy from Table 1.2 (or one of the institutions described in Section 1.4) that is most directly relevant to addressing their challenge.

Discussion Questions

1. Figure 1.1 shows that Canada's GDP per capita is approximately 37% lower than that of the United States, despite Canada having a comparable education system, similar institutions, and vast natural resource wealth. What do you think explains this gap? Is it a problem that Canada should try to close? What would closing it require?
2. Should the Bank of Canada be fully independent from the elected federal government? What would you gain, and what would you lose, by giving Parliament more direct control over monetary policy? Draw on the Policy Debate box in Section 1.5 in your answer.
3. Canada's regional economic inequality — illustrated in Figure 1.3 — means that national economic policies have unequal regional consequences. Is this a problem unique to Canada, or a challenge faced by all federations? How should the federal government balance national economic objectives with regional equity?

4. The opening of this chapter describes four people whose lives are shaped by macroeconomic forces they cannot individually control. Do you think most Canadians understand how economics affects their daily lives? What are the consequences — political, social, and economic — of widespread economic illiteracy?

CHAPTER 2

Inflation, Monetary Policy, and the Cost of Living

In January 2021, Fatima filled her grocery cart in Saskatoon and paid \$183. By January 2022, the same cart cost \$206. By June 2022, it cost \$228. In eighteen months, her weekly grocery bill had risen by \$45 — a 25% increase on food alone. Gasoline at the pump had gone from \$1.04 per litre to \$1.85. Her landlord had issued a rent increase notice citing rising property taxes and maintenance costs. Her variable-rate mortgage payment had climbed \$390 per month since March 2022, tracking the Bank of Canada's overnight rate almost in lockstep.

Fatima had not changed her lifestyle. She had not moved to a more expensive neighbourhood or started eating at restaurants. She had the same job, the same apartment, and the same spending habits. But the purchasing power of her income was shrinking, month by month, in a way she could not control and could barely predict.

Statistics Canada would later record that Canada's Consumer Price Index rose 8.1% in June 2022 — the highest reading in 40 years. That number meant something very specific: an average Canadian household needed \$108.10 in June 2022 to buy what \$100 bought in June 2021. For Fatima, whose budget had less slack than the average, the real erosion was larger.

What caused inflation this severe? Why did it emerge so suddenly after three decades of low, stable prices? How does the Bank of Canada fight it, and at what cost? And who bears the heaviest burden when prices rise faster than wages? These are the questions this chapter answers.

Learning Objectives

- Define inflation, calculate the CPI inflation rate from price data, and identify the four main biases in CPI measurement.
- Distinguish headline CPI from core inflation measures and explain why the Bank of Canada monitors both.
- Apply the Fisher equation to calculate real interest rates from nominal rates and inflation, and derive real wage growth from nominal wage and CPI data.
- Use the aggregate demand–aggregate supply framework to explain demand-pull and cost-push inflation, and apply both to Canada’s 2021–2023 episode.
- Describe the short-run and long-run Phillips Curve, interpret Canadian data on the inflation–unemployment relationship, and define the sacrifice ratio.
- Explain how the Bank of Canada’s inflation-targeting framework works, trace the channels of monetary policy transmission, and evaluate the 2022–2024 rate-hike cycle.
- Analyse the distributional consequences of inflation across income groups, between debtors and creditors, and between workers with indexed and non-indexed compensation.

2.1 Why Inflation Matters

Inflation is a sustained increase in the general price level of goods and services in an economy over time. Three words in that definition deserve emphasis. **Sustained:** a single price spike — gasoline surging one month and falling the next — is not inflation. **General:** inflation refers to the average price level, not any specific good. A single price rising while others fall is a *relative price* change, not inflation. **Over time:** inflation is a dynamic phenomenon measured as a rate of change, typically expressed as a percentage per year.

The opposite of inflation — a sustained *decrease* in the general price level — is **deflation**. Deflation may sound attractive (prices falling means money buys more), but it is one of the most dangerous macroeconomic conditions an economy can experience. When prices are expected to fall, households postpone purchases (why buy today what will be cheaper tomorrow?), firms delay investment, profits shrink, layoffs rise, and the economy can fall into a self-reinforcing downward spiral. Japan’s “Lost Decade” of the 1990s is

the modern cautionary example.

Low, stable, *positive* inflation — in Canada’s case, the 2% target — is the policy goal. Why 2% rather than 0%? Three reasons:

1. **Measurement bias:** The CPI tends to overstate true inflation by approximately 0.3–0.6 percentage points because it cannot fully capture quality improvements, new goods, and consumer substitution toward cheaper alternatives. A 2% measured target therefore implies roughly zero true inflation.
2. **Policy buffer:** If the equilibrium real interest rate is approximately 2% and inflation is targeted at 2%, the Bank of Canada can set a nominal rate of approximately 4% at equilibrium. This leaves 400 basis points of room to cut rates before hitting the zero lower bound in a recession. A 0% inflation target would leave only 200 basis points of room — dangerously thin.
3. **Wage adjustment:** Inflation allows real wages to fall without requiring nominal pay cuts, which workers strongly resist. In a world of 2% inflation, a firm that needs to reduce labour costs can simply freeze wages — real wages fall while nominal wages stay constant.

Think Like an Economist

If 2% inflation is the goal, should the Bank of Canada have started raising rates in late 2021 when inflation was already at 4.7%, rather than waiting until March 2022 when it had reached 5.7%? This is one of the most debated questions in recent Canadian monetary policy history. Your answer requires a theory of how quickly monetary policy works (the transmission lags), an assessment of whether the inflation was supply-side or demand-side, and a judgement about the risk of over-tightening into recession. We return to this question in the Policy Debate at the end of this chapter.

2.2 Measuring Inflation: The Consumer Price Index

2.2.1 The CPI Basket Approach

Canada’s primary measure of inflation is the *Consumer Price Index (CPI)*, produced monthly by Statistics Canada. The CPI measures the change in the total cost of a fixed *representative basket* of goods and services purchased by a typical Canadian urban household.

Definition: Consumer Price Index (CPI)

The **Consumer Price Index** measures the average change over time in the prices paid by urban consumers for a fixed basket of goods and services.

Economic intuition: The CPI asks: “How much does it cost today, relative to a base period, to maintain a fixed standard of living?” If the basket cost \$1,000 in the base period and costs \$1,081 today, the CPI is 108.1, implying 8.1% cumulative inflation since the base.

Formal definition: Statistics Canada periodically determines the basket weights from the Survey of Household Spending. Prices are collected from approximately 30,000 retail outlets, service providers, and landlords across Canada each month.

Canadian note: Statistics Canada publishes the CPI monthly in Table 18-10-0004-01. The June 2022 CPI reading was the highest year-over-year change since September 1983, when Canada was in the final stages of fighting the 1970s inflation surge.

Construction of the CPI proceeds in three steps:

1. **Determine basket weights.** Statistics Canada’s Survey of Household Spending identifies what Canadian urban households buy and in what proportions. These proportions are the *expenditure weights* w_i for each category i .
2. **Collect prices.** Each month, approximately 100,000 price observations are gathered from retailers, landlords, and service providers across Canada.
3. **Construct the index.** The CPI compares current-period prices to base-period prices, weighted by expenditure shares.

The formal CPI formula is a Laspeyres price index:

$$\text{CPI}_t = \frac{\sum_i w_i \cdot P_{i,t}}{\sum_i w_i \cdot P_{i,0}} \times 100 \quad (2.1)$$

where $P_{i,t}$ is the price of good i in period t , $P_{i,0}$ is the price of good i in the base period, and w_i is the expenditure weight of good i .

The *inflation rate* between two consecutive periods is:

$$\pi_t = \frac{\text{CPI}_t - \text{CPI}_{t-1}}{\text{CPI}_{t-1}} \times 100 \quad (2.2)$$

Worked Example 2.1 — Calculating the CPI from a Simple Basket

Suppose a household’s basket consists of three goods:

Good	Weight w_i	Base Price $P_{i,0}$	Current Price $P_{i,t}$	Price Relative $P_{i,t}/P_{i,0}$
Food	0.16	\$120	\$131.40	1.095
Shelter	0.30	\$900	\$963.00	1.070
Transport	0.20	\$300	\$342.00	1.140
<i>(Other categories omitted for simplicity)</i>				

Step 1: Calculate the weighted sum of price relatives:

$$\begin{aligned}\text{CPI}_t &= (0.16 \times 1.095) + (0.30 \times 1.070) + (0.20 \times 1.140) + \dots \\ &= 0.1752 + 0.3210 + 0.2280 + \dots = 0.7242 \text{ (for these three)}\end{aligned}$$

Full CPI (with all categories at assumed price relatives): Scaling up with remaining categories at an assumed average relative of 1.060:

$$\text{CPI}_t \approx 107.5 \implies \pi_t \approx 7.5\%$$

Interpretation: The basket now costs 7.5% more than in the base period. Because shelter (the largest weight) rose only 7.0% while transport (a smaller weight) rose 14.0%, the shelter-weighted average moderates what would otherwise be a higher measured rate.

Table 2.1. Canada CPI Inflation by Component, 2020–2024 (Annual Average, %)

CPI Component	2020	2021	2022	2023	2024
All-items (headline)	0.7	3.4	6.8	3.9	2.6
Food	2.4	3.9	9.7	5.9	1.5
Food purchased from stores	2.8	3.1	10.3	6.5	1.2
Food from restaurants	1.5	5.0	8.3	5.1	2.1
Shelter	2.4	4.6	6.9	6.0	5.7
Mortgage interest costs	0.8	1.3	11.5	28.5	19.4
Rent	3.0	1.4	4.5	8.1	8.9
Transportation	−3.6	8.3	10.6	2.4	0.8
Gasoline	−14.3	32.2	31.7	−4.2	−2.1
Energy (total)	−8.5	19.5	28.5	−8.5	−2.7
Services	1.4	2.5	5.5	4.7	4.0
Goods (excl. food/energy)	0.3	4.3	7.2	2.7	0.5

Source: Statistics Canada, Table 18-10-0004-01, CPI by component. Values are annual averages; 2024 values are estimates consistent with published data.

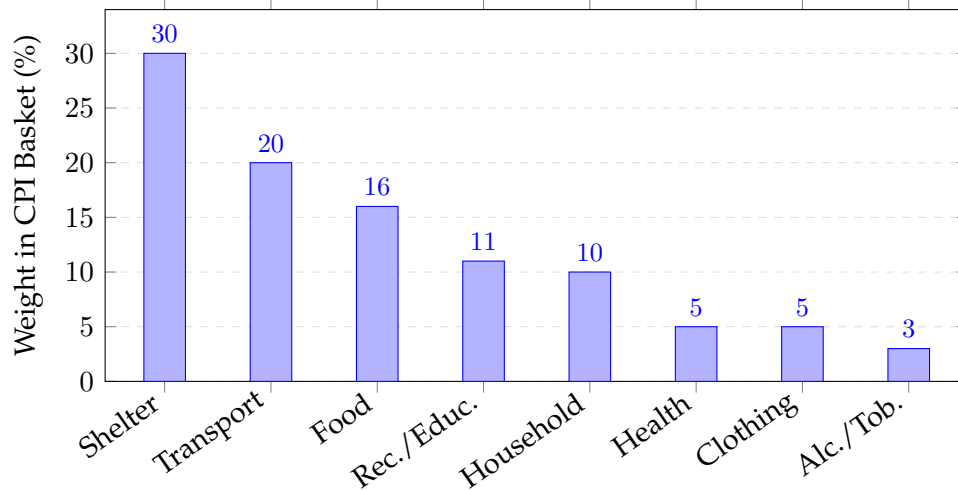


Figure 2.1. Weights in Canada's CPI Basket, 2022. Shelter and transportation together account for 50% of the CPI basket. This structural weighting means that even moderate increases in rent or mortgage costs — both classified under Shelter — move the headline inflation rate substantially. The dominance of Shelter also means that the Bank of Canada's rate hikes, which raised mortgage costs sharply, paradoxically *increased* one component of the CPI (mortgage interest costs) even as they reduced the underlying demand driving inflation.

Source: Statistics Canada, CPI Basket Weights, 2022 reference basket.

2.2.2 Biases in CPI Measurement

The CPI is a Laspeyres index: it holds the basket fixed at its base-period composition. This creates systematic upward biases in measured inflation:

- **Substitution bias:** When the price of good A rises relative to good B, consumers substitute toward B. The fixed CPI basket does not reflect this — it still assumes the old, pre-increase quantities of A are purchased. Result: the CPI overstates the true cost-of-living increase.
- **Quality bias:** If a new car costs \$5,000 more than last year's model but is also significantly better (safer, more fuel-efficient, with better technology), not all of the \$5,000 increase is inflation. Some of it is more quality per dollar. Statistics Canada adjusts for some quality changes, but imperfectly.
- **New goods bias:** When a completely new product (say, smartphones in 2007) enters the market, it is not immediately included in the basket. The large initial price declines as new products diffuse through the market are missed.
- **Outlet substitution bias:** When consumers shift purchases from traditional retailers to

lower-priced online sellers, the CPI may not fully capture the effective price reduction.

Studies by Statistics Canada and the Bank of Canada suggest the CPI overstates true inflation by approximately 0.3–0.5 percentage points per year on average. This is one reason the Bank of Canada’s 2% inflation target is set above zero rather than at zero.

2.2.3 Canada’s Annual Inflation Record, 2010–2026

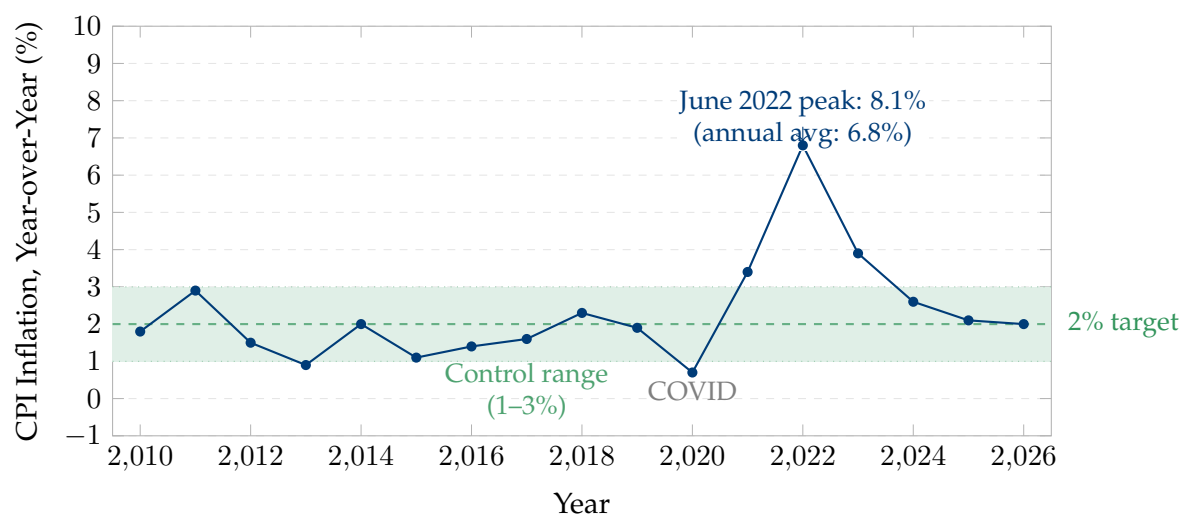


Figure 2.2. Canada Annual CPI Inflation Rate, 2010–2026. The shaded band shows the Bank of Canada’s 1–3% control range; the dashed line marks the 2% midpoint target. For most of 2012–2021, inflation remained within or near the band, reflecting credible monetary policy. The 2020 drop to 0.7% (pandemic demand collapse) and the subsequent surge to a peak of 8.1% in June 2022 (annual average 6.8%) represent the most dramatic deviation from target in the Bank’s 30-year history of inflation targeting. By 2024, inflation had returned to the upper portion of the target range.

Source: Statistics Canada, Table 18-10-0004-01; Bank of Canada. Annual averages; 2025–2026 are estimates consistent with Bank of Canada projections.

Key Takeaway

The CPI is the standard measure of inflation but contains systematic upward biases of approximately 0.3–0.5 percentage points per year. The Bank of Canada targets 2% measured CPI inflation, which is roughly consistent with near-zero *true* inflation once biases are accounted for. Understanding CPI biases is essential for interpreting economic data and evaluating policy.

2.3 Core Inflation: Seeing Through the Noise

Headline CPI measures the price level experienced by the average household. But for *monetary policy*, headline CPI is a problematic guide. The reason is that a central bank can only affect demand conditions — it cannot directly lower the price of crude oil, ease a shipping container shortage, or replace a failed harvest. When inflation is driven by temporary supply shocks (a spike in gasoline prices, a drought-induced food price jump), a central bank that responds aggressively risks *policy error*: tightening into a supply-driven recession.

Core inflation measures remove the most volatile components from the headline CPI to reveal the underlying, persistent inflation trend that monetary policy can actually address. The Bank of Canada monitors three official core measures:

Definition: Bank of Canada Core Inflation Measures

CPI-trim: Removes the tails of the monthly price change distribution — the highest and lowest 20% of components by price change magnitude — and computes the weighted average of the remaining 60%.

CPI-median: The price change at the 50th percentile of the weighted distribution of component price changes. Exactly half of the CPI basket (by weight) is changing faster than this rate, and half slower.

CPI-common: Extracts the common variation in price changes across CPI components using statistical factor analysis. It captures inflation that is economy-wide rather than sector-specific.

Why three measures? Each measure has different statistical properties. By monitoring all three and looking for consensus, the Bank reduces the risk of any single measure sending a misleading signal. When all three core measures are elevated, the Bank is confident that inflation is broad-based, persistent, and amenable to a policy response.

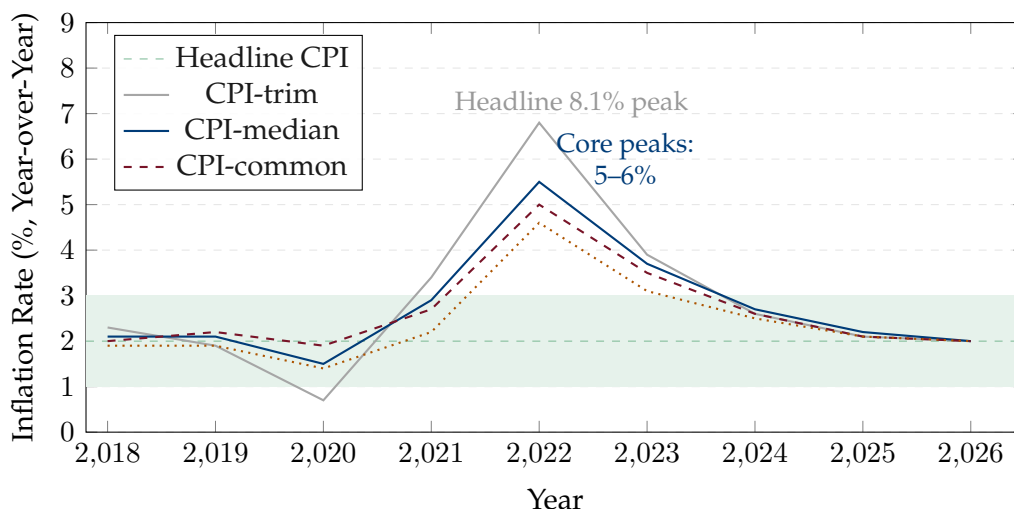


Figure 2.3. Headline CPI vs. Bank of Canada Core Inflation Measures, 2018–2026. The core measures (CPI-trim, CPI-median, CPI-common) peaked approximately 2–3 percentage points below the 8.1% headline peak, reflecting the contribution of volatile energy and some food prices to the headline figure. Crucially, all three core measures rose substantially above the 3% upper bound of the control range in 2022 — confirming that the inflation surge was broad-based, not merely a transitory spike in a few volatile categories. This broad-based elevation justified the Bank of Canada’s aggressive tightening.

Source: Bank of Canada, Measures of Core Inflation database; Statistics Canada, Table 18-10-0004-01. Values for 2025–2026 are estimates.

A critical insight from Figure 2.3: the gap between headline and core inflation *narrowed* in 2023–2024. As energy prices fell back from their 2022 peaks, headline inflation declined sharply. But core measures remained elevated well into 2024 because *services inflation* — wages, rents, and labour-intensive services — proved “sticky” downward. This stickiness reflects the second-round effects of inflation: when workers see prices rise, they demand higher wages; higher wages increase firms’ costs, pushing up prices further; and the cycle reinforces itself. Breaking this cycle is the primary rationale for the Bank of Canada’s rate-hike strategy.

2.4 Real vs. Nominal Interest Rates: The Fisher Equation

When a bank offers you a savings account paying 5% interest, is that a good deal? The answer depends on inflation. If prices are rising at 3% per year, your 5% nominal return only delivers 2% more purchasing power — the real return is just 2%. If inflation is running at 6%, your “5%” nominal return is actually a loss of 1% in purchasing power.

This insight is captured by the *Fisher equation*, one of the most important relationships in monetary economics:

$$r \approx i - \pi \quad (2.3)$$

where r is the *real interest rate*, i is the *nominal interest rate*, and π is the *inflation rate*.

The exact version is:

$$(1 + r) = \frac{(1 + i)}{(1 + \pi)} \iff r = \frac{1 + i}{1 + \pi} - 1 \quad (2.4)$$

For the values typically encountered in Canada, the approximation $r \approx i - \pi$ (Equation 2.3) is sufficiently accurate for analysis.

Definition: Real vs. Nominal Interest Rates

The **nominal interest rate** i is the rate stated in the loan contract or savings agreement — the number on the Bank of Canada website.

The **real interest rate** $r = i - \pi^e$ is the nominal rate adjusted for (expected) inflation. It represents the true cost of borrowing or the true return to saving in terms of purchasing power.

When making economic decisions, it is the *real* interest rate that matters. A firm deciding whether to invest in new equipment compares the real borrowing cost against the real rate of return on the equipment. A household deciding whether to take out a mortgage compares the real cost of the loan against the real appreciation of the home.

Ex ante vs. ex post: Before the fact (*ex ante*), borrowers and lenders agree on a nominal rate based on their expectation of future inflation π^e . After the fact (*ex post*), the real rate reflects actual inflation π . Unexpected inflation benefits debtors (who repay in cheapened dollars) and hurts creditors (who receive less purchasing power than expected).

Table 2.2. Real Interest Rate Calculation, Canada: 2018–2026

Year	Nominal Rate i (Bank Rate, %)	CPI Inflation π (%)	Real Rate $r \approx i - \pi$ (%)	Real Rate Interpretation	Policy Context
2018	1.75	2.3	−0.6	Mildly stimulative	Late expansion
2019	1.75	1.9	−0.2	Roughly neutral	Pre-COVID
2020	0.25	0.7	−0.5	Stimulative	COVID emergency
2021	0.25	3.4	−3.2	Highly stimulative	Post-COVID stimulus
2022	4.25*	6.8	−2.6*	Still negative	Tightening cycle
2023	5.00	3.9	+1.1	Mildly restrictive	Tightening complete
2024	3.75†	2.6	+1.2	Mildly restrictive	Easing begins
2025	3.00	2.1	+0.9	Near neutral	Easing continues
2026	2.75	2.0	+0.8	Near neutral	Soft landing

*End-of-year Bank Rate; mid-year 2022 average rate was approximately 2.0%, making the real rate even more negative at −4.8%. †December 2024 rate after four cuts in the second half of 2024.

Source: Bank of Canada key interest rates; Statistics Canada Table 18-10-0004-01; author calculations.

The Key Lesson from Table 2.2: In 2021 and through most of 2022, the real interest rate was deeply *negative* — borrowers were effectively being paid to borrow in real terms. At a mid-2022 Bank Rate of $i = 2.5\%$ and inflation of $\pi = 7\%$, the real rate was approximately $r = -4.5\%$. This meant that anyone borrowing at the prime rate to purchase real assets (housing, stocks, commodities) was receiving a massive implicit subsidy from inflation. Negative real rates were a powerful engine of the demand boom that itself contributed to inflation.

2.4.1 Real Wages: The Fisher Equation Applied to Labour

The Fisher relationship applies equally to wages. The *nominal wage* W is the dollar amount a worker earns. The *real wage* is the nominal wage deflated by the price level:

$$w_{\text{real},t} = \frac{W_t}{\text{CPI}_t/100} \quad (2.5)$$

The growth rate of real wages is approximately:

$$g_{w,\text{real}} \approx g_{W,\text{nominal}} - \pi \quad (2.6)$$

Worked Example 2.2 — Real Wage Growth, Canada 2021–2023

Average hourly wages in Canada grew from \$28.57 in January 2021 to \$32.36 in January 2023.

- Nominal wage growth: $(32.36 - 28.57)/28.57 \times 100 = 13.3\%$
- CPI in January 2021: 138.8; January 2023: 155.6
- CPI change: $(155.6 - 138.8)/138.8 \times 100 = 12.1\%$
- Real wage growth: $13.3\% - 12.1\% = +1.2\%$ over two years

Despite nominal wage gains that looked impressive, real wages grew by only 0.6% per year on average — barely keeping pace with productivity. For workers in the bottom two income quintiles, where wage gains were smaller and inflation in food and shelter (which loom larger in their budgets) was higher, real wages actually fell.

2.5 The Aggregate Demand–Aggregate Supply Framework

The *Aggregate Demand–Aggregate Supply (AD-AS) model* is the workhorse framework for analysing macroeconomic fluctuations, including inflation. It explains how the overall price level and real output are jointly determined in the economy, and why they change over time.

2.5.1 Aggregate Demand

The *Aggregate Demand (AD) curve* shows the combinations of price level and real output at which the goods market and money market are simultaneously in equilibrium. It slopes downward for three reasons:

- **Real balance effect:** A higher price level reduces the real value of household wealth (including cash and bonds held at face value), reducing consumption spending.
- **Interest rate effect:** A higher price level increases the demand for money (people need more nominal money to buy the same real quantities), pushing up interest rates and reducing investment.
- **Exchange rate effect:** Higher domestic prices make Canadian goods less competitive internationally, reducing exports and increasing imports — reducing net exports NX .

A simplified aggregate demand relationship can be written:

$$Y = \alpha_0 - \beta \cdot r + \gamma \cdot G - \delta \cdot P \quad (2.7)$$

where Y is real output, r is the real interest rate, G is government spending, and P is the price level. The parameters β and δ capture the sensitivity of spending to interest rates and prices, respectively.

Shifts in AD: The AD curve shifts rightward (increasing output demand at every price level) when:

- Government spending G increases (fiscal expansion)
- The central bank cuts interest rates r (monetary expansion)
- Household wealth or confidence rises
- Export demand increases (trading partner growth or currency depreciation)

2.5.2 Aggregate Supply

The *Short-Run Aggregate Supply (SRAS) curve* shows the total quantity of goods and services firms are willing to produce at each price level, given that input prices (especially wages) are fixed in the short run. It slopes upward because higher output prices increase profit margins, incentivizing firms to expand production.

A simple SRAS relationship:

$$P = P^e + \frac{1}{\alpha}(Y - Y_f) \quad (2.8)$$

where P is the current price level, P^e is the expected price level (determining the position of the SRAS curve), Y_f is potential output (full-employment output), and α is a positive parameter governing how responsive prices are to output gaps.

The *Long-Run Aggregate Supply (LRAS) curve* is vertical at $Y = Y_f$ — potential output. In the long run, all prices are flexible, the economy returns to Y_f , and the price level adjusts to whatever is required for this equilibrium. Monetary policy cannot permanently raise output above Y_f ; it can only affect the price level in the long run.

2.5.3 Two Types of Inflation

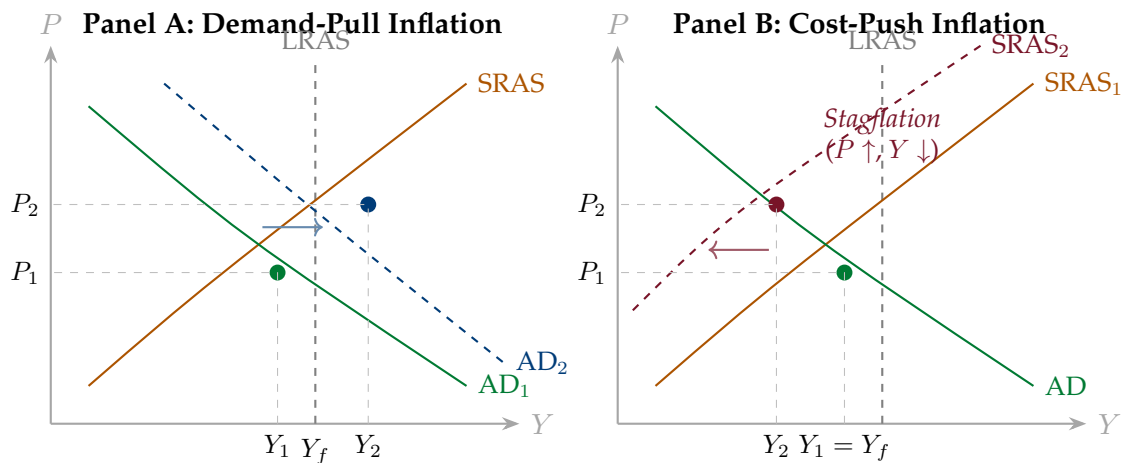


Figure 2.4. Demand-Pull vs. Cost-Push Inflation in the AD-AS Framework. **Panel A (Demand-Pull):** A rightward shift of AD (from AD_1 to AD_2) — driven by fiscal stimulus, low interest rates, or rising household wealth — moves the economy from equilibrium (Y_1, P_1) to (Y_2, P_2) , raising both output and prices. If $Y_2 > Y_f$, the economy is overheating. **Panel B (Cost-Push):** A leftward shift of SRAS — driven by higher energy prices, supply chain disruptions, or rising input costs — produces the combination of higher prices *and* lower output known as **stagflation**. Canada’s 2021–2023 inflation episode involved both mechanisms simultaneously: massive AD expansion from COVID stimulus and near-zero rates, combined with supply chain bottlenecks and the Russian-invasion energy shock.

Canadian Example: Canada's 2021–2023 Inflation: Both Demand and Supply

Canada's inflation surge was unusual because both AD and AS shocks occurred simultaneously and reinforced each other.

Demand side: The federal government injected approximately \$350 billion in emergency spending between 2020 and 2022. The Bank of Canada held its policy rate at 0.25% until March 2022. Household savings rates surged during lockdowns (\$250+ billion in excess savings accumulated), creating pent-up demand that exploded when the economy reopened. These forces shifted AD rightward continuously from mid-2020 through 2021.

Supply side: Global shipping container costs rose tenfold. Semiconductor shortages idled automotive production lines — used car prices in Canada rose over 40% in 2021. Russia's invasion of Ukraine (February 2022) triggered an energy and food commodity shock, shifting SRAS leftward. Labour shortages in food processing, transportation, and construction added further upward cost pressure.

Result: In the AD-AS diagram, Canada experienced both a large rightward AD shift and a leftward SRAS shift. Prices rose sharply. Output expanded beyond potential in 2021 (generating an “output gap” of approximately +2% of potential GDP) before the rate hikes began cooling demand. This combination — demand exceeding supply on both sides — is precisely what generates the most acute inflationary episodes.

2.6 The Phillips Curve: Inflation and Unemployment

The *Phillips Curve* describes the empirical and theoretical relationship between inflation and unemployment. The original version, identified by economist A.W. Phillips (1958) using UK data from 1861–1957, showed a stable negative relationship: when unemployment was low, wages (and thus prices) tended to rise faster.

2.6.1 The Short-Run Phillips Curve

The short-run Phillips Curve can be expressed as:

$$\pi_t = \pi_t^e - \alpha(u_t - u_n) \quad (2.9)$$

where:

- π_t is the actual inflation rate in period t

- π_t^e is the expected inflation rate (which shifts the curve)
- u_t is the actual unemployment rate
- u_n is the *natural rate of unemployment* (the NAIRU — Non-Accelerating Inflation Rate of Unemployment)
- $\alpha > 0$ is a parameter measuring how sensitive inflation is to labour market slack

The term $(u_t - u_n)$ is the *unemployment gap*. When $u_t < u_n$ (labour markets are tight, unemployment below natural rate), there is upward pressure on wages and prices: $\pi_t > \pi_t^e$. When $u_t > u_n$ (slack in labour markets), inflation tends to run below expectations: $\pi_t < \pi_t^e$.

For Canada, the Bank of Canada estimates the natural rate of unemployment u_n at approximately 5.5–6.0% in the mid-2020s (accounting for structural factors such as the geographic mismatch between job vacancies and unemployed workers, and skills mismatches). When unemployment fell to 4.9% in mid-2022, the gap $(u_t - u_n) \approx -1.0$ to -1.5 percentage points, contributing upward pressure on wages and prices.

2.6.2 The Long-Run Phillips Curve

If workers have *rational expectations* — they correctly anticipate inflation on average — then any attempt by the central bank to exploit the inflation–unemployment trade-off will be defeated. If the Bank tolerates higher inflation to keep unemployment below the natural rate, workers eventually demand higher nominal wages to compensate for the higher expected inflation, shifting the SRAS curve upward and returning the economy to u_n — but now at a permanently higher inflation rate.

The long-run Phillips Curve is therefore **vertical at** $u = u_n$: there is no long-run trade-off between inflation and unemployment. This result, developed independently by Edmund Phelps and Milton Friedman in 1968, is one of the most important insights in macroeconomics. It implies:

- Monetary policy cannot permanently lower unemployment below u_n .
- Attempting to do so only generates accelerating inflation.
- The only permanent gains in employment come from structural policies that reduce u_n itself (better job matching, education, reduced geographical barriers to labour mobility).

2.6.3 The Sacrifice Ratio

When a central bank wants to *reduce* inflation that has become embedded in expectations, it must push unemployment above the natural rate — accepting a period of higher unemployment. The *sacrifice ratio* measures the cost of this disinflation:

$$\text{Sacrifice Ratio} = \frac{\text{Cumulative excess unemployment (\%)}}{\text{Reduction in inflation (pp)}} \quad (2.10)$$

Empirical estimates for Canada typically place the sacrifice ratio between 1.5 and 3.0: to permanently reduce inflation by 1 percentage point requires approximately 1.5–3.0 percentage points of excess unemployment sustained for one year. For the 2022–2024 disinflation of approximately 4 percentage points (from 6.8% to 2.6%), the implied employment cost was significant.

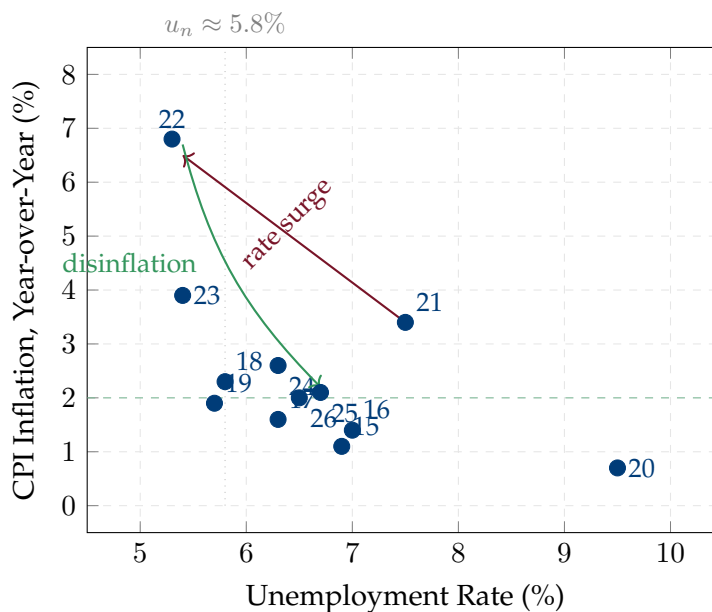


Figure 2.5. Phillips Curve: Canada, 2015–2026. Each point plots the annual unemployment rate against the annual average CPI inflation rate, labelled by the last two digits of the year. The broad negative relationship — lower unemployment associated with higher inflation — is visible in 2015–2022, consistent with the short-run Phillips Curve. The 2020 point (year 20) is an outlier: unemployment spiked to 9.5% while inflation collapsed to 0.7%, reflecting the demand destruction of the COVID lockdowns. The 2022 point (year 22) sits at the extreme top-left: very low unemployment (5.3%) and very high inflation (6.8%), reflecting a severely overheated economy. The red arrow traces the surge and the green arrow traces the subsequent disinflation path, along which unemployment rose as inflation fell.

Source: Statistics Canada, Tables 14-10-0287-01 and 18-10-0004-01. Annual averages; 2025–2026 estimated.

2.7 Canada's Inflation-Targeting Framework

Canada was the second country in the world (after New Zealand in 1990) to adopt a formal *inflation target*. The framework was introduced in February 1991 under a joint agreement between the Bank of Canada and the federal government. It has been renewed multiple times, most recently in 2021 with modifications that reflected the lessons of the COVID era.

The key features of the current framework:

- **Target:** 2% measured CPI inflation.
- **Control range:** 1–3% (the Bank uses the range as an indicator of acceptable outcomes, not as hard bounds).
- **Horizon:** 18–24 months — the time over which the Bank aims to return inflation to the 2% target following a shock. This reflects the lags in monetary policy transmission.
- **Instrument:** The overnight rate (policy rate).
- **Communication:** Eight pre-announced Monetary Policy Announcement (MPA) dates per year, accompanied by a quarterly Monetary Policy Report (MPR) with detailed economic forecasts.
- **Accountability:** The Bank Governor must write an open letter to the Minister of Finance explaining why inflation has moved more than 1 percentage point above or below the 2% target.

Why targeting works — the role of credibility: Inflation targeting is more than a rule; it is a commitment device. When firms, workers, and financial markets believe the Bank will keep inflation at 2%, they set wages and prices consistent with 2% inflation. This self-fulfilling property — “credibility” — makes disinflation less costly (smaller sacrifice ratio) and makes it easier to keep inflation at target even when shocks occur. If the Bank loses credibility, inflation expectations become de-anchored, requiring more aggressive tightening to achieve the same disinflationary effect.

The 2021–2022 episode tested this credibility severely. By maintaining the policy rate at 0.25% while inflation surpassed 5% — citing concerns about premature withdrawal of pandemic support — the Bank allowed long-term inflation expectations to creep above 2%. By the time rates rose aggressively, some of the benefit of credibility had been partially eroded, requiring more tightening than if the Bank had acted earlier.

2.8 Monetary Policy Transmission

When the Bank of Canada changes its overnight rate, the effects on inflation and output do not occur immediately. There are multiple channels through which the policy rate change propagates through the economy, and each channel operates with different lags. The full effect on inflation is typically estimated to take **18–24 months**.

Definition: The Four Channels of Monetary Policy Transmission

- 1. Credit Cost Channel:** Higher overnight rates raise the cost of borrowing for households and firms. Mortgage rates, car loan rates, business credit lines, and corporate bond yields all rise. This reduces investment spending (firms borrow less to expand) and consumer spending (households reduce debt-financed purchases). This is the fastest and most direct channel.
- 2. Asset Price Channel:** Higher interest rates reduce the present value of future cash flows, lowering stock prices, bond prices, and real estate valuations. Lower household wealth reduces consumption through the *wealth effect*. A fall in equity prices also raises firms' cost of capital, further reducing investment.
- 3. Exchange Rate Channel:** Higher Canadian interest rates attract foreign capital seeking superior returns, increasing demand for Canadian dollars and appreciating the exchange rate. A stronger Canadian dollar makes imports cheaper (reducing the domestic price of imported goods) and makes Canadian exports less competitive internationally (reducing net exports). Both effects reduce domestic price pressure.
- 4. Expectations Channel:** When the Bank credibly signals its inflation intentions, households and firms adjust their price- and wage-setting behaviour immediately, *before* the other channels fully operate. If workers and firms believe the Bank will achieve 2% inflation, they set wages and prices consistent with 2% — effectively doing the Bank's job for it. This is why central bank communication (*forward guidance*) is itself a monetary policy instrument.

The relative importance of these channels in Canada reflects the structure of the Canadian economy. The credit cost channel is particularly powerful because:

1. Canada has one of the highest household debt-to-income ratios among OECD countries (180% as of 2023), making households extremely sensitive to interest rate changes.
2. Approximately 35% of Canadian mortgages were in variable-rate products as of 2022, meaning rate increases transmitted to mortgage payments with minimal lag.

3. Housing wealth constitutes approximately 40% of total Canadian household net worth, making the asset price channel powerful.

2.9 Case Studies

Case Study: The Bank of Canada's 2022–2024 Rate-Hike Cycle

Background: By early 2022, Canada's inflation had risen to levels not seen since the early 1980s. The Bank of Canada, which had kept its policy rate at the emergency floor of 0.25% since March 2020, faced the challenge of rapidly tightening monetary policy without triggering a severe recession.

The Timeline: The tightening cycle began on March 2, 2022, with a 25-basis-point increase — the Bank's first rate hike since October 2018. What followed was the most rapid tightening cycle in the Bank's modern history.

Table 2.3. Bank of Canada Rate Decisions, March 2022 – December 2024

Date	Change (bps)	New Rate (%)	Bank's Primary Rationale
Mar 2, 2022	+25	0.50	Inflation exceeding target; recovery solid
Apr 13, 2022	+50	1.00	Inflation broad-based; front-loading needed
Jun 1, 2022	+50	1.50	CPI 7.7% in Apr; labour markets tight
Jul 13, 2022	+100	2.50	CPI 8.1% June peak; unconventional move
Sep 7, 2022	+75	3.25	Core measures still rising
Oct 26, 2022	+50	3.75	Moderate; economy showing some cooling
Dec 7, 2022	+50	4.25	Inflation still above target
Jan 25, 2023	+25	4.50	Conditional pause signalled
Mar 8, 2023	+25	4.75	Core inflation re-accelerated
Jun 7, 2023	+25	5.00	GDP and CPI stronger than expected
Jul–Dec 2023	<i>Held</i>	5.00	<i>Assessment period</i>
Jun 5, 2024	–25	4.75	Inflation returning to target
Jul 24, 2024	–25	4.50	Labour market easing
Sep 4, 2024	–25	4.25	CPI at 2.5%; output gap opened
Oct 23, 2024	–50	3.75	Downside risks growing
Dec 11, 2024	–50	3.25	Inflation at target; growth below potential

Source: Bank of Canada, Monetary Policy Announcements, 2022–2024. “bps” = basis points (1 basis point = 0.01 percentage point).

The July 2022 Anomaly: The July 13, 2022 decision — a 100-basis-point increase, the largest single-meeting rate hike since 1998 — was remarkable. The Bank cited the June 2022 CPI print of 8.1% and the risk that inflation expectations would become “de-anchored” from the 2% target. At that point, the Bank was willing to accept a sharp slowdown rather than allow inflation to become entrenched.

Outcomes: By June 2024, inflation had returned to 2.7% — near the 2% target. Canada avoided a technical recession (two consecutive quarters of negative real GDP growth) in 2022–2023, though real GDP per capita did decline. The unemployment rate rose from a 50-year low of 4.9% in mid-2022 to 6.5–6.8% by late 2024.

Evaluation: Did the Bank of Canada succeed? On price stability, yes: inflation returned to target within two years of the June 2022 peak. On the real economy,

the outcome was relatively benign by historical standards — no deep recession, modest unemployment increase, and nominal wage growth that largely kept pace with declining inflation. However, mortgage holders who renewed into 5% rates experienced the most severe payment shock in a generation, and some economists argue the Bank held rates too high for too long in 2023–24, unnecessarily slowing growth.

Case Study: Shelter Inflation: The CPI's Most Powerful Component

The Issue: Table 2.1 shows that shelter inflation remained elevated — and in some measures *accelerating* — into 2023 and 2024, even as energy and goods prices eased. Mortgage interest costs rose 28.5% in 2023 and 19.4% in 2024 year-over-year, making shelter the single largest driver of measured inflation in the post-peak period.

The Paradox: The Bank of Canada raised interest rates to fight inflation. But because mortgage interest costs are included in the CPI shelter component (they are part of the cost of “owning” a home), the rate hikes directly increased the CPI they were supposed to reduce. This creates a short-run paradox: tighter monetary policy mechanically raises measured inflation through the mortgage channel, even as it slows the economy and reduces demand-side inflation pressure.

Why Does This Happen in Canada? Canada’s CPI shelter methodology, unlike that of many other countries, includes mortgage interest cost (MIC) directly. The US CPI, by contrast, uses *owners’ equivalent rent* — an estimate of what homeowners would pay if they rented their homes — which is more insulated from short-run interest rate movements.

Policy Implication: This is one reason the Bank of Canada monitors core inflation measures (particularly CPI-trim and CPI-median) that tend to dampen the direct mortgage cost effect. It is also why economists sometimes argue that the headline CPI in Canada gives a misleading picture of “true” cost-of-living inflation during monetary tightening cycles.

Quantitative impact: In 2023, mortgage interest costs contributed approximately +1.4 percentage points to Canada’s headline inflation of 3.9% — meaning that without the mortgage component, headline inflation would have been closer to 2.5%. The same rate hikes that were cooling inflation were adding roughly 1.4 points to the inflation number, creating both an economic paradox and a political communications challenge for the Bank.

Case Study: Who Pays for Inflation? Distributional Consequences Across Income Groups

Background: The CPI measures inflation for the “average” household. But average households do not exist — spending patterns, asset holdings, debt levels, and income sources vary enormously across the income distribution. The 2022 inflation surge was not experienced equally.

The Regressivity of Inflation: Low-income households devote larger shares of their budgets to food, shelter, and energy — the three categories that rose fastest in 2022. Statistics Canada analysis estimates that the effective inflation rate experienced by the bottom income quintile in 2022 was approximately 8.5%, compared to approximately 6.8% for the top quintile. A gap of 1.7 percentage points may seem small in absolute terms, but for a household spending 95% of its income, the difference between an 8.5% and 6.8% inflation rate is the difference between severe hardship and moderate difficulty.

Debtors and Creditors: Unexpected inflation redistributes wealth from creditors to debtors. Anyone holding a fixed-rate loan from before the inflation surge found that the real value of their debt was being eroded by inflation. Mortgage holders with 5-year fixed rates locked in at 2% in 2020 enjoyed negative real borrowing costs throughout 2022 — their 2% nominal rate minus 6%+ inflation implied a real borrowing cost of approximately -4% . Retirees and pension funds holding fixed-income securities saw the real value of their assets eroded.

Wage-earners vs. Capital owners: Workers whose wages did not fully adjust to inflation experienced real pay cuts. Workers in unionized sectors with cost-of-living adjustment (COLA) clauses in their contracts were partially protected. Non-unionized, part-time, and gig workers — disproportionately low-income, racialized, and young — had the least wage protection. Capital owners who held real assets (housing, stocks, commodities) saw their nominal values rise with inflation, preserving real wealth while nominal wealth grew.

Policy Implications: The distributional consequences of inflation strengthen the case for price stability as a distributive goal, not just an efficiency goal. Inflation is not a politically neutral phenomenon: it systematically favours debtors over creditors, real asset holders over cash savers, and the flexible-wage over the fixed-wage workforce.

2.10 Distributional Effects of Inflation

The case study above is worth formalising. Figure 2.6 presents estimated effective inflation rates by income quintile for Canada in 2022, constructed by reweighting CPI components using the expenditure patterns of each quintile from the Survey of Household Spending.

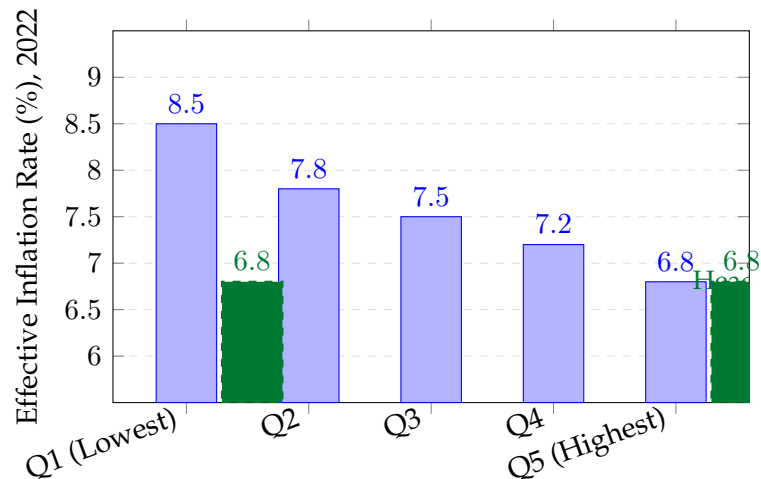


Figure 2.6. Estimated Effective Inflation Rate by Income Quintile, Canada, 2022. Low-income households (Q1) experienced approximately 8.5% effective inflation — 1.7 percentage points above the headline CPI of 6.8% — because they spend a larger share of their budgets on food, shelter, and energy, the three categories with the sharpest price increases in 2022. The dashed line shows the official headline CPI for reference. Inflation is not a neutral tax: it falls disproportionately on those least able to absorb it.

Source: Author calculations using Statistics Canada Survey of Household Spending and CPI component data. Methodology follows Argente and Lee (2021) and Statistics Canada analytical studies.

2.11 The Rate-Hike Cycle: Nominal and Real Rates

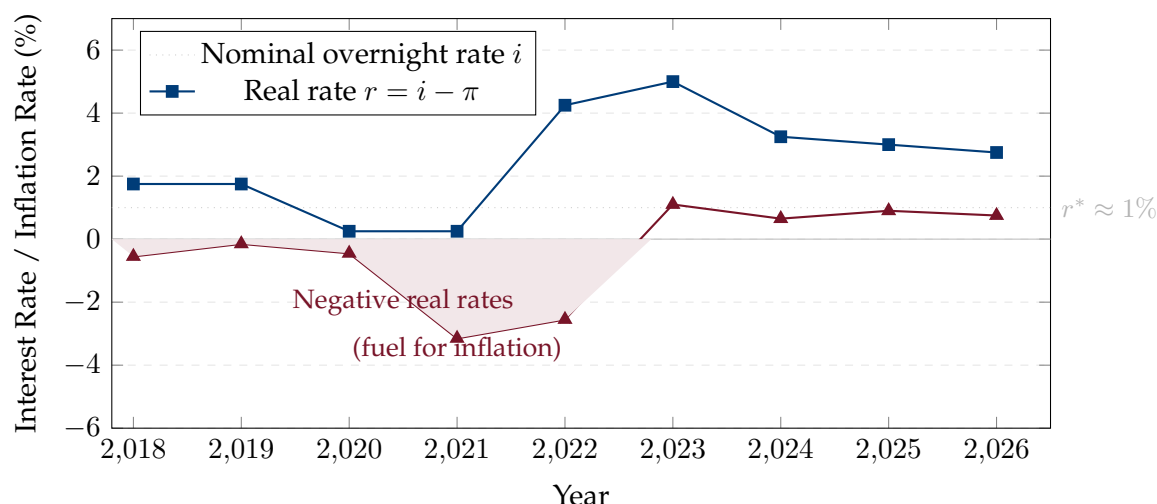


Figure 2.7. Nominal Overnight Rate vs. Real Interest Rate, Canada, 2018–2026. The shaded region marks the period of negative real interest rates (2018–2022). In 2021, the real rate fell to approximately -3.2% — deeply negative — as the nominal rate was held at 0.25% while inflation rose to 3.4% . By mid-2022, the combination of rapid nominal rate increases and peak inflation meant the real rate was still close to -4.5% even as nominal rates rose to 2.5% . Only in 2023, when nominal rates reached 5.00% and inflation fell to 3.9% , did the real rate finally turn positive. The return to positive real rates was the mechanism through which monetary policy finally cooled aggregate demand.

Source: Bank of Canada key interest rates; Statistics Canada Table 18-10-0004-01; author calculations using Fisher approximation $r \approx i - \pi$.

Policy Debate

Policy Debate: Did the Bank of Canada Hike Too Far, Too Fast — or Not Far Enough, Too Late?

The Issue: The Bank of Canada’s 2022–2024 rate-hike cycle was the most aggressive in its modern history. Whether it was the right response depends critically on the diagnosis of what caused the inflation and on judgements about the costs of different types of policy error.

Case 1 — Too Slow to Act (The Hawkish Critique): Critics argue the Bank should have raised rates earlier — potentially as early as the summer of 2021, when CPI was already at 3.7% . By maintaining emergency-level rates through 2021, the Bank:

- Allowed real interest rates to fall to -3% , massively stimulating borrowing and spending.
- Failed to “lean against” surging housing prices, which added to the wealth effect and the inflation spiral.
- Let inflation expectations drift above 2% , requiring more aggressive tightening later.

The implicit sacrifice ratio of this delayed response was higher than it needed to be: more unemployment was ultimately required to achieve the same disinflation because expectations had partially de-anchored.

Case 2 — Too Aggressive (The Dovish Critique): Others argue the Bank moved too fast and too far, particularly after mid-2022. By the time the July 2022 100-basis-point hike was made, global supply chains were already improving, energy prices were beginning to fall, and the inflation peak was likely already past. The aggressive tightening:

- Imposed mortgage payment shocks on millions of Canadians who had no role in causing inflation.
- Slowed real GDP per capita growth below zero for five consecutive quarters in 2023–24.
- Risked pushing unemployment unnecessarily high, creating real human costs (job loss, bankruptcy, reduced business investment) to achieve a disinflation that was partly happening on its own as supply chains normalized.

The Evidence:

- Canada’s peak inflation (8.1%) was lower than the UK (11.1%) and comparable to the US (9.1%).
- The Bank achieved its inflation target by mid-2024 without a technical recession.
- Real GDP per capita nonetheless fell for several quarters — a “per capita recession” even without the formal definition being triggered.
- Variable-rate mortgage holders saw monthly payments increase by an average of approximately \$700–\$900.
- Business insolvencies rose to near-record levels in 2024.

The Analytical Framework: Evaluating central bank decisions requires distinguishing what the Bank *knew* at the time from what we know now. Using today's data to critique 2021–2022 decisions is “Monday morning quarterbacking.” The correct standard is: given the information available in real time, were the Bank's decisions within the range of reasonable policy choices? Most economists conclude the answer is yes — though many would have preferred earlier action in 2021 and possibly a slightly lower terminal rate in 2023. This is a case where the difference between a good outcome and a great outcome was perhaps 50–75 basis points of peak rate — a reminder of how consequential seemingly small numerical differences in monetary policy can be.

Your Position: After reading this chapter, where do you stand? Use the Fisher equation, the AS-AD model, and the Phillips Curve to structure your argument.

Economics in Everyday Life: How Inflation Affects Your Grocery Bill — and What You Can Do About It

When Statistics Canada reports that food prices rose 9.7% in 2022, that average conceals enormous variation. Fats and oils: +23.3%. Pasta and other grain products: +19.6%. Fresh vegetables: +9.5%. Restaurant meals: +8.3%.

As a consumer, you can partially escape measured food inflation through *substitution* — exactly the behaviour that creates the CPI's substitution bias. Switching from branded cereal to store-brand, choosing chicken over beef, buying a larger format item or catching a sale — all of these reduce your *personal* inflation rate below the measured CPI for food.

This is not just a survival tip. It illustrates a deep economic point: the CPI measures the cost of a *fixed* basket. Real households are not fixed-basket consumers — they respond to price signals. Your ability to substitute gives you more real purchasing power than the headline inflation rate implies. The gap between your personal inflation rate and the CPI, averaged across all consumers, is precisely the *substitution bias* in CPI measurement discussed in Section 2.2. Estimated at 0.3–0.5 percentage points per year on average, this bias was likely larger in 2022 when relative price changes were unusually large — giving households more scope for substitution than normal.

Chapter Summary

- **Inflation** is a sustained increase in the general price level; **deflation** is a sustained decrease. Low positive inflation (2% in Canada) is preferred to zero because it provides a policy buffer, accommodates relative wage adjustments, and partially offsets CPI measurement biases.
- The **CPI** is a Laspeyres index measuring the cost of a fixed basket of goods and services. It contains upward biases (substitution, quality, new goods, outlet) of approximately 0.3–0.5 percentage points per year.
- **Core inflation measures** (CPI-trim, CPI-median, CPI-common) remove volatile components to reveal the underlying trend. All three rose above 5% in 2022, confirming that the inflation surge was broad-based, not merely energy-driven.
- The **Fisher equation** ($r \approx i - \pi$) distinguishes nominal from real interest rates. In 2021, the real overnight rate in Canada was approximately -3.2% — the most stimulative monetary stance in modern Canadian history.
- The **AD-AS model** explains two types of inflation: *demand-pull* (rightward AD shift, rising prices and output) and *cost-push* (leftward SRAS shift, stagflation). Canada's 2021–2023 surge involved both simultaneously.
- The **short-run Phillips Curve** ($\pi = \pi^e - \alpha(u - u_n)$) shows a negative relationship between inflation and unemployment. The **long-run Phillips Curve** is vertical at the natural rate u_n : monetary policy cannot permanently reduce unemployment below u_n .
- Canada's **inflation-targeting framework** (since 1991) targets 2% CPI inflation within a 1–3% control range. The Bank of Canada conducts policy primarily through the overnight rate.
- Monetary policy transmits through four channels: credit cost, asset prices, exchange rate, and expectations. Full effects take 18–24 months.
- The 2022–2024 **rate-hike cycle** raised the overnight rate 475 basis points to 5.00% in 16 months — the fastest tightening in modern Canadian history. Inflation returned to target by 2024 without a technical recession, but at a significant cost to mortgage holders.
- **Inflation is regressive**: low-income households experienced approximately 8.5% effective inflation in 2022 vs. 6.8% for the top quintile, because they spend disproportionately more on food, shelter, and energy.

Key Terms

CPI (Consumer Price Index)

A Laspeyres price index measuring the cost of a fixed basket of goods; Canada's primary inflation measure.

CPI-common

Bank of Canada core measure using factor analysis to extract economy-wide price movements.

CPI-median

The price change at the 50th percentile of the weighted component distribution.

CPI-trim

Core inflation measure that removes the top and bottom 20% of price-change outliers.

Core inflation

Inflation measures that exclude the most volatile CPI components to reveal the underlying trend.

Cost-push inflation

Inflation arising from a leftward shift of the SRAS curve (rising input costs, supply shocks).

Deflation

A sustained fall in the general price level.

Demand-pull inflation

Inflation arising from a rightward shift of the AD curve (excess demand).

Ex ante real rate

The real interest rate based on expected future inflation; $r^e = i - \pi^e$.

Ex post real rate

The real interest rate based on actual realised inflation; $r = i - \pi$.

Fisher equation

The relationship $r \approx i - \pi$ linking real rates, nominal rates, and inflation.

Inflation targeting

A monetary policy framework committing to keep inflation near a numerical target.

Laspeyres index

A price index using base-period quantity weights; the methodology underlying Canada's CPI.

Long-run Phillips Curve

The vertical relationship between inflation and unemployment at the natural rate u_n .

NAIRU

Non-Accelerating Inflation Rate of Unemployment; the unemployment rate consistent with stable inflation.

Nominal interest rate

The stated rate on a financial instrument; not adjusted for inflation.

Overnight rate

The Bank of Canada's policy rate; the rate at which major financial institutions lend to each other overnight.

Phillips Curve

The relationship between inflation (or wage inflation) and unemployment.

Real interest rate

The nominal interest rate adjusted for inflation; $r = i - \pi$.

Real wage

Nominal wage deflated by the price level; measures purchasing power of labour income.

Sacrifice ratio

Cumulative excess unemployment required per percentage-point reduction in inflation.

Short-run Phillips Curve

The negative relationship between inflation and unemployment at a given inflation expectation.

Stagflation

Simultaneous high inflation and high unemployment, typically arising from an adverse supply shock.

Substitution bias

CPI overstatement due to ignoring consumer substitution away from goods with rising relative prices.

Zero lower bound

The practical floor on nominal interest rates (approximately 0%); limits the downward scope of monetary stimulus.

Review Questions

1. **(Concept)** Define inflation and explain why a 2% inflation target is preferred to 0% by most central banks. In your answer, discuss the zero lower bound and the measurement bias arguments.
2. **(Measurement)** Canada's CPI basket assigns a 30% weight to Shelter and 20% to Transportation. In 2022, Shelter rose 6.9% and Transportation rose 10.6% year-over-year.
 - (a) What is the combined contribution of these two categories to headline inflation, assuming the remaining categories average 6.5%?
 - (b) If all other categories fell to zero growth, what would headline inflation be?
 - (c) What does this illustrate about the sensitivity of CPI to its largest components?
3. **(Fisher Equation)** In January 2021, the Bank of Canada overnight rate was 0.25% and CPI inflation was 1.0%. By December 2022, the overnight rate had risen to 4.25% and inflation was 6.3%.
 - (a) Calculate the real interest rate in January 2021 and December 2022.
 - (b) In which period was monetary policy more accommodative (stimulative)? Explain using the Fisher equation.
 - (c) How does your answer change the interpretation of the rate hike cycle as "aggressive"?
4. **(AD-AS)** Use the AD-AS model to explain Canada's 2022 inflation surge. Draw a diagram showing:
 - (a) The starting equilibrium at potential output Y_f .
 - (b) The rightward shift of AD resulting from COVID fiscal stimulus.
 - (c) The leftward shift of SRAS resulting from supply chain disruptions.
 - (d) The new equilibrium showing both higher prices and a different level of output.Label all curves and equilibria clearly.
5. **(Phillips Curve)** Using Equation 2.9 and estimates $\alpha = 0.5$ and $u_n = 5.8\%$:
 - (a) What does the model predict for inflation if the unemployment rate is 4.8% and expected inflation is 2%?

- (b) What does the model predict if unemployment is 7.3% and expected inflation is 3%?
- (c) If the Bank of Canada wants to reduce inflation from 5% to 2% and the sacrifice ratio is 2.0, how much cumulative excess unemployment (above u_n) would be required?
6. **(Policy)** Why does the Bank of Canada monitor core inflation measures (CPI-trim, CPI-median) rather than simply responding to headline CPI? Under what circumstances would a supply shock cause the Bank to *tolerate* higher headline inflation without raising rates?

Quantitative Problems

1. **CPI Calculation.** A household's monthly expenditure basket consists of four categories:

Category	Weight (%)	Price (Base Year)	Price (Current Year)
Food	20	\$600	\$672
Housing	35	\$1,200	\$1,272
Transport	25	\$400	\$460
Other	20	\$300	\$318

- (a) Calculate the Laspeyres CPI for the current year (base year = 100) using Equation 2.1. Show all steps.
- (b) Calculate the inflation rate π from base year to current year.
- (c) Which category contributed most to the measured inflation rate? Calculate each category's contribution to π .
- (d) Suppose the household switches entirely from Transport to a category priced at base-year levels. What is the household's *true* cost-of-living increase? How does it compare to the Laspeyres CPI answer? What bias does this illustrate?

2. **Fisher Equation Applications.**

- (a) A 5-year Government of Canada bond yields 4.8% nominal. If expected inflation over the next 5 years is 2.2%, what is the expected real yield? Is this above or below the historical average neutral real rate of approximately 1%?
 - (b) A student takes a variable-rate student loan at prime + 1.5%. In January 2020, the prime rate is 3.45% (overnight + 2.2%). In December 2022, the prime rate is 6.45%. The student's income is \$42,000 per year. CPI in 2020 was 137.0; in 2022 it was 151.9. Calculate the real interest rate on the student loan in each year and the real value of \$42,000 income in each year (in 2020 dollars).
 - (c) Canada runs a current account deficit when nominal interest rates are below trading partners' rates (capital flows out). In 2020, the Bank of Canada cut rates to 0.25% while the ECB kept rates at -0.50% and the Fed cut to 0-0.25%. Who had the relatively highest real rates given the respective inflation differentials? What does this imply for capital flows?
3. **Real Wages and Cost of Living.** Average weekly earnings in Canada were \$1,020 in January 2020 and \$1,147 in December 2022. CPI (all-items, 2002=100) was 137.9 in January 2020 and 156.0 in December 2022.
- (a) Calculate real average weekly earnings in each period (in January 2020 dollars) using Equation 2.5.
 - (b) Calculate the nominal and real growth rates of weekly earnings over the period.
 - (c) If low-income workers (bottom quintile) earned \$720/week in January 2020 and \$780/week in December 2022, but experienced an effective inflation rate of 13.8% over that period, what was the change in their real wages?
 - (d) What economic term describes a situation where nominal wages rise but real wages fall?
4. **Phillips Curve and Sacrifice Ratio.** Suppose the short-run Phillips Curve for Canada is: $\pi_t = \pi_t^e - 0.6(u_t - 5.8\%)$
- (a) If expected inflation is 2% and actual inflation is 4.5%, what does the Phillips Curve imply about the unemployment rate at that time?
 - (b) The Bank of Canada wishes to reduce inflation from 4.5% to 2% over three years. In each year, the Bank accepts unemployment 1.2 percentage points above the natural rate. What is the implied sacrifice ratio? Is this within the range of estimates in the literature (1.5-3.0)?

- (c) If inflation expectations become fully rational (people correctly anticipate the Bank's actions), what happens to the sacrifice ratio? What does this tell us about the value of central bank credibility?

5. **Distributional Analysis.** Use the following simplified data to analyse distributional inflation:

Category	CPI Weight	2022 Rise	Q1 Budget Share	Q5 Budget Share
Food	16%	9.7%	22%	10%
Shelter	30%	6.9%	38%	22%
Energy	8%	28.5%	12%	4%
Transport	12%	7.8%	8%	16%
Services	20%	5.5%	12%	30%
Other	14%	5.2%	8%	18%

- (a) Calculate the headline CPI inflation rate using official CPI weights and 2022 price changes.
- (b) Calculate the effective inflation rate for Q1 (lowest-income) households using their budget shares as weights.
- (c) Calculate the effective inflation rate for Q5 (highest-income) households.
- (d) By how many percentage points did Q1 experience higher inflation than Q5? What drives this difference?
- (e) Propose one policy instrument that could specifically compensate low-income households for this burden. What would be its fiscal cost if Canada has approximately 4 million households in Q1, each spending \$40,000 per year?
6. **Data Exercise.** Visit the Bank of Canada website (www.bankofcanada.ca) and find:
- (a) The current overnight rate target.
- (b) The most recent annual CPI inflation reading.
- (c) Using these two numbers, calculate the current real overnight rate. Is monetary policy currently stimulative, neutral, or restrictive (assume the neutral real rate is approximately 1%)?

- (d) Download the CPI-trim and CPI-median series from the Bank of Canada's core inflation page. Report whether core measures are above or below the 2% target. What does this imply about the appropriate monetary policy stance?

Essay Questions

1. **(600–800 words)** “The Bank of Canada’s aggressive rate-hike cycle in 2022–2023 was necessary and ultimately successful.” Using the AD-AS model, the Fisher equation, and the Phillips Curve, evaluate this claim. Your essay should discuss: (a) the causes of the 2022 inflation surge; (b) the mechanism through which rate hikes reduce inflation; (c) the costs of the tightening cycle; and (d) your overall assessment.
2. **(500–700 words)** The CPI reports a single number for inflation, but inflation is experienced differently across income groups, age cohorts, homeowners vs. renters, and variable-rate vs. fixed-rate mortgage holders. Using data from this chapter, write an essay arguing that “the CPI is a measure of average inflation, but Canada needs complementary distributional inflation measures to make good policy.” What additional measures would you recommend Statistics Canada publish?
3. **(400–600 words)** Japan maintained near-zero inflation (and frequently deflation) for much of the 1990s–2010s despite aggressive monetary stimulus. Canada experienced 8.1% peak inflation in 2022 despite similar initial monetary conditions (near-zero rates). Using the AS-AD framework, propose at least two hypotheses that explain why the same ultra-loose monetary policy produced such different inflation outcomes in the two countries. What does this imply about the limits of the quantity theory of money?

Discussion Questions

1. Fatima, from our opening vignette, is a variable-rate mortgage holder who experienced both high inflation (eroding her real income) and sharply higher mortgage payments (from the rate hikes). Is she a net winner or loser from the sequence of events? What if she had locked into a 5-year fixed rate at 2% in 2020 instead? What if she were a renter with no mortgage?
2. The Bank of Canada held rates at emergency levels (0.25%) throughout 2021, even as inflation rose above 4%, citing concerns about pandemic economic scarring. Was this

the right call? What information would you need to evaluate the Bank's decision in real time, rather than in hindsight?

3. "Inflation is a hidden tax on savings, a transfer from creditors to debtors, and a regressive burden on low-income households." Do you agree? Who are the winners and who are the losers from an unexpected inflation surge? Does your answer change if the inflation is anticipated vs. unexpected?
4. Canada's CPI Shelter component includes mortgage interest costs, which rose sharply as the Bank of Canada raised rates. Should mortgage interest costs be in the CPI? What are the arguments for and against the current Canadian methodology? How would switching to "owners' equivalent rent" (the US approach) have changed the reported inflation numbers in 2023?

CHAPTER 3

Canada's Housing Crisis: Markets, Affordability, and Policy

In February 2022, a detached house in East Vancouver sold for \$2.1 million. The winning offer was \$400,000 above the asking price, waived all conditions, and was accepted over 22 competing bids. The buyers — two engineers in their mid-thirties with combined household income of approximately \$220,000 — put down \$420,000 (20%) and took on a mortgage of \$1,680,000. At that month's prevailing 5-year fixed rate of 3.2%, their monthly payment was \$8,067.

They were the lucky ones. They could afford it.

Across the city, a hospital nurse earning \$72,000 per year and a teacher earning \$65,000 per year were renting a two-bedroom apartment for \$2,800 per month — 40% of their combined take-home pay. They had been saving for a down payment for six years, but house prices had risen faster than their savings every single year. The down payment they needed was now larger than the house price they had originally targeted.

And in Prince George — 800 kilometres to the north — a single father working at the mill was paying \$1,350 per month for a two-bedroom apartment in a town where the average household income was \$78,000. He was not in a “hot market.” But the housing shortage had spread.

These three households inhabit the same economy, regulated by the same Bank of Canada, subject to the same federal tax law, and served by the same CMHC mortgage insurance program. Yet their housing experiences are radically different — determined by the intersection of global capital flows, local zoning bylaws, demographic waves, and macroeconomic policy.

Understanding why Canadian housing became so unaffordable — and what could actually fix it — is one of the most important applied questions in contemporary Canadian economics.

Learning Objectives

- Apply the supply-and-demand framework to housing markets, explain why the short-run housing supply curve is nearly vertical, and predict the price effects of demand shifts.
- Derive the user cost of housing formula and use it to evaluate the rent-versus-buy decision under different interest rate and price appreciation assumptions.
- Calculate standard housing affordability measures (price-to-income ratio, shelter cost burden, CMHC affordability index) and apply the standard mortgage payment formula to scenarios with different interest rates.
- Explain the supply-side constraints on Canadian housing production — including zoning, land costs, construction labour, and permitting — and evaluate their relative importance.
- Analyse the demand-side forces driving Canadian house prices, including immigration, financialization, low interest rates, and demographic trends.
- Describe Canada's rental market dynamics, define the rental vacancy rate, and evaluate the economic arguments for and against rent control.
- Evaluate major federal and provincial housing policy instruments using supply-and-demand analysis, and identify which interventions address the root cause of unaffordability.

3.1 Housing as an Economic Good

Housing is an unusual commodity. It is simultaneously a **consumption good** (shelter is a basic human need), a **durable asset** (houses last 50–100 years), a **financial investment** (households hold most of their net worth in housing), and a **location-specific good** (a house in Vancouver cannot substitute for a house in Saskatoon).

These characteristics give housing markets properties that distinguish them from markets for ordinary goods:

- **Highly inelastic short-run supply.** It takes 18–36 months to plan, permit, and build new housing. The existing stock of homes changes slowly. A sudden increase in demand therefore causes prices to rise sharply in the short run, even if the long-run supply response would eventually moderate prices.
- **Spatially fixed.** Housing supply cannot respond to demand by shipping inventory

from low-demand to high-demand regions. Land in downtown Toronto is permanently scarce in a way that land in Kansas is not. This creates permanent scarcity rents in desirable locations that compound over time.

- **Heterogeneous.** No two homes are identical. Differences in location, size, vintage, condition, and neighbourhood make “the housing market” really an overlapping set of segmented submarkets that interact but behave differently.
- **Highly leveraged.** Most housing purchases are debt-financed. A household making a 10% down payment is borrowing 90 cents of every dollar of housing value. This leverage amplifies both gains and losses, makes housing demand extremely sensitive to interest rates, and connects housing markets tightly to the credit system.
- **Regulated at every level.** Zoning bylaws (municipal), building codes (provincial), mortgage insurance rules (federal), rent regulations (provincial), environmental assessments (federal/provincial), and development charges (municipal) all constrain the supply of, and demand for, housing simultaneously.

Definition: The Rental Market and the Ownership Market

The housing market has two interconnected segments.

The **ownership market** prices houses as assets. The equilibrium price reflects the expected present value of future housing services, including the option value of residing in a particular location.

The **rental market** prices housing as a flow of services per unit of time. Renters pay for the right to occupy a dwelling without acquiring the asset.

The two markets are linked: when owning becomes more expensive relative to renting (e.g. because mortgage rates rise), demand shifts from ownership to rental, pushing rental prices up. Conversely, when rents are very high, the yield on owning rental property rises, incentivising investors to buy.

In Canada’s current crisis, both markets have tightened simultaneously — a phenomenon explained by the coincidence of record immigration, low rental vacancy, high mortgage rates, and constrained new supply.

3.2 Supply and Demand in Housing Markets

3.2.1 The Demand for Housing

The demand for housing is driven by:

- **Population growth and household formation:** More people require more dwellings. Canada's record immigration flows (431,645 permanent residents in 2022; over 1 million people added to the population including non-permanent residents) have been the single largest driver of housing demand in the mid-2020s.
- **Income:** Higher household income increases both the ability and willingness to pay for housing. Housing is a *normal good* with an income elasticity of demand estimated between 0.5 and 1.0.
- **Interest rates:** The monthly cost of servicing a mortgage depends directly on the interest rate. Lower rates reduce the flow cost of owning a given quantity of housing, increasing demand. The 2020–2021 rate cuts to 0.25% dramatically reduced mortgage payments and turbocharged demand.
- **Expected capital gains:** If buyers expect house prices to rise, the expected capital gain reduces the user cost of owning and shifts demand rightward. Extrapolative expectations can generate self-fulfilling price bubbles.
- **Investor demand:** A significant share of housing demand comes from investors who do not occupy the units. In some markets, investor demand competes directly with owner-occupiers and pushes prices above what the fundamental rental income stream would justify.

3.2.2 The Supply of Housing

The supply of housing in any given market is determined by:

- **Existing stock:** The vast majority of housing supply in any year is the existing stock of homes. In Canada, approximately 15 million private dwellings are occupied. New construction in a typical year (200,000–260,000 units) adds only 1.3–1.7% to the stock.
- **New construction:** *Housing starts* measure the number of new residential units on which construction has begun. Their determinants include land prices, labour costs, material costs, developer financing costs, development charges, permitting timelines, and zoning regulations.
- **Conversion and densification:** Basement suites, laneway houses, and conversions of

single-family homes to multiplexes can increase supply without requiring greenfield land.

- **Demolition and obsolescence:** A small fraction of the existing stock is demolished or becomes uninhabitable each year. Net new supply = New construction – Demolitions.

3.2.3 Supply Elasticity and Its Consequences

The *housing supply elasticity* measures how responsive new construction is to price increases:

$$\varepsilon_s = \frac{\% \Delta Q_s^{\text{housing}}}{\% \Delta P_{\text{housing}}} \quad (3.1)$$

Studies for major Canadian cities consistently find supply elasticities well below 1.0 — typically between 0.2 and 0.6 for the short run. This means that a 10% increase in house prices produces only a 2–6% increase in housing supply. The reasons for this inelasticity are explored in Section 3.6.

Figure 3.1 illustrates the consequences of low supply elasticity for housing price dynamics.

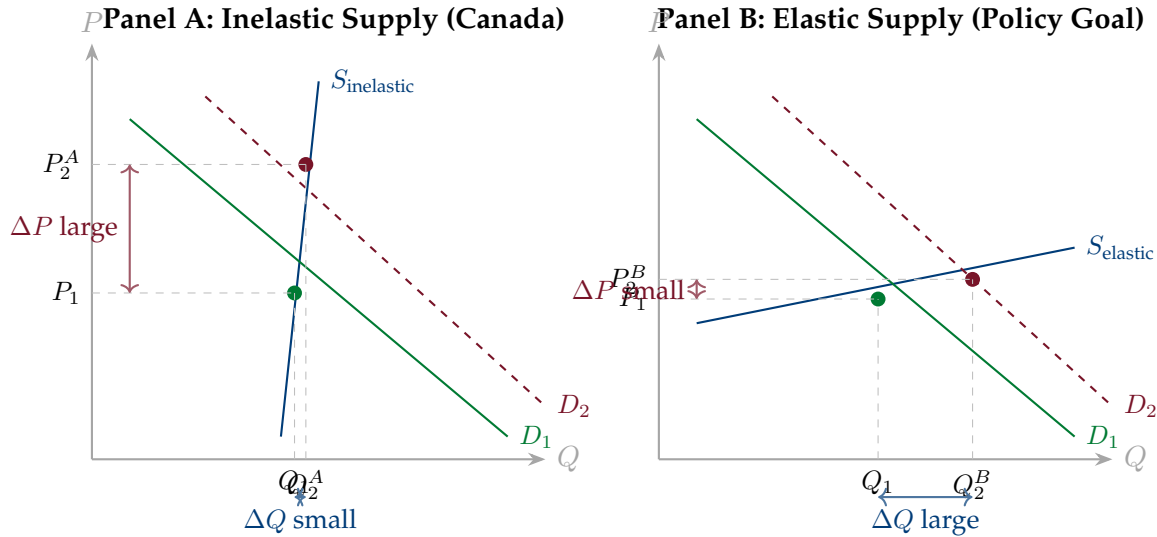


Figure 3.1. Housing Supply Elasticity and Price Dynamics. **Panel A** shows Canada's current reality: the supply curve is nearly vertical (inelastic) because zoning, land scarcity, and slow permitting prevent new housing from responding quickly to demand. When demand shifts from D_1 to D_2 , prices rise sharply while quantity barely changes. **Panel B** shows the policy goal: a more elastic supply curve absorbs the same demand shift with a much smaller price increase and a much larger quantity response. The fundamental insight is that housing affordability is primarily a supply problem, and the supply curve's slope is the critical policy parameter.

3.3 The User Cost of Housing: Buy or Rent?

The decision to buy or rent a home is analysed using the *user cost of housing* framework. The user cost is the annual economic cost of occupying a home you own, expressed as a fraction of its market value:

$$UC = (r + \delta + \tau - g) \times P \quad (3.2)$$

where:

- r = real mortgage interest rate
- δ = annual depreciation rate (approximately 1.5–2.5%)
- τ = property tax rate (approximately 0.5–1.5%)
- g = expected real capital gains rate (annual price appreciation above inflation)
- P = current price of the home

Renting is more economical when the annual rent is less than UC; buying is more economical when annual rent exceeds UC.

Worked Example 3.1 — User Cost Under Different Conditions

Consider a home priced at \$700,000 in Ottawa under three scenarios:

Parameter	2021 (Low Rate)	2023 (High Rate)	2026 (Normal)
Mortgage rate r (%)	2.0	5.8	4.5
Depreciation δ (%)	2.0	2.0	2.0
Property tax τ (%)	1.0	1.0	1.0
Expected real gain g (%)	4.0	0.0	1.5
User cost rate (%)	1.0	8.8	6.0
Annual user cost (\$)	\$7,000	\$61,600	\$42,000
Monthly equivalent	\$583	\$5,133	\$3,500

Interpretation: In 2021, the monthly user cost of owning a \$700,000 home was only \$583 — far below any comparable rent. This explains the demand surge of 2020–2022: owning was far cheaper than renting in user cost terms. By 2023, the monthly user cost had risen to \$5,133, making owning economically irrational relative to renting in many markets.

Think Like an Economist

Why did house prices continue rising in 2021 even though prices were already “unaffordably high” by price-to-income measures? The user cost framework answers this. When $g > r$ (expected capital gains exceed the interest rate), the user cost can approach zero or become negative. The more expensive the house, the more capital gain the owner earns — so higher prices make buying *more* attractive, not less. This creates a feedback loop that sustains price bubbles: rising prices increase expected future gains, reduce user cost, attract more buyers, push prices higher, and so on. The loop breaks when interest rates rise sharply (as in 2022), expected capital gains collapse, and user costs suddenly jump.

3.4 Measuring Housing Affordability

3.4.1 The Price-to-Income Ratio

The *price-to-income ratio (PTI)* is the most widely reported affordability benchmark:

$$PTI = \frac{\text{Median House Price}}{\text{Median Annual Household Income}} \quad (3.3)$$

A PTI of 3.0 is traditionally considered “affordable.” Above 5.0 is “seriously unaffordable.” Above 9.0 is “severely unaffordable.”

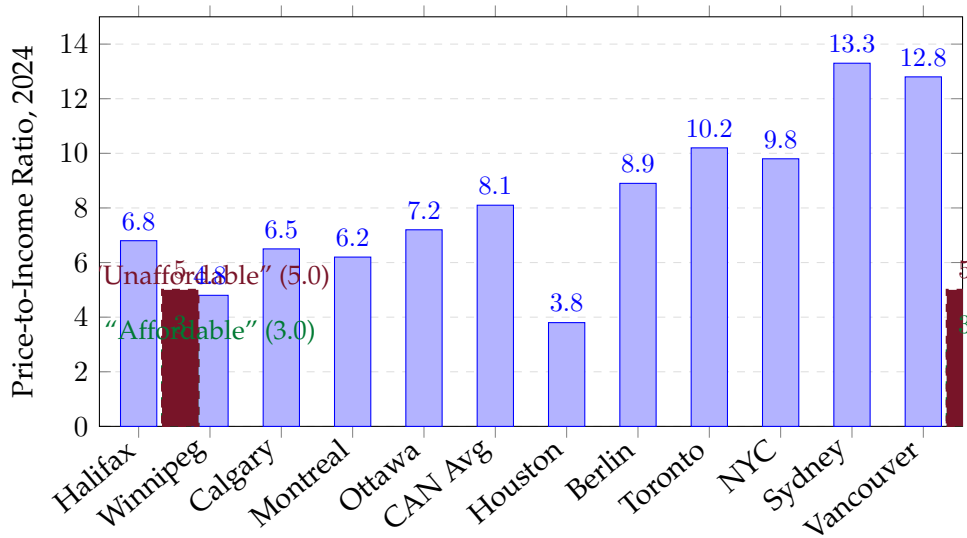


Figure 3.2. Price-to-Income Ratio, Selected Canadian and International Cities, 2024. Vancouver (12.8) and Toronto (10.2) rank among the least affordable cities globally. Only Winnipeg (4.8) of the major Canadian cities approaches the historic “affordable” norm of 3.0 (green dashed line). The international comparison is instructive: Houston (3.8) is a major North American city with permissive land-use rules; Berlin (8.9) has high prices but also a large social housing sector that insulates lower-income households from full market exposure. Canada’s affordability problem is real, severe, and primarily concentrated in its two largest metropolitan areas.

Source: Demographia International Housing Affordability Survey, 2024; CMHC Housing Market Outlook, 2024.

3.4.2 The Mortgage Payment Formula

A second affordability measure is the *shelter cost burden*: the fraction of income required to service housing costs. The conventional threshold is 30% of gross income.

The standard Canadian mortgage payment formula is:

$$M = P_0 \times \frac{r_m(1 + r_m)^n}{(1 + r_m)^n - 1} \quad (3.4)$$

where M is the monthly payment, P_0 is the principal (price minus down payment), r_m is the effective monthly rate derived from the annual rate under the Canadian semi-annual compounding convention ($r_m = (1 + r_{\text{annual}}/2)^{1/6} - 1$), and n is total monthly payments (amortization years $\times 12$).

Table 3.1. Monthly Mortgage Payment and Affordability, \$700,000 Home, 20% Down (\$140,000), \$560,000 Principal, 25-Year Amortization

Rate (%)	Monthly Payment	Annual Cost	Income at 30% Rule	Total Interest
2.0	\$2,370	\$28,440	\$94,800	\$150,800
3.0	\$2,646	\$31,752	\$105,840	\$233,600
4.0	\$2,936	\$35,232	\$117,440	\$320,800
5.0	\$3,263	\$39,156	\$130,520	\$418,900
5.8	\$3,531	\$42,372	\$141,240	\$499,300
7.0	\$3,921	\$47,052	\$156,840	\$616,600

Source: Author calculations using Equation 3.4 under Canadian semi-annual compounding. “Income at 30% rule” is the annual gross income required for the mortgage payment to represent exactly 30% of income. Total interest assumes full 25-year amortization.

At 5.8% — the approximate 5-year fixed rate prevailing through 2023 — a household buying a \$700,000 home needed gross income of at least \$141,240 per year to meet the 30% threshold. In Vancouver, where the benchmark price exceeded \$1.1 million through 2024, the required income at 5.8% exceeded \$225,000.

3.4.3 The CMHC Affordability Index

CMHC publishes a *Housing Affordability Index* measuring mortgage payment, property taxes, and utilities as a percentage of median household income for a median-priced home.

Table 3.2. CMHC Housing Affordability Index, Selected Cities, 2015–2024

City	2015	2018	2020	2022	2023	2024
Vancouver	80.2	87.6	61.4	93.8	90.2	92.1
Toronto	56.8	74.2	67.3	89.4	82.7	75.3
Ottawa	38.2	42.1	44.5	62.3	58.7	48.2
Calgary	35.6	32.1	31.8	42.5	45.1	38.4
Montréal	34.0	38.4	41.2	55.2	49.8	42.3
Halifax	26.1	28.3	30.2	49.3	46.5	44.2
Winnipeg	25.8	26.5	27.3	33.2	31.5	29.4
National avg	40.2	50.1	46.8	65.3	58.4	53.2

Source: Canada Mortgage and Housing Corporation, *Housing Market Outlook*, 2024. Index = (mortgage payment + taxes + utilities) as % of median household income. Values above 30% indicate unaffordable housing.

Vancouver's 2024 index of 92.1% means the median Vancouver household buying the median-priced home would spend 92% of gross income on housing costs — mathematically impossible for most families. The national average peaked at 65.3% in 2022 and has partially improved since, but remains far above the pre-crisis levels of the early 2010s.

3.5 The Historical Price Record

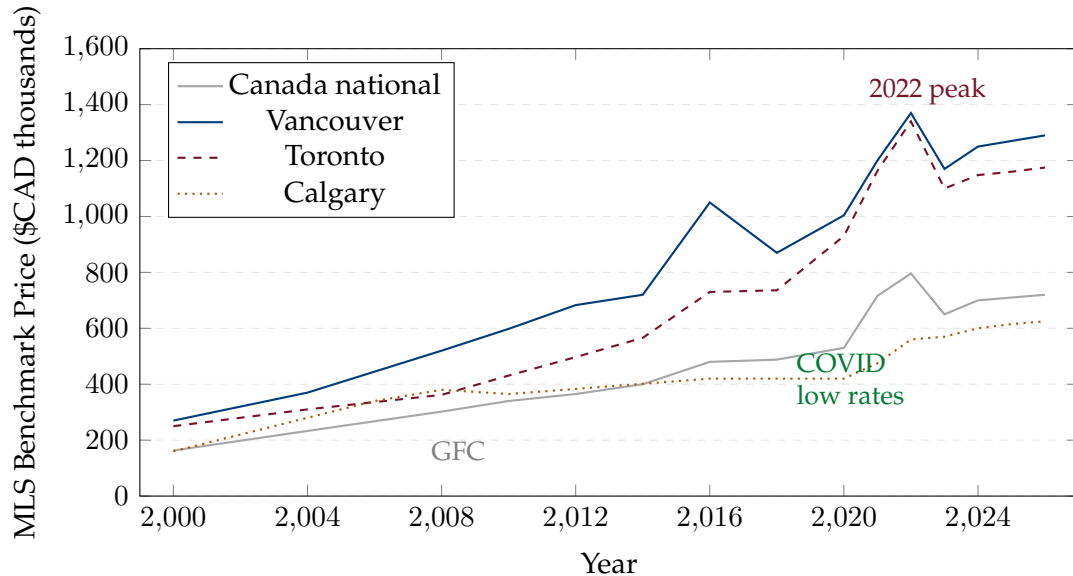


Figure 3.3. MLS Composite Benchmark House Prices, Canada and Selected Cities, 2000–2026. Canadian house prices have risen more than four-fold in nominal terms since 2000; Vancouver and Toronto five- to six-fold. Key episodes: the 2008 Global Financial Crisis caused only a brief pause; the 2016 BC Foreign Buyer Tax caused a temporary Vancouver correction; COVID-era ultra-low rates (2020–2021) triggered the most explosive price surge in history; and the 2022–2023 rate hikes caused the first significant correction, though prices remain far above pre-pandemic levels. Calgary’s flatter profile reflects oil price cycle sensitivity.

Source: Canadian Real Estate Association (CREA), MLS Home Price Index. 2025–2026 estimated.

3.6 Supply-Side Constraints on Canadian Housing

The central analytical question in housing economics is whether price increases are primarily driven by demand or supply. The academic evidence increasingly points to **supply constraints as the primary long-run driver**. The same demand pressures exist in many countries without producing the same price outcomes — what differs is how quickly supply responds.

3.6.1 Zoning and Land Use Regulation

The most important supply-side constraint is **zoning** — the municipal regulation of how land may be used. In most Canadian cities, the majority of residential land is

zoned exclusively for single-family detached housing. This “Euclidean” zoning model prevents the construction of apartments, townhouses, or duplexes on most residential land regardless of market demand.

The economic consequences are profound:

- **Land value inflation:** When zoning prevents densification, the value of a single-family lot captures the entire surplus from excluding all alternative, potentially more productive uses.
- **Reduced supply elasticity:** Zoning directly reduces ε_s (Equation 3.1). Developers who want to build apartments cannot do so without lengthy, costly, and uncertain rezoning processes.
- **Parking minimums and setback requirements:** Many municipalities require 1–2 parking spaces per unit, minimum lot sizes, setbacks, and maximum lot coverage. Each constraint reduces the number of units buildable per parcel.

Canadian Example: British Columbia's 2023 Zoning Reform

In December 2023, BC passed legislation requiring all municipalities in Metro Vancouver and most municipalities with populations over 5,000 to permit “small scale multi-unit housing” — duplexes, triplexes, fourplexes, and sixplexes — on lots previously zoned exclusively for single-family homes. This *upzoning* legislation was the most significant provincial override of municipal zoning in Canadian history. Economists estimate the reform could permit hundreds of thousands of additional units over the next decade if complemented by appropriate infrastructure investment and streamlined permitting. Early evidence from 2025 suggests modest but positive effects on building permit applications in affected areas.

3.6.2 Development Costs and Construction Capacity

Even where zoning permits higher density, additional cost factors constrain supply:

- **Development charges:** In Toronto, development charges on a new condo unit reached approximately \$150,000–\$200,000 in 2023 — representing 15–25% of the unit's total cost, ultimately borne by buyers through higher prices.
- **Construction labour shortages:** Canada's construction sector faces significant skilled labour shortages — carpenters, electricians, plumbers, ironworkers. Construction wages have risen faster than overall wages since 2020.

- **Permitting timelines:** The time from a development application to a building permit in major Canadian cities averages 12–36 months. Each month of delay costs developers carrying charges on their land.
- **High-rise construction costs:** Tall concrete and steel construction costs \$350–\$600 per square foot in Canada, compared to \$150–\$250 for wood-frame low-rise. The economics only work at very high price levels.

3.6.3 The Supply-Demand Gap: Housing Starts vs. Population Growth

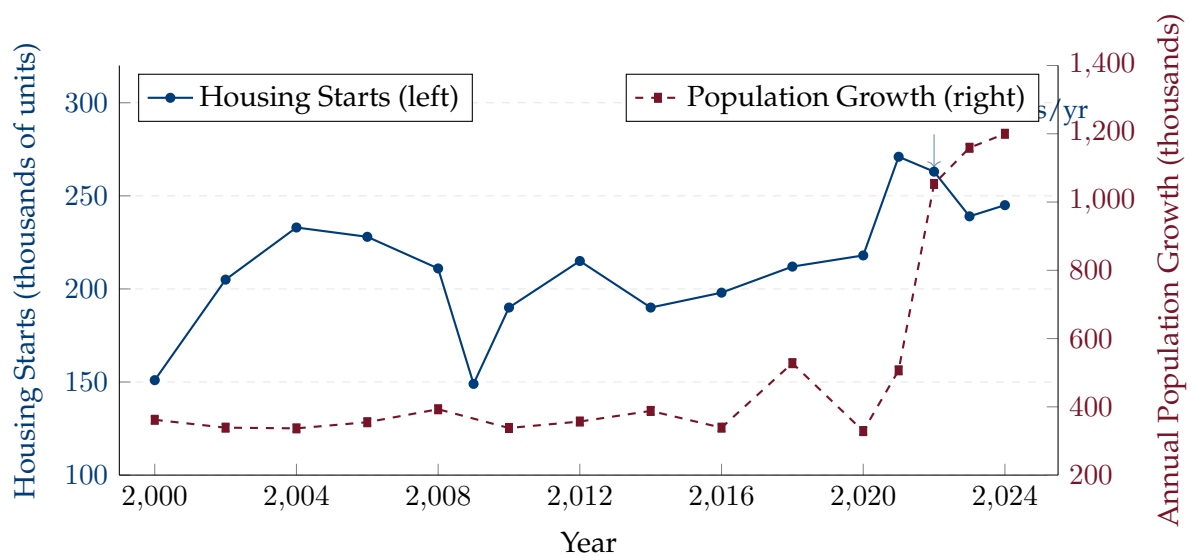


Figure 3.4. Housing Starts vs. Annual Population Growth, Canada, 2000–2024. Until approximately 2017, housing starts and population growth moved in rough proportion. From 2022 onwards, Canada’s record immigration levels produced population growth of over 1 million persons per year — requiring approximately 400,000–480,000 new housing units annually (at 2.5 persons per household) — while actual construction remained at 240,000–265,000 starts. This structural supply shortfall is the primary driver of the post-2021 rental vacancy crisis and rent inflation.

Source: Statistics Canada, Tables 34-10-0035-01 (housing starts) and 17-10-0008-01 (population growth estimates).

3.7 Demand-Side Forces: Immigration, Finance, and Demographics

While supply constraints are the primary structural explanation for high prices, several demand-side forces have amplified and accelerated the affordability crisis.

3.7.1 Immigration and Population Growth

Canada's record immigration of 431,645 permanent residents in 2022 (and approximately 1 million total population additions including non-permanent residents) represents the single largest year-over-year demand shock to housing in Canadian history. The housing consequences are significant because:

1. Newcomers must immediately find accommodation.
2. Newcomers concentrate geographically: 57% of 2022 permanent residents settled in Ontario or British Columbia — precisely the markets with lowest vacancy rates.
3. Non-permanent residents (international students, temporary foreign workers) are disproportionately renters, adding pressure to already strained rental markets.

Immigration does not make immigration bad policy — Chapter 6 presents the strong evidence for its economic benefits. But it does require that housing supply policy be coordinated with immigration policy. Canada set record immigration targets without simultaneously implementing the supply-side reforms needed to house the newcomers. This policy coordination failure contributed significantly to the 2022–2024 rental crisis.

3.7.2 The Interest Rate Channel

As established in Chapter 2, the Bank of Canada's decision to hold the overnight rate at 0.25% through 2020–2021 created deeply negative real interest rates. Through the user cost mechanism (Equation 3.2), this dramatically reduced the annual cost of owning housing:

- At $r = 2\%$ and $g = 5\%$ expected capital gains, the user cost rate was approximately -0.5% to $+1.0\%$.
- This meant the annual cost of owning a \$1,000,000 home was \$0–\$10,000, while comparable rental costs ran \$36,000–\$48,000 per year in Vancouver or Toronto.
- The rent-vs-buy calculation was decisively in favour of buying, generating enormous ownership demand.

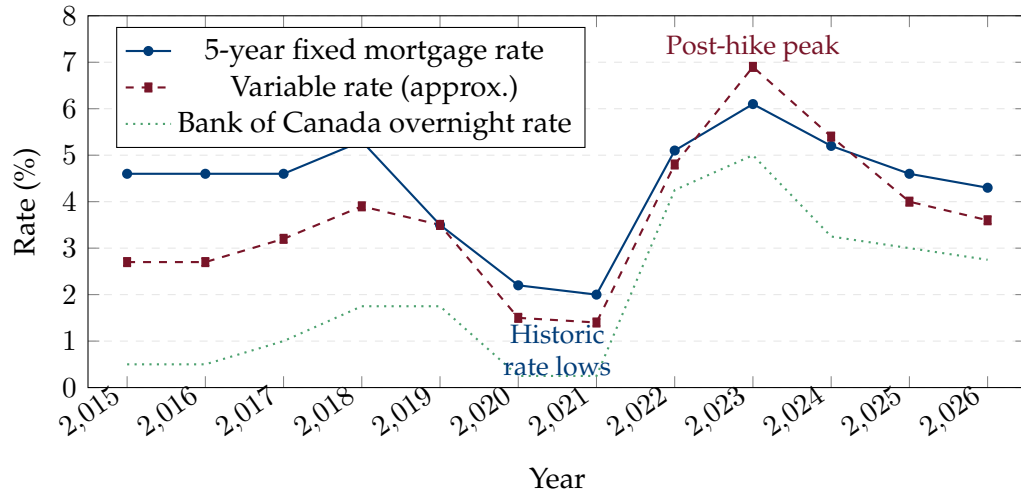


Figure 3.5. Mortgage Rates vs. Bank of Canada Overnight Rate, 2015–2026. The 5-year fixed mortgage rate fell to a historic low of approximately 2.0% in 2021, driving a massive surge in housing demand. The rate-hike cycle drove the 5-year fixed rate above 6% by mid-2023 — more than tripling the cost of new mortgage borrowing. Variable-rate holders saw their payments rise almost instantly, tracking the overnight rate with minimal lag.

Source: Bank of Canada; Statistics Canada. 2025–2026 estimated.

3.7.3 Financialization of Housing

A third demand-side factor is the *financialization of housing* — the increasing treatment of residential real estate as a financial asset class:

- **Investor purchasing:** CMHC estimates that investors accounted for 20–30% of condominium purchases in Toronto and Vancouver in recent years, competing with first-time buyers.
- **Short-term rentals:** Airbnb and similar platforms converted approximately 31,000 long-term rental units into tourist accommodation in Canada’s 10 largest cities.
- **Vacancy and speculation:** Some buyers hold units vacant as stores of value. Several Canadian cities have introduced vacancy taxes (Vancouver Empty Homes Tax, BC Speculation and Vacancy Tax) that address the problem directly and generate data to measure its scale.

3.8 The Rental Crisis

As buying became impossible for an increasing share of Canadians, demand for rental housing surged — but rental supply grew slowly, pushing vacancy rates to historic lows

and rents to historic highs.

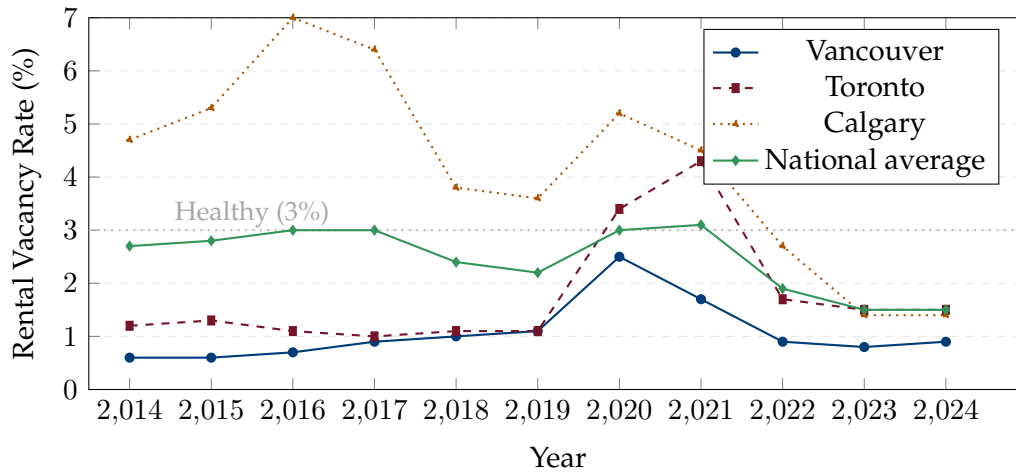


Figure 3.6. Rental Market Vacancy Rates, Selected Canadian Cities, 2014–2024. The “healthy” vacancy rate of 3% provides a benchmark for a functioning rental market. Vancouver has been below 3% for all but one year since at least 2014. The COVID-19 pandemic temporarily elevated Toronto’s vacancy to 4.3% in 2021, but this relief was transitory: by 2022, all major markets had returned to near-zero vacancy. Calgary’s trajectory is distinctive — the oil price collapse of 2015–2016 created a temporary glut, but the city subsequently tightened as energy employment recovered and immigration accelerated post-2021.

Source: Canada Mortgage and Housing Corporation, Rental Market Report, 2024.

When vacancy rates fall below 2%, the rental market becomes a landlords’ market. Average asking rents for 2-bedroom apartments reached \$3,200/month in Vancouver and \$2,750/month in Toronto in late 2024. For a household earning the provincial median income, these rents consume 40–55% of after-tax income.

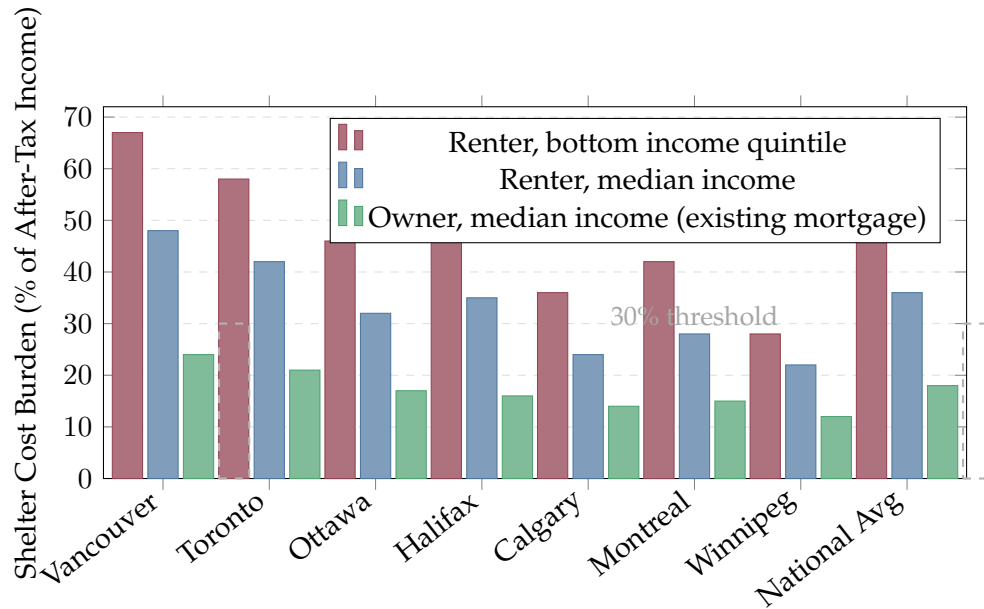


Figure 3.7. Shelter Cost Burden by Tenure and Income Group, Major Canadian Cities, 2024. Shelter cost burden is annual shelter costs as a percentage of annual after-tax income. The 30% threshold (grey dashed line) is the conventional affordability standard. Low-income renters (bottom quintile) spend over two-thirds of after-tax income on shelter in Vancouver. Even median-income renters exceed the 30% threshold in Vancouver and Toronto. By contrast, median-income homeowners with existing mortgages — often locked in at pre-2022 rates — face a far smaller shelter cost share, illustrating the growing economic divide between housing “haves” (established owners) and “have-nots” (renters and prospective first-time buyers).

Source: Statistics Canada, Survey of Household Spending, 2023–2024; CMHC Rental Market Report, 2024. Author calculations.

3.9 Case Studies

Case Study: Vancouver: From Affordable to Global Outlier, 1980–2026

Background: In 1980, the average house price in Metro Vancouver was approximately \$120,000. Average household income was approximately \$28,000. The price-to-income ratio was 4.3. By 2022, the average price had reached \$1.37 million while average household income was approximately \$107,000. The PTI had reached 12.8 — among the highest in the world.

What Changed?

- **Geographic constraints:** Metro Vancouver is squeezed between mountains, the US border, the ocean, and the Agricultural Land Reserve. The ALR, established in 1973 to protect farmland, constrains the urban growth boundary and reduces the supply of developable land.
- **Low supply elasticity:** Single-family zoning covers approximately 70% of Metro Vancouver's residential land. Height limits restrict building up; the ALR restricts building out.
- **Capital flows and international demand:** Beginning in the late 1980s, Metro Vancouver became a destination for substantial capital inflows from Asia — Hong Kong in the 1990s, mainland China in the 2000s–2010s. The correlation between capital account openness and Vancouver price acceleration is visible in Figure 3.3.
- **Interest rate cycles:** Vancouver prices are highly sensitive to real interest rates via the user cost mechanism, amplifying each low-rate period.

Policy Responses: BC and Vancouver have deployed virtually every policy tool: 15–20% Foreign Buyer Tax (2016), BC Speculation and Vacancy Tax (2018), mortgage stress test (federal, 2018), transit-oriented development zoning (2023), Small Scale Multi-Unit Housing legislation (2023), and short-term rental restrictions (2024).

Outcome: Despite these interventions, Vancouver remains among the three least affordable cities in the world. The 2022–2023 rate-hike correction reduced prices approximately 14%, but prices remained 25–30% above 2019 levels. The fundamental supply constraint has not been resolved. The lesson: demand-side interventions alone cannot solve a problem rooted in supply inelasticity and decades of under-building.

Case Study: Canada vs. Germany: What Different Housing Policies Produce

Background: Germany provides one of the most instructive international comparisons. Both countries are wealthy, democratic, urbanised societies with significant immigrant populations. Yet their housing outcomes differ dramatically.

German Housing Outcomes (2024):

- Berlin price-to-income ratio: approximately 8.9

- National homeownership rate: approximately 50% (the lowest in the EU)
- Long-term rental tenure: approximately 60% of households rent, many in the same unit for 10–20+ years
- Purpose-built rental stock: approximately 55% of total dwellings

What Germany Does Differently:

1. **Tenant protection:** Strong legal protection against eviction allows households to plan long-term without fear of displacement, reducing the “premium” on homeownership as the only path to residential stability.
2. **Large social housing sector:** Germany’s social housing stock is 5–6 times larger (as a share of dwellings) than Canada’s.
3. **Cooperative housing:** Cooperatives own approximately 7% of German housing stock; members pay below-market cost rents and have tenure security without speculative price exposure.
4. **Transfer taxes on resale:** Germany charges 3.5–6.5% land transfer taxes on property sales, reducing speculative trading.

Implication for Canada: Canada’s high homeownership aspiration and underdeveloped rental sector are not inevitable features of a wealthy economy — they are the product of specific policy choices. The German comparison suggests that sustained, multi-decade commitments to building non-market housing and protecting tenant tenure can produce meaningfully better affordability outcomes.

Case Study: Rent Control: The Economics of a Popular Policy

Background: Several Canadian provinces impose rent control — limits on rent increases for existing tenants. Ontario’s rent increase guideline caps annual increases at the Ontario CPI rate for most units. BC has similar provisions.

The Economic Case Against Rent Control: A binding rent ceiling $\bar{R}$ set below market rent R^* creates:

- **Excess demand:** At $\bar{R} < R^*$, quantity demanded exceeds quantity supplied. Non-price allocation mechanisms (personal connections, waiting lists) replace the price mechanism.

- **Reduced supply:** If developers anticipate future capped rents, expected returns on purpose-built rental projects fall, reducing new investment. Ontario explicitly exempts units built after November 2018 from rent control to address this.
- **Misallocation:** Tenants have strong incentives to remain in controlled units even when their circumstances change, reducing labour market mobility.
- **Reduced maintenance:** Landlords facing controlled rents may reduce maintenance spending to cut costs.

The Economic Case For Rent Control: When vacancy rates are near zero, landlords have market power and can extract rents above competitive levels. Rent control can correct a market power problem, provide tenure security, and prevent displacement of established low-income communities.

The Evidence: A 2019 Stanford study found that San Francisco's rent control reduced rental housing supply by 15% as landlords converted units to condos or redeveloped to escape controls. Ontario's pattern of low purpose-built rental construction throughout the 1990s–2000s (when rent control was broadly applied), followed by a surge in purpose-built starts after the 2018 exemption, is consistent with the supply-reduction effect.

The Policy Synthesis: Rent control protects *existing* tenants at the expense of *future* tenants and new supply. Where tenant protection and tenure security are the goals, alternatives such as vacancy controls, tenant right-of-first-refusal, and anti-eviction legislation may achieve those goals with less supply distortion. Where affordability is the goal, housing allowances and social housing expand supply rather than constrain it.

3.10 Social Housing: Canada in International Context

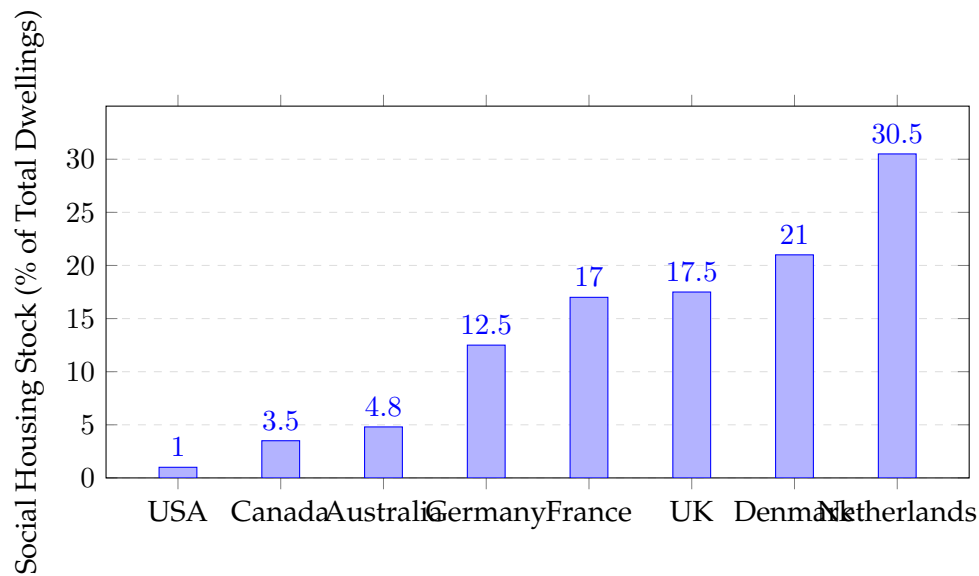


Figure 3.8. Social Housing as a Share of Total Dwellings, Selected Countries. Canada’s social housing stock (approximately 3.5%) is among the smallest of comparable wealthy nations. The Netherlands (30.5%) and Denmark (21%) maintain large non-market housing sectors that insulate a significant share of their populations from private market volatility. The OECD recommends that countries targeting housing affordability increase non-market supply to at least 10–15% of total dwellings. Achieving this in Canada would require building approximately 750,000–1,000,000 additional social and affordable units — a multi-decade project requiring sustained political commitment and federal investment.

Source: OECD Affordable Housing Database, 2023; CMHC National Housing Strategy, 2024.

3.11 Federal Housing Policy: What Has Been Tried

Table 3.3. Major Federal Housing Policy Measures, 2015–2026

Policy		Year	Type	Mechanism	Assessment
Mortgage stress test		2018	Demand (regulatory)	Borrowers qualify at contract rate + 2%, reducing borrowing capacity	Effective at limiting leverage; worsens access for marginal buyers
First Home Savings Account		2023	Demand (tax)	\$40K tax-free savings for first-time buyers; contributions tax-deductible	Modest benefit; may capitalise into higher prices
FTHBI (Shared Equity)		2019	Demand (subsidy)	CMHC takes 5–10% equity stake; disbanded 2024	Limited uptake; demand subsidy in supply-constrained market
National Housing Co-Investment Fund		2018	Supply (subsidy)	Low-interest loans for affordable rental construction	Positive but underfunded; units produced below target
GST Removal on Purpose-Built Rentals		2023	Supply (tax)	Eliminates 5% GST on new purpose-built rental construction	Positive; reduces effective construction cost by 3–5%
Housing Accelerator Fund		2023	Supply (incentive)	\$4B fund to municipalities contingent on zoning reform	Promising; targets root constraint; early results positive
Foreign Buyer Ban		2023	Demand (regulatory)	Prohibits non-residents from buying residential property	Marginal effect; foreign buyers were small share of transactions

Source: Department of Finance Canada; CMHC; Parliamentary Budget Officer, *Housing Affordability Assessment*, 2024.

The dominant pattern in Table 3.3 is instructive: most federal housing policies through 2022 were **demand-side** measures — tax credits, subsidies, and regulatory restrictions. With a near-vertical supply curve (Panel A of Figure 3.1), demand-side interventions cannot improve affordability. A demand subsidy to a fixed-supply market raises prices; a demand restriction may lower prices briefly but also reduces ownership opportunity for the households the policy was meant to help.

The shift from 2022 onwards toward supply-side measures — the Housing Accelerator Fund, the GST removal on purpose-built rentals, federal pressure on municipalities to upzone — reflects a belated recognition of this basic supply-and-demand reality.

Policy Debate: Is Foreign Ownership Driving Canadian Housing Unaffordability?

The Issue: Few housing policy questions generate more political heat and less empirical consensus than the role of foreign ownership. Advocates argue that offshore capital has driven Vancouver and Toronto prices beyond what domestic incomes could support. Sceptics argue foreign ownership is a convenient scapegoat for domestic policy failures.

Evidence Supporting a Significant Foreign Effect:

- CMHC's non-resident ownership data showed that non-residents owned approximately 2.4% of residential properties in Ontario and 4.6% in British Columbia, with concentrations in condominium markets.
- The 2016 BC Foreign Buyer Tax caused an immediate and dramatic (though temporary) cooling of Vancouver prices, particularly in the detached segment most favoured by offshore buyers.

Evidence Against a Large Foreign Effect:

- Aggregate non-resident ownership of 2–5% of total housing stock is insufficient to explain prices 3–5 times higher than domestic income fundamentals alone would support. If foreign ownership were the primary driver, the 2023 foreign buyer ban should have made housing affordable — it did not.
- Canadian housing has appreciated across *all* markets, including cities (Halifax, Saskatoon, Prince George) with essentially zero foreign ownership activity. A supply-constrained explanation applies everywhere; a foreign-buyer explanation applies only to a few markets.

- Many studies find that domestic investors and domestic demographic pressure are quantitatively more important drivers.

Policy Implication: The evidence is clear: Canada cannot tax its way to housing affordability. Foreign buyer restrictions are not irrelevant, but they address a marginal contributor to a problem driven primarily by supply constraints and domestic demand. Supply must grow.

Economics in Everyday Life: The Mathematics of Your First Home

Suppose you graduate from UNBC at age 22 earning \$58,000 per year in Prince George, with student debt of \$28,000. The current benchmark price for a two-bedroom home in Prince George is approximately \$380,000.

Down payment: A 5% minimum down payment requires \$19,000, plus CMHC insurance of approximately \$7,600 added to your loan. A 20% down payment (\$76,000) eliminates the insurance.

Saving timeline: After tax, rent (\$1,500/month), loan payments (\$400/month), and living costs (\$1,800/month), you might save \$600–\$1,000/month. At \$800/month, you accumulate \$76,000 in about 8 years — but if prices appreciate at 3% annually while you save, the 20% down payment target grows from \$76,000 to \$96,200. You are on a treadmill.

At purchase (age 30): A 25-year mortgage on a \$304,000 principal at 4.5% costs approximately \$1,655/month — 28.6% of your gross monthly income of \$5,800. This is within the 30% guideline, but tight.

Key insight from the user cost formula: Whether buying is rational depends on the gap between r (your mortgage rate) and g (expected future appreciation). If prices are flat or falling, the user cost rises sharply. If rates fall and prices appreciate, the user cost makes owning attractive. You cannot control either variable — they are set by the Bank of Canada and the housing market. What you can control is your savings rate and your First Home Savings Account contributions. The mathematics of homeownership in modern Canada is more uncertain, more leveraged, and more dependent on external forces than at any previous point in the country's history.

Chapter Summary

- Housing is simultaneously a consumption good, a durable asset, a financial investment, and a location-specific good. These characteristics make housing markets unusually sensitive to interest rates, supply constraints, and demographic shifts.
- The short-run housing supply curve is nearly vertical in Canadian cities (supply elasticity $\varepsilon_s \approx 0.2\text{--}0.6$) due to zoning restrictions, slow permitting, development charges, and construction labour shortages. When supply is inelastic, demand shocks produce large price increases with minimal quantity response.
- The **user cost formula** ($UC = (r + \delta + \tau - g) \times P$) explains the rent-vs-buy decision. In 2021, deeply negative real rates and high expected capital gains drove user costs near zero, generating enormous ownership demand. By 2023, user costs had risen six-fold, making renting more economical in many markets.
- The **mortgage payment formula** (Equation 3.4) shows that at 5.8% for a \$700,000 home with 20% down, the monthly payment is \$3,531, requiring gross income of at least \$141,240 to meet the 30% affordability threshold.
- Vancouver (PTI 12.8) and Toronto (PTI 10.2) rank among the least affordable cities globally. The CMHC affordability index for Vancouver reached 92.1% in 2024.
- House prices have risen more than four-fold nationally since 2000. The 2020–2022 COVID-era rate cuts triggered the most explosive price surge in Canadian history. The 2022–2023 rate hike cycle produced the first significant correction but left prices far above pre-pandemic levels.
- The supply-demand gap is structural: Canada builds 240,000–265,000 units per year but needs 400,000–480,000 at current immigration levels. This shortfall drives both ownership price appreciation and the rental vacancy crisis.
- The rental vacancy rate has fallen below 2% in most major Canadian cities, producing double-digit rent inflation and acute affordability problems for low-income renters.
- Most federal housing policies through 2022 were demand-side. Economic analysis demonstrates that demand subsidies in supply-constrained markets raise prices rather than improving affordability. Post-2022 policy has shifted toward supply-side instruments.
- International comparisons show that large non-market housing sectors, strong tenant

protections, and permissive zoning produce significantly better affordability outcomes in comparable democracies.

Key Terms

Development charges

Fees levied by municipalities on new construction to fund infrastructure costs.

Financialization The increasing treatment of residential real estate as a financial asset rather than primarily as shelter.

Housing starts New residential units on which construction has begun in a given period.

Housing supply elasticity

$\varepsilon_s = \% \Delta Q_s / \% \Delta P$; how responsive new construction is to house price changes.

Normal good A good for which demand rises with income; housing has income elasticity of approximately 0.5–1.0.

Mortgage stress test

Federal requirement to qualify at contract rate + 2% or the Bank of Canada benchmark rate, whichever is higher.

Price-to-income ratio (PTI)

Median house price divided by median annual household income; the primary affordability benchmark.

Purpose-built rental

Buildings constructed specifically for the long-term rental market.

Rent control Government regulations limiting annual rent increases.

Rental vacancy rate

Percentage of purpose-built rental units that are unoccupied and available.

Shelter cost burden

Shelter costs as a share of household income; above 30% is conventionally “unaffordable.”

Social housing

Housing owned by government or non-profits and rented at below-market rates to income-qualifying households.

Upzoning

A land-use regulation change permitting higher- density development than was previously allowed.

User cost of housing

Annual economic cost of owning a home: $UC = (r + \delta + \tau - g) \times P$.

Vacancy tax

A tax on vacant residential properties, incentivising owners to rent units out.

Zoning

Municipal regulation specifying permitted land uses and development standards.

Review Questions

- (Concept)** Explain why the short-run housing supply curve is nearly vertical in major Canadian cities. What are the three most important constraints on rapid housing supply expansion? Which is most amenable to policy reform?
- (Application)** Using the user cost formula (Equation 3.2), calculate the annual user cost of owning a \$900,000 home in each scenario. In which is buying most attractive relative to paying \$2,800/month rent?
 - $r = 2\%, \delta = 2\%, \tau = 0.8\%, g = 5\%$
 - $r = 5.5\%, \delta = 2\%, \tau = 0.8\%, g = 1\%$
 - $r = 4.0\%, \delta = 2\%, \tau = 0.8\%, g = 3\%$
- (Measurement)** Canada’s average house price rose from approximately \$530,000 in 2020 to \$796,000 in 2022.
 - What was the nominal percentage price increase?
 - If median household income rose from \$84,000 to \$90,000 over the same period, what happened to the PTI?

- (c) What does the PTI change tell us that the nominal price increase alone does not?
4. **(Policy)** A policymaker proposes a \$30,000 grant to all first-time homebuyers. Using Figure 3.1, explain what happens to house prices in: (a) a market with highly inelastic supply; (b) a market with elastic supply. Which scenario reflects most Canadian urban markets, and what does this imply about the policy's effectiveness?
 5. **(Rent control)** Using supply-and-demand analysis, explain the short-run and long-run effects of a binding rent ceiling set below the market rent. What is the key difference between the short-run and long-run outcomes, and why does this matter for policy evaluation?
 6. **(International comparison)** Canada's social housing stock is approximately 3.5% of total dwellings; the Netherlands is 30.5%. Explain the mechanism by which a larger social housing sector would affect equilibrium rents in the private market. What are two obstacles to Canada rapidly expanding its social housing stock?

Quantitative Problems

1. **Mortgage Payment Calculations.** Use Equation 3.4 with Canadian semi-annual compounding ($r_m = (1 + r_{\text{annual}}/2)^{1/6} - 1$) and 25-year amortization.
 - (a) A \$600,000 home, 10% down, at 4.0% annual rate. Monthly payment?
 - (b) Same home with 20% down at 4.0%. Monthly payment?
 - (c) Same home with 20% down at 6.5%. Monthly payment?
 - (d) For each scenario, what gross annual income is required for the payment to equal exactly 30% of monthly gross income?
2. **User Cost Analysis.** A household in Kelowna, BC is choosing between renting and buying. The home costs \$650,000; a comparable rental unit rents for \$2,400/month (\$28,800/year).
 - (a) Calculate the user cost under 2021 conditions ($r = 2\%$, $\delta = 2\%$, $\tau = 0.8\%$, $g = 6\%$) and under 2023 conditions ($r = 5.8\%$, $\delta = 2\%$, $\tau = 0.8\%$, $g = 0\%$). Is renting or buying more economical in each year?
 - (b) What annual capital gain rate g (holding 2023 parameters constant) would make the household indifferent between renting and buying? Show your algebra.

- (c) If the home price falls to \$560,000 but the rate remains at 5.8%, calculate the new user cost. Has affordability improved?
3. **Housing Supply Elasticity.** In Market A (Toronto), a 20% price increase led to a 4% increase in housing starts. In Market B (Houston), the same 20% increase led to a 30% increase in starts.
 - (a) Calculate the price elasticity of housing supply in each market.
 - (b) Draw the supply curves, starting from common initial conditions of \$400,000 and 50,000 units.
 - (c) Suppose demand increases by 30% in both markets (demand elasticity = -0.8). What is the new equilibrium price in each?
 - (d) What policy changes would move Market A toward Market B's supply elasticity?
4. **Affordability Index Calculation.** A city has the following characteristics: median benchmark price \$820,000; 20% down payment (\$164,000); 5.0% mortgage rate; 25-year amortization; 0.65% annual property tax rate; \$250/month utilities; median annual household income \$102,000.
 - (a) Calculate the monthly mortgage payment.
 - (b) Calculate the monthly property tax payment.
 - (c) Calculate the CMHC Affordability Index (total monthly housing costs as % of median monthly gross income).
 - (d) Compare to Table 3.2. Which Canadian city does this most closely resemble?
5. **Supply-Demand Gap.** Canada's population grew by 1,053,000 persons in 2022.
 - (a) At an average household size of 2.5 persons, how many new households were formed in 2022?
 - (b) Adding a demolition rate of 0.5% of the existing stock ($\approx 75,000$ units), what total units were needed?
 - (c) Actual starts were 263,000. What was the shortfall?
 - (d) If this shortfall persists for 5 years, how large a housing deficit accumulates? What effect would you expect on rents?
6. **Data Exercise.** Visit the CMHC website (www.cmhc-schl.gc.ca) and find the most recent Housing Market Outlook for your city.

- (a) Report the current rental vacancy rate and year-over-year change in average asking rents.
- (b) Report housing starts over the past 12 months.
- (c) Compare starts to CMHC's estimated required units. Is supply growing faster or slower than demand?
- (d) Identify one supply-side policy your municipal or provincial government has announced. Using Figure 3.1, evaluate whether it targets the right constraint.

Essay Questions

1. **(600–800 words)** “Canada’s housing crisis is fundamentally a supply problem, and demand-side policies cannot solve it.” Evaluate this claim using the supply-and-demand model. Your essay should: (a) explain why the short-run supply curve is nearly vertical; (b) use a diagram to show why demand subsidies raise prices in supply-constrained markets; (c) describe at least two supply-side policies and evaluate their effectiveness; and (d) discuss whether demand-side interventions could complement effective supply-side reforms.
2. **(500–700 words)** Compare Canada’s housing outcomes to those of Germany (Case Study 2). What are the three most important institutional differences between the two countries? For each, assess the feasibility and political obstacles to implementing a similar approach in Canada. Would you recommend that Canada move toward the German model? Defend your answer.
3. **(400–600 words)** The user cost of housing in 2021 was near zero in many Canadian markets, while the same calculation in 2023 yielded a cost above 8% of home value. Write an essay explaining what drove this change, what its economic consequences were for the housing market, and what it tells us about the relationship between monetary policy and housing affordability. Connect your analysis explicitly to Chapter 2’s discussion of the Bank of Canada’s rate decisions.

Discussion Questions

1. The opening vignette describes three households: Vancouver engineers who bought for \$2.1M, a nurse and teacher who cannot afford to buy, and a Prince George mill worker paying high rent. Using concepts from this chapter, explain why these three

households face such different situations despite the same national macroeconomic policies. Which economic force is most responsible for each household's situation?

2. BC's 2023 upzoning legislation allows multiplexes on formerly single-family lots. Who wins and who loses from this policy? Consider existing homeowners, would-be first-time buyers, renters, developers, and local municipalities. Does the aggregate economic benefit outweigh the distributional costs?
3. Should Canada maintain the Agricultural Land Reserve in BC, given that it constrains the supply of developable land around Metro Vancouver and contributes to housing unaffordability? Or does food security and environmental preservation justify the housing cost? How would an economist structure the argument for each side?
4. The Case Study on rent control concludes that it protects existing tenants at the expense of future tenants and new supply. Is this an acceptable trade-off? Consider the perspective of: (a) a long-term renter in a controlled unit in Toronto; (b) a newcomer searching for an apartment in the same city; and (c) a developer deciding whether to build new rental units. Is there a policy that achieves rent control's goals without its supply-side costs?

CHAPTER 4

Labour

This chapter is under development.

CHAPTER 5

Inequality

This chapter is under development.

CHAPTER 6

Immigration

This chapter is under development.

CHAPTER 7

Fiscal

This chapter is under development.

CHAPTER 8

Trade

This chapter is under development.

CHAPTER 9

Technology

This chapter is under development.

CHAPTER 10

Climate

This chapter is under development.

Glossary of Key Terms

This glossary defines key economic terms introduced in this textbook. Terms are listed alphabetically, with cross-references to the chapter in which each term first appears.

Adaptation (climate): Adjustments in economic activity, infrastructure, and behaviour to reduce the harm caused by climate change that is already occurring or locked in. *See* Chapter 10.

Aggregate Demand (AD): The total quantity of goods and services demanded across the entire economy at each price level; the sum of consumption, investment, government expenditure, and net exports. *See* Chapter 2.

Aggregate Supply (AS): The total quantity of goods and services that firms in an economy are willing and able to produce at each price level. The short-run aggregate supply (SRAS) slopes upward; the long-run aggregate supply (LRAS) is vertical at potential output. *See* Chapter 2.

Automation: The replacement of human labour with machines, software, or robots in the performance of specific tasks. *See* Chapter 9.

Bank of Canada: Canada's central bank, established in 1934 as a Crown corporation. It conducts monetary policy with the primary objective of maintaining low and stable inflation within a 1–3% target range. *See* Chapter 2.

Built-in inflation: Inflation arising from the expectation of future inflation; wage and price increases made in anticipation of continued price rises, creating a self-fulfilling wage-price spiral. *See* Chapter 2.

Canada Carbon Rebate (CCR): The payment made by the federal government to eligible Canadian households to return revenue from the consumer fuel charge. Paid quarterly as a flat amount per household. *See* Chapter 10.

Canada Child Benefit (CCB): A non-taxable, income-tested monthly benefit for families with children under 18. Launched in 2016 to replace previous child benefit programs; credited with significantly reducing child poverty in Canada. *See* Chapter 5.

Cap-and-trade: An emissions reduction policy that sets a maximum (cap) on total greenhouse gas emissions and allows firms to buy and sell emissions permits. *See*

Chapter 10.

Capital gains: The profit realized from selling an asset (real estate, shares, bonds) for more than its purchase price. *See* Chapter 5.

Carbon leakage: The displacement of emissions-intensive production to jurisdictions with lower or no carbon prices as a result of domestic carbon pricing policy. *See* Chapter 10.

Carbon tax: A direct fee levied on greenhouse gas emissions, proportional to the carbon content of fossil fuels burned. A Pigouvian tax designed to internalize the externality cost of carbon emissions. *See* Chapter 10.

Central bank independence: The principle that a central bank has the authority to set monetary policy without direct political interference from the elected government. *See* Chapter 2.

Climate mitigation: Actions that reduce greenhouse gas emissions or enhance carbon sinks, thereby slowing or reducing future climate change. *See* Chapter 10.

Consumer Price Index (CPI): A measure of the average change over time in the prices paid by urban consumers for a representative fixed basket of goods and services, produced monthly by Statistics Canada. *See* Chapter 2.

Core inflation: Measures of inflation that exclude or reduce the influence of volatile food and energy prices, providing a cleaner signal of underlying inflationary trends. The Bank of Canada uses CPI-trim, CPI-median, and CPI-common. *See* Chapter 2.

Cost-push inflation: Inflation caused by an increase in the cost of inputs to production (e.g., oil, wages), which reduces aggregate supply. *See* Chapter 2.

Data network effects: The process by which a platform with more users generates more data, improving its AI models and service quality, which attracts more users—a self-reinforcing competitive advantage. *See* Chapter 9.

Deflation: A sustained decrease in the general price level. Deflation can trap an economy in a downward spiral of postponed spending and rising real debt burdens. *See* Chapter 2.

Demand-pull inflation: Inflation caused by an increase in aggregate demand, resulting in “too much money chasing too few goods.” *See* Chapter 2.

Development charges: Fees levied by municipal governments on builders and developers to fund the infrastructure (roads, water, parks, transit) required to support new residential and commercial development. *See* Chapter 3.

Exclusionary zoning: Land-use regulations that prohibit multi-unit residential buildings (apartments, duplexes, townhouses) in residential zones, permitting only detached single-family homes. A primary driver of housing supply constraints in Canadian cities. *See* Chapter 3.

Fiscal policy: The use of government spending and taxation to influence aggregate demand and economic output. Controlled by the elected government through the Department of Finance Canada. *See* Chapter 1.

Fisher equation: The relationship between nominal interest rates, real interest rates, and inflation: $r \approx i - \pi$, where r is the real rate, i is the nominal rate, and π is inflation. *See* Chapter 2.

Free Prior and Informed Consent (FPIC): A principle under the United Nations Declaration on the Rights of Indigenous Peoples (UNDRIP) stating that Indigenous peoples have the right to give or withhold consent to resource development on their traditional territories. *See* Chapter 5.

Frictional unemployment: Short-term unemployment arising from the process of matching workers with jobs; inherent in any dynamic labour market. *See* Chapter 4.

Gender wage gap: The difference between median earnings of men and women in the labour market, typically expressed as women's earnings as a percentage of men's. *See* Chapter 4.

General-purpose technology (GPT): An innovation that is pervasive across many sectors, capable of continuous improvement, and able to generate complementary innovations. Examples include steam power, electricity, computers, and AI. *See* Chapter 9.

Gig economy: A labour market characterized by short-term contracts and freelance work, often mediated by digital platforms. Workers are typically classified as independent contractors. *See* Chapter 4.

Gini coefficient: A summary measure of income (or wealth) inequality derived from the Lorenz curve. Ranges from 0 (perfect equality) to 1 (perfect inequality). *See* Chapter 5.

Green industrial policy: Government interventions (subsidies, tax credits, public investment, mandates) designed to accelerate the deployment of clean energy technologies and the transition to a low-carbon economy. *See* Chapter 10.

Gross Domestic Product (GDP): The total market value of all final goods and services produced within a country's borders in a given period (typically one year). *See* Chapter 1.

Housing affordability: The relationship between housing costs (prices, rents) and household income. Commonly measured by the price-to-income ratio or the share of income spent on shelter. *See* Chapter 3.

Imputed rent: The estimated rental value of owner-occupied housing—the income an owner would receive by renting their property, or equivalently, the rent they save by owning. *See* Chapter 3.

Income quintile: One-fifth of the population ranked by income. *See* Chapter 5.

Inflation: A sustained increase in the general price level of goods and services in an economy over time. *See* Chapter 2.

Inflation targeting: A monetary policy framework in which the central bank publicly commits to maintaining inflation at a specific numerical target (in Canada, 2% within a 1–3% range). *See* Chapter 2.

Kuznets hypothesis: Simon Kuznets’s 1955 prediction that income inequality follows an inverted-U pattern as an economy develops: rising during early industrialization and falling as development matures. Empirically challenged by the experience of advanced economies since 1980. *See* Chapter 5.

Labour force participation rate: The percentage of the working-age population (15+) that is either employed or actively seeking work. *See* Chapter 4.

Labour productivity: Output per unit of labour input; typically measured as real GDP per hour worked. *See* Chapter 4.

Lorenz curve: A graphical representation of the income (or wealth) distribution, plotting the cumulative share of total income against the cumulative share of the population ranked from poorest to richest. *See* Chapter 5.

Low Income Cut-Off (LICO): Statistics Canada’s original poverty measure, based on the share of income devoted to food, shelter, and clothing. No longer promoted as an official measure but still published. *See* Chapter 5.

Low Income Measure (LIM): A relative poverty measure defined as income below 50% of median adjusted household income; the standard OECD cross-country poverty measure. *See* Chapter 5.

Marginal Revenue Product of Labour (MRP_L): The additional revenue generated by hiring one more unit of labour; the basis for the firm’s labour demand curve. *See* Chapter 4.

Market Basket Measure (MBM): Canada's official poverty line since 2019. Measures the cost of a specific basket of goods and services needed for a modest but adequate standard of living, with regional variation. *See* Chapter 5.

Minimum wage: A government-mandated floor on wages below which employers cannot legally pay workers. Set provincially in Canada, with a separate federal rate for federally regulated industries. *See* Chapter 4.

Monetary policy: Actions taken by a central bank to influence interest rates, the money supply, and ultimately inflation and economic activity. In Canada, conducted by the Bank of Canada through the overnight rate. *See* Chapter 1.

Monetary policy transmission: The process by which a change in the central bank's policy rate affects economic variables such as mortgage rates, the exchange rate, credit conditions, asset prices, and ultimately inflation. *See* Chapter 2.

Monopsony: A labour market in which there is only one buyer of labour (one employer), giving that employer market power to set wages below the competitive level. *See* Chapter 4.

Mortgage stress test: OSFI's regulatory requirement (B-20) that mortgage borrowers must qualify at the higher of their contract rate plus 2 percentage points, or 5.25%, ensuring borrowers have a buffer against rate increases. *See* Chapter 2.

Natural rate of unemployment: The rate of unemployment consistent with full employment—the sum of frictional and structural unemployment; the rate toward which the economy gravitates in the long run. *See* Chapter 4.

Negative externality: A cost imposed on a third party (not the buyer or seller) by an economic transaction that is not reflected in the market price. *See* Chapter 10.

Network effects: The phenomenon whereby the value of a product or service to any one user increases as more users adopt it. Network effects create tendencies toward winner-take-all market outcomes. *See* Chapter 9.

Normative economics: Economic analysis that makes value judgements about what *should* happen, rather than what *is* or *will* happen. Contrasts with positive economics. *See* Chapter 1.

Opportunity cost: The value of the next-best alternative foregone when a choice is made. The true economic cost of any decision. *See* Chapter 1.

Overnight rate: The Bank of Canada's policy interest rate—the target for the rate at

which major financial institutions lend to each other overnight. Changes in the overnight rate transmit through the financial system to affect mortgage rates, business loans, and the exchange rate. *See* Chapter 2.

Pigouvian tax: A tax set equal to the marginal external cost of a negative externality, designed to align private incentives with social costs. Named after economist Arthur C. Pigou. The carbon tax is a Pigouvian tax. *See* Chapter 10.

Platform economy: A market structure in which a digital intermediary (platform) connects two or more groups of users (e.g., buyers and sellers, drivers and riders) and generates value primarily from network effects and data. *See* Chapter 9.

Positive economics: Economic analysis that describes and explains economic phenomena without making value judgements; empirical and testable claims about how the economy works. *See* Chapter 1.

Poverty: A condition in which a household or individual lacks sufficient income or resources to meet a defined minimum standard of living. May be measured as absolute poverty (inability to meet basic needs) or relative poverty (income below a fraction of the median). *See* Chapter 5.

Price elasticity of supply: The percentage change in quantity supplied resulting from a 1% increase in price. Inelastic supply ($E_s < 1$) means supply responds weakly to price. *See* Chapter 3.

Productivity paradox: The observation that investment in information and communication technologies does not immediately produce measurable aggregate productivity gains—the lag between GPT adoption and full productivity impact. *See* Chapter 9.

Real interest rate: The nominal interest rate adjusted for inflation: $r \approx i - \pi$. The real interest rate measures the true cost of borrowing in terms of purchasing power. *See* Chapter 2.

Rent control: Government regulation limiting the rent a landlord may charge or the permissible annual rent increase for a residential unit. *See* Chapter 3.

Routine vs. non-routine tasks: In the task-based model of automation, routine tasks are governed by explicit, codifiable rules (most automatable); non-routine tasks require judgment, creativity, or physical dexterity in variable environments (less automatable). *See* Chapter 9.

Skill-biased technological change (SBTC): Technological change that increases relative

demand for skilled (typically higher-educated) workers and decreases relative demand for unskilled workers, widening the wage premium for education. *See* Chapter 5.

Social cost of carbon (SCC): The economic value of the damage caused by emitting one additional tonne of CO₂-equivalent into the atmosphere, including all present and future climate harms. *See* Chapter 10.

Stagflation: The simultaneous occurrence of high inflation and low economic output (or recession). A particularly challenging macroeconomic condition to address with conventional policy tools. *See* Chapter 2.

Structural unemployment: Long-term unemployment resulting from a mismatch between workers' skills and available jobs, or from geographic mismatch. *See* Chapter 4.

Task-based model of automation: An economic framework that classifies tasks (rather than whole jobs) by their susceptibility to automation, distinguishing routine from non-routine and cognitive from manual tasks. *See* Chapter 9.

Time inconsistency problem: The tendency for a policy-maker who has committed to a future policy to have incentives to deviate from that commitment once the relevant time arrives. The problem underlies the case for central bank independence. *See* Chapter 2.

Trade-off: The situation in which choosing more of one thing means getting less of another. A fundamental concept in economics reflecting scarcity. *See* Chapter 1.

Unemployment rate: The percentage of the labour force that is without work but actively seeking employment. *See* Chapter 4.

Union density: The share of paid workers who are members of a trade union. *See* Chapter 4.

UNDRIP (United Nations Declaration on the Rights of Indigenous Peoples): An international human rights instrument adopted by the UN General Assembly in 2007. Canada passed legislation in 2021 (Bill C-15) to align Canadian laws with UNDRIP. *See* Chapter 5.

Wage polarization: The pattern of employment and wage growth observed in advanced economies: expansion at the top (high-skill, high-wage) and bottom (low-skill, low-wage) of the distribution, with contraction in the middle (middle-skill, routine occupations). Attributed to skill-biased technological change and automation. *See* Chapter 9.

Wealth tax: An annual levy on the total net worth (assets minus liabilities) of individuals above a specified threshold, as distinct from taxes on income flows. *See* Chapter 5.

Winner-take-all market: A market in which one or a few firms capture most of the market share due to strong network effects, economies of scale, or first-mover advantages. Common in digital platform markets. *See* Chapter 9.

Bibliography
